Institutional info

Scopus

Save search

Set search alert

Advanced query

Search within: Article title, Abstract, Keywords
Search documents *: outdoor

AND

Search within: Article title, Abstract, Keywords
Search documents: therm*

AND

Search within: Article title, Abstract, Keywords
Search documents: noise OR acoust* OR sound

Add search field

Reset Search

Beta

Documents Preprints Secondary documents

674 documents found

Analyze results

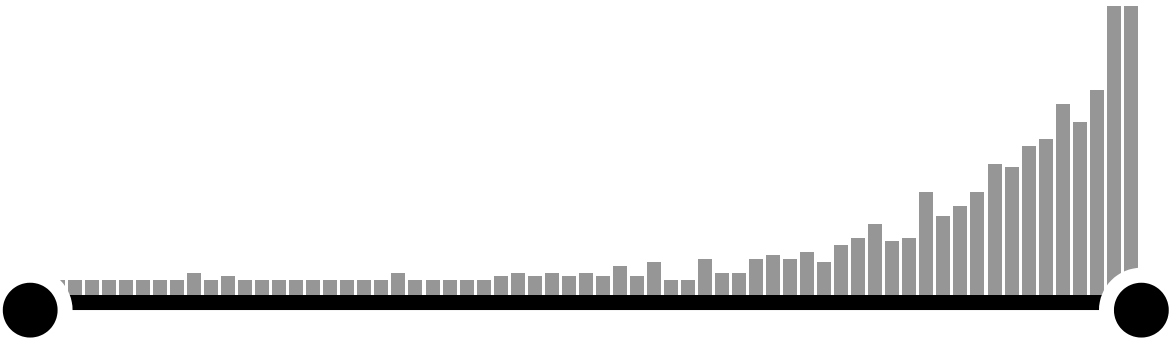
Refine search

Search within results

Filters

Year

Range Individual



from – to

Subject area

- Engineering 392
- Environmental Science 145
- Physics and Astronomy 143
- Computer Science 131
- Energy 98

Show all

Document type

- Article 409
- Conference paper 199
- Book chapter 25
- Review 21
- Conference review 18

Show all

Language

- English 640
- Chinese 30
- German 4
- Japanese 2
- French 1

Show all

Keyword

- Thermal Comfort 94
- Air Quality 82
- Ventilation 71
- Indoor Air Pollution 62
- Thermal 59

Show all

Country/territory

Source type

Source title

Author name

Publication stage

Affiliation

Funding sponsor

Open access

Export filter counts

All Export Download PDF (Beta) Citation overview ••• More Hide all abstracts Sort by Date (ne

	Document title	Authors	Source	Year	Citations
201	Article • *Open access* **Combined influence of multi-sensory comfort in winter open spaces and its association with environmental factors: Wuhan as a case study**	Li K.,Liu M.	Building and Environment , 248, 111037	2024	42

Hide abstract Find It @ UCL View at Publisher Related documents

Open space comfort significantly affects urban well-being, particularly in dense areas. This study uses drone imagery, climate measurements, and questionnaires to explore *acoustic*, aesthetic, light, *thermal* and overall comfort in Wuhan's pedestrian street, plazas, and parks, which proposing a hypothesis in both morphological factors and meteorological factors collectively influence comfort senses, and subsequently, overall comfort. The results indicate the sky view factor (SVF) contributes to all the comfort senses. Spaces with SVF over 32.9 % have 170 % higher light comfort and 160 % higher *thermal* comfort than those below 13.3 %. As SVF change from under 13.3 % to over 32.9 %, mean aesthetic comfort vote increases by 134 %, while *acoustic* comfort declines by 36 %. With Universal *Thermal* Climate Index (UTCI) from below 11.5 °C to higher than 21.5 °C, aesthetic comfort's influence weight on overall comfort rises by 85.4 %, while *thermal* comfort decreases by 32.4 %, indicating a falling importance of *thermal* but increasing that of aesthetic comfort with warmer environment. Also, the study emphasizes higher “cross-modal effects” between aesthetics and *acoustic* comfort compared to sunligt and *thermal* comfort. Research will help inform improvements to *outdoor* environments in similar climates.

202	Conference Paper **Semi-empirical model for amplitude fluctuations by atmospheric turbulence in aircraft flyover sound**	Lincke D.,Pieren R.	53rd International Congress and Exposition on Noise Control Engineering Internoise 2024, 6	2024	1

Hide abstract Find It @ UCL View at Publisher Related documents

Long-range *sound* propagation in the turbulent atmosphere leads to audible fluctuations in the *sound* amplitude. So far, the best currently available theoretical model for predicting amplitude fluctuations is limited to situations with low turbulence and short propagation distances. A semi-empirical model was recently introduced, which extends the range of applicability of the theoretical model based on a large dataset of aircraft flyover measurements. The dataset encompasses field measurements in diverse meteorological conditions and for different *sound* propagation distances. The semi-empirical model accounts for partial saturation of amplitude fluctuations. The semi-empirical model is used to generate auralizations of aircraft flyovers, considering the impact of atmospheric turbulence. The meteo- and distance-dependent characteristics of amplitude fluctuations in aircraft flyover *sound* are successfully reproduced by the model. The realism of the auralization is significantly improved compared to auralizations without consideration of amplitude fluctuations. The semi-empirical model enhances modeling techniques for *outdoor sound* propagation of elevated *noise* sources such as aircraft, drones, and wind turbines.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
203	**Auralization of atmospheric turbulence-induced amplitude fluctuations in aircraft flyover sound based on a semi-empirical model**	Lincke D.,Pieren R.	Acta Acustica, 8, 47	2024	5

Hide abstract Find It @ UCL View at Publisher Related documents

Atmospheric turbulence causes well-audible fluctuations in *sound* amplitude as it propagates through the atmospheric boundary layer. However, the best currently available theoretical model describing amplitude fluctuations is limited to relatively short propagation distances and weak turbulence. This article presents a semi-empirical model for predicting the standard deviation of amplitude fluctuations, combining the existing theoretical model of Ostashev and Wilson with empirical data. Leveraging a dataset of over 5000 aircraft flyovers across diverse meteorological conditions, the model incorporates factors such as frequency- and distance-dependent saturation of amplitude fluctuations, an effective propagation length based on the boundary layer height, turbulence decay time during transitional states of the atmosphere, and turbulence production by nocturnal low-level-jets. The mean absolute error of the semi-empirical model compared to the measurements is 0.4 dB. The application of the semi-empirical model to aircraft auralization is demonstrated for a flyover in two meteorological conditions. The findings will enhance the modeling of *outdoor sound* propagation for elevated *sound* sources like aircraft and wind turbines. Enhanced modeling of these fluctuations can considerably improve the realism of the listening experience in aircraft flyover auralization. Further, it helps estimating the measurement uncertainty in measured aircraft *noise*, particularly for maximum levels.

Discover early research ideas

View preprints published by authors to have an early idea of upcoming research documents.

View 45 preprints

		Document title	Authors	Source	Year	Citations
	Conference Paper					
204		**Investigation of the vibro-acoustic signature of a heat pump through an experimental transfer path analysis**	Neves L.,Moreira R., Herbst J.	53rd International Congress and Exposition on Noise Control Engineering Internoise 2024, 5	2024	0

Hide abstract Find It @ UCL View at Publisher Related documents

With the increasing interest on heat pumps due to their energy efficiency and environmentally favorable characteristics, concerns regarding their operational *noise* and vibration levels are becoming paramount, particularly in residential applications where occupant comfort is a priority. A heat pump consists of significantly distinct components regarding their physical properties, contributing to a complex dynamic system with respect to *noise* and vibration. To better understand its vibro-*acoustic* signature, this work presents an experimental transfer path analysis (TPA) of an *outdoor* unit of a heat pump system for residential applications. In this study, the main *noise* sources and the predominant vibration and *noise* paths are identified. Predictions of the total and partial responses at the receiver side is determined in order to quantify the contribution of each individual path. For the determination of the operational internal forces, the inversion matrix method is adopted. The findings of this study can contribute to the development of effective design guidelines and engineering solutions for minimizing the *acoustic* and vibrational footprint of residential heat pumps.

		Document title	Authors	Source	Year	Citations
	Conference Paper					
205		**Thermal Music: Exploring Sensation of Temperature as a Performance Parameter**	Snyder J.,Meggers F., Polito D.,Hu G.	Proceedings of the International Conference on New Interfaces for Musical Expression	2024	1

Hide abstract Find It @ UCL View at Publisher Related documents

	Document title	Authors	Source	Year	Citations

Thermal Music is a project to explore the possibilities of coordinating control of temperature sensation with control of *sound* and light. Unlike convective heating technologies that change the temperature of the air around us, radiative heating works by directly heating the surface of our skin via infrared. The technology is widely used in portable heaters in *outdoor* dining areas of restaurants. One interesting advantage of radiative heating is that if the source of infrared radiation is shaded by a *thermally* reflective surface, the change in perceived temperature by a person near the heater is very rapid. We worked to create a system that could quickly control robotic shades in front of a radiative heater to synchronize changes in perceived temperature with music and light. We also explored the inverse, using *thermal* camera input as a control method for audio, and presented both of these techniques at a "*Thermal* Music" concert at Princeton University in October of 2023.

Article • *Open access*

206 **Chronic impacts of natural infrastructure on the physical and psychological health of university students during and after COVID–19: a case study of Chengdu, China**

Peng Y.,He H.,Lv B., ... Song H.,Chen Q.

Frontiers in Public Health , 12, 1508539

2024

7

Hide abstract Find It @ UCL View at Publisher Related documents

Introduction: The COVID-19 pandemic has underscored the health benefits of green spaces, yet research on how specific elements of natural infrastructure affect well-being during the pandemic has been limited. Methods: This study, conducted at Sichuan Agricultural University with 300 students in 2022, investigated how urban natural infrastructure impacts physical and psychological well-being during the pandemic. Different aspects of natural infrastructure, such as *thermal* comfort, air quality (negative ion concentration), and *noise* and light levels, varied in their positive effects on students' health. Results: The findings revealed that 65.6% of university students felt reduced stress when engaging with *outdoor* spaces, and 72.8% of them renewed recognized the therapeutic value of nature. Discussion: The study emphasizes the importance of incorporating natural elements into urban planning to enhance *outdoor* activity and well-being, especially in post-pandemic settings. Recommendations are provided for future urban design to address the therapeutic needs of specific populations.

Article • *Open access*

207 **STHN: Deep Homography Estimation for UAV Thermal Geo-Localization With Satellite Imagery**

Xiao J.,Zhang N.,Tortei D., Loianno G.

IEEE Robotics and Automation Letters , 9(10)

2024

18

Hide abstract Find It @ UCL View at Publisher Related documents

Accurate geo-localization of Unmanned Aerial Vehicles (UAVs) is crucial for *outdoor* applications including search and rescue operations, power line inspections, and environmental monitoring. The vulnerability of Global Navigation Satellite Systems (GNSS) signals to interference and spoofing necessitates the development of additional robust localization methods for autonomous navigation. Visual Geo-localization (VG), leveraging onboard cameras and reference satellite maps, offers a promising solution for absolute localization. Specifically, *Thermal* Geo-localization (TG), which relies on image-based matching between *thermal* imagery with satellite databases, stands out by utilizing infrared cameras for effective nighttime localization. However, the efficiency and effectiveness of current TG approaches, are hindered by dense sampling on satellite maps and geometric *noises* in *thermal* query images. To overcome these challenges, we introduce STHN, a novel UAV *thermal* geo-localization approach that employs a coarse-to-fine deep homography estimation method. This method attains reliable *thermal* geo-localization within a 512-meter radius of the UAV's last known location even with a challenging 11% size ratio between *thermal* and satellite images, despite the presence of indistinct textures and self-similar patterns. We further show how our research significantly enhances UAV *thermal* geo-localization performance and robustness against geometric *noises* under low-visibility conditions in the wild.

Conference Paper • *Open access*

	Document title	Authors	Source	Year	Citations
208	**Review of urban water thermal environment cooling**	Qiu Y.F.F.,He B.J.	Iop Conference Series Earth and Environmental Science , 1363(1), 012073	2024	3

Hide abstract Find It @ UCL View at Publisher Related documents

The urban underlying surface undergoes constant changes due to the acceleration of urbanization. The increasingly prominent heat island phenomenon has led to higher energy consumption and worsened *thermal* conditions, significantly impacting people's daily *outdoor* activities and health. Water bodies, as a unique form of underlying surface, play a crucial role in enhancing the urban microclimate through their large heat capacity, high latent heat of evaporation, and low reflectivity. Previous research on the *thermal* comfort effect of water bodies primarily focuses on meteorology, environmental science, and landscape architecture. While quantitative studies have been conducted at both macro-scale city regions and micro-scale field domains to investigate the mechanisms behind these effects, there is a lack of translation from data into practical application strategies. In this study, we aim to adopt an interdisciplinary approach to review previous research achievements comprehensively. Firstly, we will examine the mechanisms and influencing factors involved in water body *thermal* regulation. Secondly, we will explore major methods and techniques for cooling urban water bodies while establishing an interactive model that considers landscape elements alongside physical factors affecting microclimate conditions. By developing a climate-adaptive design framework that addresses human *thermal* comfort requirements effectively, our goal is to create more livable and sustainable urban environments through scientifically *sound* strategies for cooling urban water bodies.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
209	**Actively ventilating calf hutches using solar-powered fans: Effects on hutch microclimate and calf thermoregulation**	Dado-Senn B.,Van Os J., Dorea J.,Laporta J.	Jds Communications, 5(1)	2024	12

Hide abstract Find It @ UCL View at Publisher Related documents

Although active ventilation via fans is an effective and widely adopted heat abatement method for use with adult dairy cattle, it has yet to be investigated in *outdoor* hutch-housed dairy calves despite most US calves being raised in such systems. We investigated a solar-powered fan system for *outdoor* calf hutches and its effect on hutch microclimate and calf *thermoregulation*. During summer, a 3 × 3 Latin square was replicated 4 times (n = 12 preweaning heifers) with 4-d exposure periods to minimally (CON; rear windows closed), passively (PASS; rear windows opened), or actively (ACT; solar-powered fan, activated at dry bulb temperature [T_{db}] > 21°C) ventilated hutch systems. Hutch microclimate and calf *thermoregulation* were evaluated either continuously (T_{db}, humidity, rectum surface temperature, and behavior) or after a daily 30-min inside restriction (air speed, air particle number, *noise* level, respiration, and sweating rate, and skin and rectal temperature). Active ventilation substantially increased hutch air speed relative to PASS and CON (1.76 vs. 0.19 vs. 0.05 m/s). However, PASS hutches had the lowest INT T_{db} (27.2 vs. 26.4 vs. 27.8°C), whereas ACT INT T_{db} was reduced at 0900 and 1000 h relative to CON but not PASS. Similarly, ACT reduced calf respiration rates and lowered rectum surface temperature at 0800 and 0900 h when compared with CON but not PASS. The lack of strong ACT influence on calf outcomes over PASS could partially be explained by the decreased proportion of time ACT calves spent inside their hutch (48.7 vs. 67.3 vs. 64.1% of each hour). Overall, ACT improved hutch microclimate and calf responses relative to CON but not PASS. Either ACT or PASS ventilation may be sufficient to provide heat abatement to continental hutch-housed calves.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
210	**A prenatal acoustic signal of heat reduces a biomarker of chronic stress at adulthood across seasons**	Udino E., Oscos-Snowball M.A., Buchanan K.L., Mariette M.M.	Frontiers in Physiology , 15, 1348993	2024	1

Hide abstract Find It @ UCL View at Publisher Related documents

During development, phenotype can be adaptively modulated by environmental conditions, sometimes in the long-term. However, with weather variability increasing under climate change, the potential for maladaptive long-term responses to environmental variations may increase. In the arid-adapted zebra finch, parents emit “heat-calls” when

	Document title	Authors	Source	Year	Citations

experiencing heat during incubation, which adaptively affects offspring growth in the heat, and adult heat tolerance. This suggests that heat-call exposure may adjust individual phenotype to hot conditions, potentially compromising individual sensitivity to cool weather conditions. To test this hypothesis, we manipulated individual prenatal *acoustic* and postnatal *thermal* experiences during development, and sought to assess subsequent chronic responses to *thermal* fluctuations at adulthood. We thus measured heterophil to lymphocyte (H/L) ratios in adults, when held in *outdoor* aviaries during two summers and two winters. We found that birds exposed to heat-calls as embryos, had consistently lower H/L ratios than controls at adulthood, indicative of lower chronic stress, irrespective of the season. Nonetheless, in all birds, the H/L ratio did vary with short-term weather fluctuations (2, 5 or 7 days), increasing at more extreme (low and high) air temperatures. In addition, the H/L ratio was higher in males than females. Overall, while H/L ratio may reflect how individuals were being impacted by temperature, heat-call exposed individuals did not show a stronger chronic response in winter, and instead appeared more resilient to *thermal* variability than control individuals. Our findings therefore suggest that heat-call exposure did not compromise individual sensitivity to low temperatures at adulthood. Our study also reveals that prenatal *sound* can lead to long-term differences in individual physiology or quality/condition, as reflected by H/L ratios, which are consistent with previously-demonstrated reproductive fitness differences.

Review • *Open access*

☐ 211 **A Nature-Inspired Green–Blue Solution: Incorporating a Fog Harvesting Technique into Urban Green Wall Design** — Hadba L.,Bitonto M.G.D., Oliveira M.,... Zanelli A., Silva L.T. — Sustainability Switzerland, 16(2), 792 — 2024 — 7

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

This research aims to explore the use of vegetation and nets to collect water from fog on facades to meet the needs of buildings' functional requirements, particularly *outdoor thermal* comfort, water demands, and encouraging sustainability by suggesting a new architectural green–blue wall system. The system is posited to be applicable within an urban context, given its minimal spatial requirements and adaptability to existing structures. However, similar challenges to those encountered by green walls are anticipated, wherein the provision of sustainable benefits is offset by the demands of maintenance and associated additional costs. For this reason, this paper is mainly divided into two parts: in the first part, green facades are explained, referring to their effect on urban environment, including *thermal* comfort, pollution absorption, *noise* pollution, and well-being, as well as types of plants to apply on green walls; the second part focuses on the fog collector as an irrigation system for green walls, analyzing its components, structure, and fabric, to identify its development margins in the construction industry. Fog harvesting initiatives predominantly focus on rural regions to cater to agricultural demands; however, limiting fog harvesting to agricultural settings is considered insufficient, as it represents a crucial solution for addressing water challenges in specific urban environments. Nevertheless, it is worth investigating the fog collector's potential for integrating water supply in urban environments as well. The study focuses on exploring the environmental benefits of fog harvesting and green walls, particularly through their combined implementation. The proposed review is significant for guiding the integration of a device into green facades, ensuring water self-sufficiency while concurrently addressing air purification, *noise* reduction, and *thermal* comfort for pedestrians and urban inhabitants. Nevertheless, it is worth investigating the fog collector's potential for integrating water supply in urban environments as well. The proposed review is, therefore, useful for integrating a device represented by the fog harvesting system, also identified in the text as the blue system, into the design of green facades, identified in the text as the green system, integrating the blue element in the design of the green wall to make them water self-sufficient and at the same time purifying the air, reducing *noise* pollution, or giving *thermal* comfort to pedestrians and inhabitants of the urban context.

Article • *Open access*

☐ 212 **Causal Artificial Intelligence-Driven Approach for HVAC Preventive Maintenance Explanation** — Kliangkhlao M., Haruehansapong K., Yeranee K.,Sahoh B. — IEEE Access, 12 — 2024 — 8

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

Preventive maintenance of heating, ventilation, and air conditioning (HVAC) systems aims to monitor and detect early-stage failures, thereby enhancing *thermal* comfort satisfaction and reducing energy consumption. However, it is a complex and uncertain factor, and technicians must understand the root cause of the problem to interpret

	Document title	Authors	Source	Year	Citations

prefailure events accurately. This study addresses this concern using a causal machine learning model, which encodes HVAC system behavior, indoor ambiance, and the *outdoor* environment based on random variables, with a particular focus on air conditioning units. It models their causal representation concerning prefailures using the structural causal model. It employs d-separation and d-connection to justify cause-and-effect relationships with the model and applies the expectation maximization algorithm to fit model parameters given observational data. The causal model is verified using a do-operator to rationalize the *sound* explanations (d-separated and d-connected) and employing odds ratio and confidential interval to prove their statistical strength. The model achieves high causal significance, with odds ratios of the submodel lying between 0.3-0.9 and 1.4-6.0 (exclude 1.0). The results indicate that the model and its explanations can encode human-like interpretation and achieve high causal significance aligned with real-world prefailure events. This refers to a reasoning process that is similar to that of humans, where information is included or excluded based on explanations of cause and effect. This confirms that the contribution of this study is suitable for causal-based decision-making systems for HVAC preventive maintenance.

Conference Paper

213 **Learned B-Spline Parametrization of Lattice Focal Coding for Monocular RGBD Imaging** | Guo J.,Xing Y.,Amata H., ... Dun X.,Cheng X. | 2024 IEEE International Conference on Computational Photography Iccp 2024 Proceedings | 2024 | 1

Hide abstract Find It @ UCL View at Publisher Related documents

Monocular RGBD imaging, also known as simultaneous all-in-focus (AiF) imaging and monocular depth estimation (MDE), represents a significant yet challenging task in computer vision. The crux lies in devising optical coding techniques to maximize the modulation transfer function (MTF) across various depths while minimizing their cross-correlation, all while aligning with the capabilities of image processing algorithms. End-to-end design of optics and algorithms offers a promising avenue towards achieving this holistic objective, but these approaches require solving non-convex inverse problems with millions of parameters. In this paper, we introduce a lattice-focal shape capable of nearly achieving the MTF bound as an initial solution, followed by employing B-spline parameterization for surface geometry representation to reduce the number of optimization variables. Further integration with the Restormer-based neural network, which possesses a global perspective, achieves high-performance RGBD imaging quality. Compared against state-of-the-art monocular RGBD imaging methods, our proposed approach improves the imaging peak signal-to-*noise* ratio (PSNR) by 3.0 dB and reduces the depth mean absolute error (MAE) by 39%. Experiments in real indoor and *outdoor* scenes validate the effectiveness of our method. The proposed approach paves the way for the development of monocular RGBD imaging.

Conference Paper

214 **AI based Fault Diagnosis in Power Transformer-A Deep Learning Framework using Convolutional Neural Network based on Infrared Imaging and Wasserstein Generative Adversarial Network tools** | Tanavade S.,Joy V.M., Rizwan S.M., Al Bulushi A.H. | 2024 32nd Telecommunications Forum Telfor 2024 Proceedings of Papers | 2024 | 2

Hide abstract Find It @ UCL View at Publisher Related documents

The condition of a transformer impacts the safety of electrical power networks. Transformer failures not only reduce system reliability but also cause serious accidents and financial losses. Various diagnostic techniques exist to monitor transformer operation, but most are challenging for real-time monitoring due to difficulty in online identification, *noise* interference, and high maintenance costs. This article proposes an overheating defect diagnosis method using forward-looking infrared (FLIR) *thermography* images and introduces a deep learning framework for power transformers. The proposed *thermal* camera system collects four types of FLIR images. Wasserstein Autoencoder Reconstruction (WAR) and Convolutional Neural Networks (CNN) models have been used to process these images. The trained CNN and WAR models have been combined in series to form a defect diagnosis module. The aim is to develop a cost-effective monitoring approach using remotely captured *thermal* images of a power transformer in an *outdoor* substation. The CNN model has been used to process these images to extract detailed feature parameters, preventing

	Document title	Authors	Source	Year	Citations

incomplete information issues from various feature-extraction techniques. Transformer health is classified into four categories: Monitor, Likely Defect, Repair at the Next Opportunity, and Major Defect Repair Immediately, based on temperature rise above ambient. The combined WAR-CNN technique has been used to improve remote monitoring. This approach has resulted in identifying common defects with high accuracy and has offered significant benefits over state-of-the-art methods, compact storage needs, and high diagnostic accuracy.

Article • *Open access*

215 **FM Rider: Two-FSK Modulation-Based Ambient FM Backscatter Over 100 m Distance** — Xu K.,Gong W.,Li Y., ... Song C.,Ding Y. — IEEE Transactions on Microwave Theory and Techniques, 72(9) — 2024 — 19

Hide abstract Find It @ UCL View at Publisher Related documents

Backscatter communication (BackCom) technology represents a low-cost and battery-free technology for Internet-of-Things (IoT) applications. One variant of BackCom, termed as ambient BackCom, which uses ambient signals to eliminate the need for a dedicated carrier source is more advantageous due to reduced system complexity and cost. Previous studies on ambient BackCom have primarily relied on amplitude modulation-based systems like ON OFF-keying (OOK) modulation, which achieved a maximum tag-to-receiver (Rx) distance of less than five meters. Alternatively, frequency-shift-keying (FSK)-based ambient BackCom systems have demonstrated improved distance, but with limitations. This work introduces an ambient broadcast frequency-modulation (FM) BackCom system utilizing in-band 2FSK-based modulation, designed to be compatible with both mono and stereo radio systems and extend communication distance. In order to successfully detect the backscatter 2FSK symbols, two pivotal technologies are employed: first, a quadrature demod-based in-phase/quadrature (I/Q) sample processing method enhances the signal-to-interference-plus-*noise* ratio (SINR). This enhancement has been substantiated through theoretical analysis, simulations, and cable-connected experiments. Second, a moving window-based 2FSK demodulation strategy is implemented, exhibiting resilience to tag frequency instability. The proposed system significantly enhances the tag-to-Rx communication distance, achieving an impressive range of over 100 m, a considerable improvement over the current state-of-the-art. The performance of the proposed system is evaluated in both cable-connected and *outdoor* environments, achieving a maximum bit rate of 333 bits per second (bps) at a distance of 58 m and a maximum communication distance of 107 m at a bit rate of 200 bps.

Conference Paper

216 **The Impact of the University Built Environment on Students' Mental Health and Well-Being: A Systematic Review** — Wang Y.,Zhang Y., Huang X.,Zhou Z., Cimillo M. — Lecture Notes in Civil Engineering, 393 — 2024 — 1

Hide abstract Find It @ UCL View at Publisher Related documents

Mental health and well-being of university student is critical in their academic performance, quality of life and personal growth at an important stage of their development. Among other factors, the built environment has a substantial impact, and students spend a significant part of their time in their university campus. This study reviews the relevant literature in order to synthesise the current body of knowledge and its gaps and limitations. Previous research from different fields of enquiry has demonstrated a range of effects on mental health for all the main aspects of the built environment, including its constitutive components (e.g. buildings, *outdoor* spaces), its spatial configuration (e.g. geometry, architecture), and its environmental characteristics (e.g. *thermal*, visual and *acoustic* factors). Furthermore, the current body of knowledge is still fragmented, lacking methodological consistency, systematisation and a specific framework for the educational spaces that can be translated in structured design guidelines.

Conference Paper

217 **Living Places: Healthy Homes for People & Planet. A simulation-based evaluation to ensure optimal indoor environment** — Christoffersen J., Maagaard S.,Reimer K., ... Di Santo N.,Sarran L. — PLEA 2024 Re Thinking Resilience Proceedings of 37th PLEA Conference Sustainable — 2024 — 1

	Document title	Authors	Source	Year	Citations
			Architecture and Urban Design the Book of Proceedings		

Hide abstract Find It @ UCL View at Publisher Related documents

The architectural design of Living Places, the way it is formed, shaped and designed provides the interior with plenty of daylight all year round. The design incorporates window control with its accessories to effectively mitigate overheating and create a comfortable *thermal* environment. Openable windows with high attention on easy air passages throughout the building certify high indoor air quality. Living Places and its design provide us with an indoor experience over time and space, which is an experience we enjoy and that connects us to the outside, making the *outdoor* dynamics integrated with the interior. We have used the compass model and simulated the performance, a total of 60.000 indoor climate simulations, to identify the most significant design parameters and to see if it provides the desired result. In addition, we have built a full-scale model and we are collecting insights through simple measurements of the indoor environment. Overall, the simulation shows that Living Places satisfies all main parameters of the 'healthiness' criteria: daylight, *thermal* comfort, indoor air quality and *acoustic* quality. It is a building concept that allows abundant daylight and fresh air, enhancing indoor climate quality, and showing how to build sustainability for people and the planet.

Book Chapter

218 **The Sick Building Syndrome as a Part of the Autoimmune (Autoinflammatory) Syndrome Induced by Adjuvant** — Pardo A.,Israeli E. — Autoimmune Disorders Adjuvants and Other Risk Factors in Pathogenesis — 2024 — 3

Hide abstract Find It @ UCL View at Publisher Related documents

Most people of all ages spend most of their days indoors. Today's lifestyle increases people's exposures to their indoor environments, which are sometimes of poor quality and result in adverse health effects. Sick building syndrome (SBS) is defined as a set of several clinically recognizable symptoms and ailments having no clear cause, reported by occupants of a building. The symptoms and discomfort experienced by building occupants correlate with times of their presence in the building and often interfere with their normal functioning. The syndrome has been prevalent in office buildings and evidenced in health care premises, schools, and other public and official buildings. Poor indoor environmental quality (IEQ) and especially that of indoor air quality (IAQ) appear to cause transient and long-lasting health effects among building occupants. Respiratory, skin, and eye irritation, an onset of allergies, asthma and skin rashes, nonspecific neurological symptoms such as headache and fatigue, and even multiple chemical sensitivity - all and more have been reported to be associated with SBS and poor IAQ. Many sources and conditions inherent in indoor environments have been seen as possible causal factors and triggers for the development of SBS. These have been categorized as chemical, physical, biological, psychosocial, and ergonomic. Emission of contaminants into the indoor environment, including organic and inorganic volatiles as well as particulate matter, may result from construction materials, infiltration of *outdoor* air pollution, combustion processes, *outdoor* air, the content of the building, occupants' activity inside a building, ventilation systems, and water supplies. *Thermal* discomfort and poor lighting and *acoustics*, while not related to specific morbidity, are believed to contribute to the development of SBS. Inadequate mechanical ventilation and flaws in the heating, ventilation and air conditioning (HVAC) system have been considered as key reasons for the development of SBS. Scientists regard indoor bioaerosols and especially fungal spores originating in molds as contributing factors to SBS, although evidence for a clear causality between bioaerosols and the abovementioned outcomes is not strong in the context of SBS. Researchers have suggested that personal factors and job satisfaction as well as stress and burnout, job strain, and anxiety and depression may also be associated with symptoms of SBS. Ergonomics, the interaction between the human mind and body, and the physical environment inside the building must also be considered contributing factors to the overall well-being and satisfaction of a building's occupants. The trend toward environmental sustainability and energy conservation and the emergence of green buildings have raised hopes that these will bring a decline in the incidence of SBS. Yet while less pronounced in frequency and intensity, dissatisfaction of occupants and building-related symptoms and illnesses in green buildings have continued to emerge. With technological advances and lifestyle changes, new, additional risk factors continue to develop, and the assessment of their occurrence and prevalence is warranted for different indoor environments. Unfortunately, despite the significant prevalence of poor IAQ and SBS, there is a remarkable worldwide

	Document title	Authors	Source	Year	Citations

paucity of assigned regulatory standards and provisions covering the different aspects of the issue. Shoenfeld et al. suggested that four conditions - siliconosis, macrophagic myofaciitis (MMF), the Gulf war syndrome (GWS), and postvaccination phenomena, which share clinical and pathogenic resemblances - may be included under a common syndrome entitled the “Autoimmune (Autoinflammatory) Syndrome Induced by Adjuvants” (ASIA). Comparison of the clinical manifestations, symptoms, and signs of the 4 conditions described by Shoenfeld et al. with those described for SBS shows that 9 out of 10 main symptoms are in correlation in all 5 conditions, namely myalgia, arthralgias, chronic fatigue, neurological cognitive impairment, fever, gastrointestin al and respiratory symptoms, skin manifestations, and appearance of autoantibodies. The authors further proposed several major and minor criteria, which, although requiring further validation, may aid in the diagnosis of this newly defined syndrome. We propose here that SBS may also be included as a part of “Shoenfeld’s syndrome.

Article

219 **Chemical, Physical and Microbial Parameters inside a Research Building Environment** — Awad A.H., Saeed Y., El-Gendy S.A., Hassan S.K. — Egyptian Journal of Chemistry, 66(13) — 2023 — 2

Hide abstract | Find It @ UCL | View at Publisher | Related documents

In the frame of the strategic plan Egypt 2030- a goal to create a supportive workplace environment. Indoor environment quality (IEQ) is an important factor to identify building's performance and comfort level. Key environmental parameters-related to IEQ of a multidisciplinary scientific research buildings were evaluated. Air biological (bacteria & fungi), chemical (PM, NO2, SO2, NH3, HCOH & VOCs) and physical (T°C, RH%, air velocity, *noise*, lighting & electromagnetic radiation) parameters were measured indoor/*outdoor*. The measurements were performed using integrated sampling and real-time portable data instruments. The 75th percentiles of airborne environmental bacteria, mesophilic bacteria and fungi were 5172 CFU/m^3, 1294 CFU/m^3 and 634 CFU/m^3, respectively. Indoor/*outdoor* (I/O) ratios of VOCs, NH3 and PM were ≥1. However, I/O ratios of HCOH, SO2 and NO2were ≤ 1. Indoor temperature and relative humidity ranged between 17-35.6°C and 30-66.5%, respectively. Operative temperature averaged 23°C; it located within the comfort zone ≥20 –≤24°C in only ~ 43% of the investigated sites. *Noise* value exceeded the minimum acceptable level of 60 dBA in ~77.7% of sites. Lighting intensity did not achieve the minimum acceptable level of 300 lox in~ 48.5% of sites. The electric and magnetic strengths significantly fluctuated, ranging between 1-242 V/m and 0.5-437A/m, respectively. *Thermal* comfort, indoor air quality, *noise* and lighting are used to calculate IEQ index (IIEQ). IIEQ simulation revealed that the majority of the investigated sites had good IEQ conditions. IIEQ helps identify ways to improve building environment.

Article • *Open access*

220 **The Role of Urban Vegetation in Counteracting Overheating in Different Urban Textures** — Gherri B. — Land, 12(12), 2100 — 2023 — 23

Hide abstract | Find It @ UCL | View at Publisher | Related documents

With growing global concerns about climate change, the significance of urban greenery in architecture and urban planning is becoming increasingly apparent. Urban vegetation naturally cools cities, provides comfort and clean air, and has positive social, health, and economic effects. It is essential to ensure passive *thermal* comfort and safeguard biodiversity. It is widely recognized that urban greenery not only withstands severe *outdoor* climatic events, but also symbiotically interacts with buildings and citizens. Several studies demonstrated the potential of vegetation to provide *outdoor thermal* comfort, air purification, *noise* reduction, and various other ecosystem services. To emphasize the potential of urban green spaces to interact with the local urban morphology in terms of microclimatic aspects, the research examines the dynamic connection between various urban textures and urban green spaces. This study emphasizes how urban green spaces, such as parks, green spaces, and urban greenery, respond to temperature variations in both the present scenario and the projected future. Central to this contribution is the examination of the relationship between urban vegetation and its potential to reduce and counteract urban overheating in both current and projected future scenarios. The aim is to evaluate the effectiveness of urban vegetation compared to dense urban textures. The interaction between urban block morphology, building types, vegetation, and microclimates is presented here for comparative assessment, highlighting the different *thermal* behaviour and *outdoor* comfort responses in various urban areas in current and projected scenarios. Using a microclimatic simulation tool, the research will delve deeper into the potential and constraints associated with the role of urban greens in addressing

	Document title	Authors	Source	Year	Citations

the increasing temperatures in climate change. This paper presents a comparative microclimatic evaluation of two selected green areas in Parma, Italy, within different urban contexts. The evaluation compares the current situation with a projected future scenario (2050) to determine the most effective factors for mitigating overheating phenomena in existing cities.

Article

221 **Indoor Air Quality and Indoor Environmental Quality: The role of occupational health surveillance and the cooperation between public and occupational health stakeholders** — Chirico F.,Settimo G., Magnavita N. — Journal of Health and Social Sciences, 8(4) — 2023 — 4

Hide abstract Find It @ UCL View at Publisher Related documents

This viewpoint emphasizes the growing significance of Indoor Air Quality (IAQ) in improving health and comfort in indoor environments, a crucial concern given that individuals spend approximately 90% of their time indoors. It conducts a comprehensive review of IAQ, highlighting its profound impact on overall health and quality of life, often exceeding *outdoor* environmental pollution. Our paper identifies various risks associated with IAQ, particularly for vulnerable groups such as children, the elderly, and those with chronic health conditions. It details the dual nature of indoor pollutants—external sources like traffic, heating and cooling of buildings, industrial emissions, and internal sources including combustion for food preparation and heating, building materials and furnishing, paints, cleaning products materials, and human activities. These pollutants, ranging from very volatile organic-VVOC, volatile-VOC, semi-volatile organic-SVOC compounds, particulate matter-PM10, PM2.5, and biological entities, necessitate extensive interdisciplinary research. The viewpoint discusses the severe implications of IAQ, especially in developing countries, where it contributes significantly to the global disease burden. It introduces the concept of Indoor Environmental Quality (IEQ), expanding the focus from just air quality to encompass lighting, *acoustics*, *thermal* comfort, and water quality. This holistic approach to IEQ is vital for ensuring health and well-being in indoor spaces. Furthermore, the paper explores the interconnection between IAQ and global health crises, such as climate change and the COVID-19 pandemic, and emphasizes the importance of maintaining good IAQ and visual and *thermal* comfort in workplaces. These factors are linked to improved health, productivity, and sustainable practices. The role of occupational stakeholders, employers, and medical professionals in creating a healthy work environment is critically examined. We argue for their collaboration in promoting health initiatives, ergonomic designs, and effective health surveillance. This collective approach aims to address the complexities of modern workplaces, fostering a healthier, more productive, and sustainable workforce. Take-home message: Enhancing Indoor Air and Environmental Quality lies in the collaboration between public and occupational health stakeholders. Together, they must promote healthy practices, ergonomic designs, and effective health surveillance to create spaces that support well-being, productivity, and sustainability, thereby contributing to a healthier society.

Article • *Open access*

222 **Examining the performance of thermal insulation materials used in buildings for noise insulation** — AKIN S.K.,ALPTEKIN M. — Case Studies in Thermal Engineering, 51, 103556 — 2023 — 7

Hide abstract Find It @ UCL View at Publisher Related documents

Urbanization is spreading throughout the world as a result of rising population. Accordingly, airway, seaway, road, and railway transportation networks are expanding. Advancements in transportation and population growth result in significant increases in environmental *noise*. Increased *noise* levels have a negative impact on people's health and cause a variety of issues such as sleep disorders and communication problems. Consequently, various noiseproofing (insulation) regulations and codes have appeared in various countries. However, studies conducted recently have shown that in order for providing sustainable quality of life, it is recommended that the *noise* level should not exceed certain values. On maximum *noise* levels for comfortable living places have revealed that adequate and enough insulation levels should be satisfied using the criteria specified in specifications. Nowadays, building energy savings have increased significantly with the use of *thermal* insulation materials used in buildings. This study investigates the performances of commonly used *thermal* insulation materials in buildings on *noise* insulation. Namely, the effects of various wall thicknesses and *thermal* insulation materials on *noise* insulation performance were studied. *Outdoor* and indoor *noise* levels were measured in a heavily trafficked workplace area of Konya, Turkey, which is identified as

	Document title	Authors	Source	Year	Citations

sample *noise noise* source of this study. It was found out in the study that while some *thermal* insulation materials provide adequate *noise* insulation in a specific *noise* frequency range, some others not adequate.

Article • *Open access*

223 **Developing a Feasible Integrated Framework for Occupational Heat Stress Protection: A Step Towards Safer Working Environments**

Gourzoulidis G.A.,Gofa F., Ioannou L.G., Konstantakopoulos I., Flouris A.D.

Medicina Del Lavoro, 114(5), e2023043

2023

7

Hide abstract Find It @ UCL View at Publisher Related documents

Background: Specialized occupational health and safety (OHS) issues are covered at the EU level through detailed legislation and guidelines. Unfortunately, this does not extend to occupational heat stress, not only in Greece but also (with few exceptions) internationally. One possible explanation could be the difficulty in accurately identifying the dangerous conditions, as many environmental and individualized elements are involved, and hundreds of "*thermal* stress indicators" are available. Another explanation could be the difficulty in adequately measuring hazardous conditions for workers affected more (i.e., *outdoor* and high intensity) since the biological protection framework is based on the human body's internal temperature. Methods: The Wet Bulb Globe Temperature (WBGT) has been proposed as the most efficacious *thermal* stress indicator. Since 2021, the Hellenic National Meteorological Service has provided 48-h WBGT forecast predictions to serve as a first level of alert. Real-time measurements and 48-h forecasts of WBGT are also available through a smartphone application. Additionally, as revealed when developing the occupational heat stress legislation in Cyprus and Qatar, crucial first steps are identifying the specific characteristics of worker exposure and the tripartite collaboration between employers, workers, and the State. Results: Evaluating the simplified WBGT forecasted values and the smartphone application estimates proved well-established. The *sound* scientific basis can be effectively combined with administrative measures based on the EU OHS legislative experience to produce practical solutions. Conclusions: As the climate crisis exacerbates, worker productivity and well-being will decline, underscoring the urgent need for an integrated protection framework. Such a framework is proposed here.

Article • *Open access*

224 **An integrated framework for residential layout designs: Combining parametric modeling, neural networks, and multi-objective optimization for outdoor activity space optimization**

Hu Z.,Zhang L.,Shen Q., ... Wang W.,Li K.

Alexandria Engineering Journal, 80

2023

28

Hide abstract Find It @ UCL View at Publisher Related documents

The quality of life of residents and community vitality is significantly enhanced by *outdoor* activity spaces within residential areas. Existing residential layout designs primarily focus on a restricted set of predefined layouts and most *outdoor* evaluation indices are limited to *thermal* comfort, thereby leading to an incomplete evaluation system. Furthermore, there is a scarcity of studies offering a comprehensive design process, from parameterized generation to performance evaluation. As such, there is a pressing need for an all-encompassing design framework that investigates a broad array of layout possibilities while providing multidimensional evaluations based on diverse metrics, thereby augmenting the accuracy and intelligence of residential layout designs. This research introduces a building layout generation framework that integrates parametric modeling (PD), neural network modeling (ANN), and multi-objective optimization (MOO) methods. The framework utilizes parametric design methods to construct a geometric model that adheres to building codes. Following this, it builds separate neural network models for three key metrics: sky view factor (SVF), sunshine duration, and *noise*, this approach is aimed at accelerating the evaluation process. This alternative model serves as the objective function for multi-objective optimization, identifying the optimal building layout that balances multiple metrics. Additionally, the framework utilizes the ideal point method to optimize the final Pareto solution set, thereby informing designers' and planners' program selections. Case studies of this framework indicate that the residential layouts generated can adhere to building codes while effectively balancing the three crucial metrics. Relative to the conventional modeling-evaluation process, this framework augments efficiency substantially. The design framework presented herein will offer comprehensive decision support

	Document title	Authors	Source	Year	Citations

for policy makers to optimize residential layout designs, further fostering the harmonious coexistence of people and cities, while advancing intelligent building design and sustainable urban development.

Article

225 **Field studies on indoor thermal environment and start-stop characteristics of the radiant air conditioning system during intermittent heating**

Qu M.,Yan X.,Sang X., ... Chu W.,Zhu X.

Journal of Building Engineering , 76, 107121

2023

15

Hide abstract Find It @ UCL View at Publisher Related documents

With the improvement of social economy, the heating demand of residents in Hot Summer and Cold Winter (HSCW) areas in China is increasing. The radiant air conditioning (A/C) system, is popular as a heating way due to its high comfort, low *noise* and high energy saving potential. There is still a lack of field research on the system under the intermittent heating mode in the HSCW area. To provide a guide for the operation mode of the radiant A/C system during intermittent heating in HSCW areas, this paper focused on the radiant A/C system with the fresh air system, and then studied system's practical operational parameters and dynamic characteristics in winter. The indoor and *outdoor* air temperatures and relative humidity, the water temperatures of inlet and outlet of the air source heat pump (ASHP) unit, the supply and return water temperatures of radiant terminals, as well as supply air temperature were analyzed in three stages, i.e., the initial start-up stage, the intermittent shutdown stage and the intermittent start-up stage. The results show that, it took 7 h and 50 min for the indoor air temperature to reach the set indoor temperature during the initial start-up stage. Compared to the initial start-up stage, the intermittent start-up stage reduced the time for the indoor air temperature to reach the set indoor temperature by 4 h and increased the indoor air temperature rise rate by 97%. The indoor air temperature rise rate was 0.46 °C/h and 0.91 °C/h, during the initial start-up and the intermittent start-up stage, respectively. The indoor air temperature decrease rate was 0.23 °C/h during the shutdown period. During the entire winter season, the ASHP unit switched on and off intermittently 10 times. The longest shutdown lasted for 111 h, and the indoor air temperature dropped to 17.2°C.

Article

226 **Physical phenomena of silica aerogels for vibro-acoustic applications**

Begum H.

Acoustics Bulletin, 49(5)

2023

0

Hide abstract Find It @ UCL View at Publisher

Mitigating *noise* emissions and transmission, combined with *thermal* insulation creates holistic and sustainable design for the construction of indoor and *outdoor* environments.

Article

227 **Impact of Indoor Environmental Quality on Students' Comfort in High School Buildings during the Summer Season in an Extreme Climate**

Amoatey P.,Al-Jabri K., Al-Saadi S.,Al-Harthy I., Al-Khuzairi M.

Journal of Architectural Engineering , 29(3), 04023014

2023

14

Hide abstract Find It @ UCL View at Publisher Related documents

Maintaining a satisfactory indoor environmental quality (IEQ) is an integral element of improving occupant health and well-being. In particular, IEQ is an essential factor in some facilities, such as school buildings, to improve student academic performance. This study employed an objective and subjective measurement approach to assess IEQ in a government high school (GHS) and a private high school (PHS) located in Oman, which is characterized by its extreme hot climate. A total of 212 high school students were surveyed during the summer season across 10 different classrooms to ascertain their perception of IEQ situations. The results suggested that most of the observed IEQ factors, including air temperature, relative humidity, and lighting levels across the classrooms in GHS and PHS, satisfied the ASHRAE standard, except the indoor air temperature, which was slightly above the maximum threshold of the European Standard (EN). The average indoor *sound* levels (>60 dB) exceeded critical limits of 35 dB for school buildings, which may pose the risk of high annoyance (HA) to 30% of students. Satisfactory predicted mean vote (PMV) and predicted percentage of dissatisfied (PPD) scores were obtained from both schools when they were evaluated against the ASHRAE standard compared with the EN standard. The maximum proportion of the students reported their classroom's air temperature (33%), relative humidity (25%), and air movement (47%) levels as not

	Document title	Authors	Source	Year	Citations

acceptable, and *thermal* comfort (33%), visual comfort (22%), and air quality (41%) as bad ($P < 0.001$). Multiple regression analysis revealed a strong linear relationship between indoor air temperature ($R2 = 0.89$) and other IEQ parameters, while PPD ($R2 = 0.45$) and PMV ($R2 = 0.17$) showed a weak relationship with *outdoor* environmental factors. *Outdoor* air temperature and humidity levels accounted for greater changes in indoor air temperature levels in the PHS than in the GHS. Providing increased urban vegetation, green façade, and retrofitting and the utilization of passive cooling could help improve IEQ in Oman schools.

Article • *Open access*

228 **Development of a doppler asymmetric spatial heterodyne interferometer for ground-based wind field detection at the 557.7 nm wavelength | 557.7 nm 波段地基探测风场的多普勒非对称空间外差干涉仪研制**

Liu H.,Jiang L.,Zhang X.-F., ... Tong S.-F.,Liu X.-Z.

Chinese Optics, 16(5)

2023

4

Hide abstract Find It @ UCL View at Publisher Related documents

A ground-based Doppler Asymmetric Spatial Heterodyne (DASH) interferometer with a high Signal-to-*Noise* Ratio (SNR) and large etendue (AΩ) with *thermal* compensation was developed to detect wind field information in the middle atmosphere. The detailed parameters and index of the DASH interferometer were developed for the 557.7 nm oxygen airglow spectral line. The system was designed with an expanded Field Of View (FOV) and *thermal* compensation. The half-FOV angle reached 2.815°, the etendue was 0.09525 cm^2sr, and the system's SNR was approximately 113.75. Through the *thermal* compensation design, the final optical path difference with temperature variation ($d\Delta d_0/dT$) was only 2.224×10^{-7}mm/°C. The optical system was designed and optimized according to the corresponding parameters. Image-side telecentric and bilateral telecentric optical system structures were used in the entrance optics and exit optics, respectively, and parameters such as telecentricity and distortion met the detection requirements. To verify the design results, a ground-based DASH interferometer experimental platform was constructed, and indoor and *outdoor* ground-based experiments were conducted. In the final experiment, clear interference fringes were obtained, which proves that the system design results of the DASH interferometer are reasonable, and the system's SNR and etendue meet the detection requirements.

Article • *Open access*

229 **Physics-based scintillations for outdoor sound auralization**

Bresciani A.P.C.,Maillard J., de Santana L.D.

Journal of the Acoustical Society of America, 154(2)

2023

12

Hide abstract Find It @ UCL View at Publisher Related documents

The *sound* propagating in a turbulent atmosphere fluctuates in amplitude and phase. This phenomenon, known as *acoustic* scintillation, is caused by random fluctuations in the *acoustic* refractive index of the air induced by atmospheric turbulence. Auralization techniques should consider this phenomenon to increase the realism of the synthetic *sound*. This paper proposes a physics-based formulation to model sequences of log-amplitude and phase fluctuations of a *sound* propagating in a turbulent atmosphere. This method applies to slanted and vertical propagation of the *sound*, which is useful for simulating elevated *noise* sources such as aircraft, drones, and wind turbines. The theoretical framework is based on the spatial correlation functions for the log-amplitude and phase fluctuations for spherical waves, the von Kármán spectrum, and similarity theories to model atmospheric turbulence. Two applications with audio files are presented to demonstrate the applicability of this method to tonal and broadband *noise*.

Article

230 **Integration of recycled waste paper with phase change material in building enclosure**

Abdulmunem A.R., Hussein N.F.,Samin P.M., ... Hussien H.A.,Ghazali H.

Journal of Energy Storage, 64, 107140

2023

47

Hide abstract Find It @ UCL View at Publisher Related documents

In this work, the use of recycled waste paper (RWP) in weight ratios of 25 %, 50 %, and 75 %, with phase change material (PCM) within polyvinyl chloride (PVC) panels was investigated experimentally as inner cladding for buildings.

	Document title	Authors	Source	Year	Citations

Five identical prototype rooms were built in Baghdad City (33.31° N latitude, 44.45° E longitude), the tests were done on the 21st of July 2022 with a minimum and maximum range of climate conditions by 30 °C and 41.1 °C of *outdoor* temperature, 10 % and 22 % of *outdoor* humidity, and 1.3 m/s and 5.2 m/s *outdoor* wind speed. Integration of PCM with 75 % of RWP within PVC panels resulted in enhancement by 7 % of the *acoustical* insulation, and reductions in the required of cooling load and electricity cost by 19 % and 16.3 % respectively, compared with filled panels by PCM only. Experimental results showed that these combinations of new insulation materials in buildings provided good *acoustic* and *thermal* insulation, and contributed to the reduction in the buildings' cooling and heating load.

Conference Paper • *Open access*

231 **On the Application of foam Glass in Interior Design** — Wang W.,Chi J.,Niu S. — E3s Web of Conferences , 406, 04033 — 2023 — 2

Hide abstract Find It @ UCL View at Publisher Related documents

foam glass is the waste glass under the action of other materials to make a new type of mixed material, its itself has good stability, corrosion resistance, heat resistance, not combustion, remain normal at high and low temperature, exposed to oxidation in the air, in building indoor and *outdoor* insulation, *noise* reduction, waterproof moisturizing [1], etc. In this paper, through the classification, performance, use and other aspects of foam glass, and through the method of data comparative analysis, the foam glass and various materials are compared, the data show that the foam glass has a good effect than traditional glass and *thermal* insulation materials.

Article

232 **Experimental study on comparison of indoor and multiple outdoor thermal environments excluding visual and acoustic interference** — Liu S.,Cao B.,Zhu Y. — Sustainable Cities and Society , 94, 104564 — 2023 — 32

Hide abstract Find It @ UCL View at Publisher Related documents

To make holistic assessment of the urban environment, it is important to consider both indoor and *outdoor thermal* environment. *Outdoor thermal* parameters play key roles in high quality urban life. This study concentrates on differences between indoor and *outdoor thermal* comfort excluding visual and *acoustic* interference. An experiment was carried out in indoor climate chamber and three types of *outdoor* spaces to explore subjective evaluation and physiological parameters of personnel. Subjects can achieve higher *thermal* comfort and are less sensitive to the change of *thermal* parameters outdoors. In a wide range of operative temperature and wind speed in *outdoor thermal* environment, subjects can be more inclined to stay outdoors. With VR glasses on between 27.5 and 33.5 ℃ under direct sunlight, people feel significantly cooler. In this study, VR glasses and *noise*-cancelling earphones were used in a pioneering way to exclude the influence of visual and auditory factors and comprehensively compare indoor and *outdoor thermal* comfort. According to the results, *outdoor thermal* environment can bring more comfort than indoor air conditioning environment, which highlights the importance of improving *outdoor thermal* environment. The benefits of comfortable *outdoor thermal* environments compared with indoor neutral environments can be further explored.

Article • *Open access*

233 **Influence of ground blocking on the acoustic phase variance in a turbulent atmosphere** — Ostashev V.E.,Wilson D.K., Hart C.R. — Journal of the Acoustical Society of America , 154(1) — 2023 — 5

Hide abstract Find It @ UCL View at Publisher Related documents

Sound propagation through atmospheric turbulence is important in many applications such as localization of low flying aircraft, sonic boom disturbances, and auralization of aircraft during takeoff and landing. This article extends an isotropic turbulence model in the atmospheric boundary layer to account for ground blocking of buoyancy-produced velocity fluctuations. The extended, anisotropic turbulence model is needed to correctly predict the effect of the largest velocity eddies on the statistical characteristics of *sound* signals. This model and geometrical *acoustics* are then employed to derive a closed-form expression for the variance of the phase fluctuations of a spherical *sound* wave for vertical and slanted propagation, without the use of the Markov approximation. A numerical analysis of this

	Document title	Authors	Source	Year	Citations

expression indicates significant anisotropy of the phase variance due to the buoyancy-produced velocity fluctuations with ground blocking such that it decreases in the vertical direction and increases in the near-horizontal directions. The newly formulated phase variance is compared with data from an *outdoor* experiment on vertical and slanted *sound* propagation. By accounting for ground blocking, much better agreement is obtained between the theoretical predictions and experimental data.

Conference Paper • *Open access*

234 **Indoor Environmental Quality assessment of mixed-mode ventilation with ceiling fans in the tropics**

Lei Y.,Peng Y.,Chong A.

E3s Web of Conferences , 396, 01086

2023

2

Hide abstract Find It @ UCL View at Publisher Related documents

Mixed-mode ventilation in buildings has a potential for significant energy savings. However, previous attempts in the tropics were unsuccessful due to the hot and humid weather. Indeed, there is little to no natural ventilation potential in tropical climates if conventional air conditioning systems are used. Additionally, existing studies typically do not consider the *noise* and air quality constraints while the performance of mixed-mode buildings is susceptible to *outdoor* conditions. Using elevated air movement has proven to be an effective strategy to offset the rise in the room temperature setpoint without sacrificing occupants' *thermal* comfort as it helps dissipate body heat through the convective cooling effect. Hence, this study aims at investigating the total indoor environment quality (IEQ) performance of mixed-mode ventilation operations with various air speeds based on tropically acclimatized subjects. To this end, a testbed was designed and constructed in Singapore, consisting of a variable air volume (VAV) system, ceiling fans with continuous speed control, personal ventilation (PV) desks, and plenum *acoustic* windows. Five operating modes have been tested: fully air-conditioned (AC) mode using VAV alone, hybrid air conditioned (AC+PV) mode using VAV and PV desks, concurrent ventilation (CV) mode using PV desks with the windows partially opened, natural ventilation (NV) mode with the windows fully opened, and *acoustic* natural ventilation (*Acoustic* NV) mode with the windows partially opened. 57 subjects were invited to the experiment and asked to complete a comprehensive IEQ survey under 16 different indoor conditions, which are combinations of these operating modes and air speeds ranging from 0.10 to 1.15 m/s. Experiment results show that NV and CV in Singapore can achieve more than 90% *thermal* acceptability with adequate air movement. More importantly, the median standard effective temperature (SET∗) for *thermal* neutrality is approximately 3 °C higher when operating in NV as compared to AC+PV or CV modes, indicating adaptation even when the modes switches are within the same day.

Article

235 **Thermal Load Prediction Algorithm Integrating Singular Spectrum Analysis and Neural Network | 集成奇异谱分析与神经网络的热负荷预测算法**

Wang Y.,Song Z.,You S., ... Wang K.,Song B.

Tianjin Daxue Xuebao Ziran Kexue Yu Gongcheng Jishu Ban Journal of Tianjin University Science and Technology , 56(6)

2023

5

Hide abstract Find It @ UCL View at Publisher Related documents

： Under the background of implementing the “double carbon” policy in China and upgrading the “intelligent district heating”in the heating industry， the operational control of the secondary network heating system was studied. By organically combining the singular spectrum analysis and the BP neural network algorithm from the mathematical field， an efficient *thermal* load prediction algorithm of the secondary heating network was formed. First， the original data were preprocessed， and the *noise* was filtered using the singular spectrum analysis algorithm. Then， the data features could be studied， and the load prediction model could be built by the neural network algorithm. Finally， the operation control strategy of the heat exchange station in the secondary network could be guided better with accurate load prediction. Using the building *thermal* load prediction method in 14 heating systems of a *thermal* power company in Tianjin in the heating season of 2018—2019， the singular spectrum analysis effectively filtered *noise* information from the original data. The *noise* of the *outdoor* temperature data， whose cumulative feature contribution rate was ＜9.3%， was eliminated. The time when the data fluctuation amplitude exceeded 10%

	Document title	Authors	Source	Year	Citations

decreased from 17.56% to 1.79%. Furthermore， it was proved that the BP neural network algorithm could help build an accurate model to predict the *thermal* load and the return temperature of the secondary network. The overall maximum absolute error of predicted values of the return temperature within 24 h was 0.60 ℃， and the maximum absolute errors of systems A and B were 0.43 ℃ and 0.65 ℃， respectively. It could provide a new idea for optimizing and adjusting the secondary network heating systems.

Article

236 **Combined effects of thermal and acoustic environments on outdoor human comfort in university campus** — Zhen M.,Chen Z.,Zou Q. — Urban Climate, 49, 101566 — 2023 — 34

Hide abstract Find It @ UCL View at Publisher Related documents

Urban *outdoor* public spaces are complex systems and the *thermal* and *acoustic* environments are the two main factors that affect their overall comfort. Many studies have analyzed the effects of the *thermal* environment and *acoustic* environment separately from different perspectives, while further research is required to understand the combined effects between two or more factors on the comfort in *outdoor* public spaces. In this study, we chose a university campus and assessed the combined effects of *thermal* and *acoustic* environments on human comfort at several spaces. Results showed that *thermal* evaluation is mainly affected by PET and *acoustic* evaluation is mainly affected by LAeq at most of the sites. Apart from that, interactive effects are only significant on TSV at PA, SQ and on OCV at AC. Prediction made by linear regression model revealed that there're obvious discrepancy on both *thermal* and *acoustic* evaluation between predictions and actual conditions. For overall comfort evaluations, linear regression analysis suggest that people feel “moderate” under ideal conditions where there're “no *thermal* stress” and LAeq =50 dB(A). The OCVeq chart analysis showed that the variation trend was separated by 0 with linear and centralized on each side, giving the hints that people judge their overall comfort and discomfort differently. This study concludes that it's possible to predict human comfort based on knowledge about the current *thermal* and *acoustic* environment, but due to limitations of this study, the reliability and accuracy requires further researches to confirm.

Article • *Open access*

237 **Heat vs. Health: Home Office under a Changing Climate** — Schaffernicht S.K.,Turk A., Kogler M.,... Konig B., Haluza D. — Sustainability Switzerland , 15(9), 7333 — 2023 — 8

Hide abstract Find It @ UCL View at Publisher Related documents

Stressors are especially widespread in urban agglomerations. Common themes of built environment interventions that support health and well-being are blue and green infrastructure, indoor and *outdoor* air quality, *thermal* comfort, access to natural lighting, and *acoustics*. Given the current megatrends of increasing summer temperatures and the high popularity of home offices, we aimed at modeling *thermal* comfort changes of people working at home in three Austrian cities (Vienna, Innsbruck, and Graz) during the next decades until 2090. We present findings based on (I) an inter-disciplinary literature search and (II) indoor and *outdoor* climate simulations for actual and future climate scenarios. Based on the results, we discuss the potential impacts for work and human health and well-being, and we suggest a framework for the home office in “post-COVID-19 Austria” that integrates social, ecological, and economic aspects. The results of our study indicate that, in future climate scenarios, overheating of the interior can no longer be prevented without active cooling measures and nature-based solutions. Recommendations on the adjustment of behavior under climate change, including greening, adequate ventilation, and cooling techniques, are thus urgently needed for employees who are working from home in order to maintain physical and mental health and wellbeing.

Article • *Open access*

238 **Summer weather perception and preferences in Powsin Culture Park (Warsaw, Poland)** — Rozbicka K.,Rozbicki T. — International Journal of Biometeorology , 67(5) — 2023 — 5

Hide abstract Find It @ UCL View at Publisher Related documents

Green areas situated inside the city play a very important role for many aspects. Among other, from the social point of view, they have a positive impact on the life of city habitants: directly improve well-being and health, reduce *noise*, provide opportunities for activity and recreation, improve the tourist attractiveness of the city, etc. The objective of

Document title	Authors	Source	Year	Citations

this study was to assess *thermal* sensations and preferences of people staying outdoors for recreational purposes in the area of the city park during summer 2019, as well as to identify how bioclimate perceptions are modified by personal factors (physical and physiological). In order to determine the optimum *thermal* zone for recreation and urban tourism during the summer, the regression model for mean *thermal* preferences (MTPV) every 1 °C in PET value intervals was calculated, and according to this procedure, the preferable spectrum of *thermal* conditions for tourism and recreation in Warsaw which is related to the PET value range between 27.3 and 31.7 °C. All age groups indicated the highest frequency of neutral *thermal* sensation vote, which decreased with feeling more extreme *thermal* conditions. In the case of classification by the gender, the men more than the women indicated the *thermal* conditions as neutral, “slightly warm,” and “warm.” Studies have shown that women were more sensitive than men to extreme *thermal* sensations especially “hot” and men more often than women indicated greater acceptance for comfortable and warmer *thermal* conditions.

Article • *Open access*

239 **Superhydrophobic and easy-to-clean full-packing nanopatterned microlens array with high-quality imaging** — Fang C., Xu W., Zhu L., Zhuang Y., Zhang D. — Optics Express, 31(9) — 2023 — 9

Hide abstract Find It @ UCL View at Publisher Related documents

The high-quality imaging and easy cleaning property of microlens array (MLA) are two very important factors for its *outdoor* work. Herein, a superhydrophobic and easy-to-clean full-packing nanopatterned MLA with high-quality imaging is prepared by *thermal* reflow together with sputter deposition. Scanning electronic microscopy (SEM) images demonstrate that the sputter deposition method can improve 84% packing density of MLA prepared by *thermal* reflow to 100% and add nanopattern on the surface of microlens. The prepared full-packing nanopatterned MLA (npMLA) possess clear imaging with a significant increase of signal-to-*noise* ratio and higher transparency compared with the MLA prepared by *thermal* reflow. Besides for excellent optical properties, the full-packing surface displays a superhydrophobic property with a contact angle of 151.3°. Further, the full-packing contaminated by chalk dust become easier to be cleaned by nitrogen blowing and deionized water. As a result, the prepared full-packing is considered to be potential for various applications in the *outdoor*.

Article • *Open access*

240 **Natural ventilation as a healthy habit during the first wave of the COVID-19 pandemic: An analysis of the frequency of window opening in Spanish homes** — Navas-Martin M.A., Cuerdo-Vilches T. — Journal of Building Engineering, 65, 105649 — 2023 — 18

Hide abstract Find It @ UCL View at Publisher Related documents

Since SARS-CoV-2 spread worldwide in early 2020, many countries established lockdowns for protection. With a main transmission by aerosols, ventilation was promoted. This article analyses natural ventilation of Spanish housing during the spring 2020. An online questionnaire was launched, obtaining for this study 1502 responses. The comparative window opening before and during confinement, and households, dwellings and home activity variables, were analysed. The binary logistic regression model before pandemic indicated that ventilating properly related to: a worse perceived IAQ (OR = 1.56); *thermal* adaptation measures, especially those that prefer to open/close windows (OR = 1.45); not having heating system (OR = 1.15); and using power to heat water (OR = 1.60). For the confinement period, the model highlighted: being an employee (OR = 1.88); using heavy clothing in the home (OR = 2.36); and again, open/close windows for adaptation (OR = 2.24). According to specific tasks in quarantine, frequent ventilation was boosted by: an increasing use of oven (OR = 14.81); and alteration of work-habits (OR = 2.70), sport-habits (OR = 1.79), and *outdoor*-activities (OR = 1.60). Thus, an adequate natural ventilation pattern during the quarantine was linked to low environmental comfort in general, by virtue of indoor air quality. This is corroborated by less *acoustic-thermal* insulation, worse indicators of heating use, and the adaptive response to opening/closing windows when external temperature changed.

Review

	Document title	Authors	Source	Year	Citations
241	**Acoustic tomographic reconstruction of temperature and flow fields with focus on atmosphere and enclosed spaces: A review**	Othmani C., Dokhanchi N.S.,Merchel S., ... Voelker C.,Takali F.	Applied Thermal Engineering , 223, 119953	2023	35

Hide abstract Find It @ UCL View at Publisher Related documents

Acoustic travel-time TOMography (ATOM) technique is being widely used as a remote sensing method to determine the air temperature and air flow velocity distribution for both *outdoor* and most recently indoor applications. The first process of ATOM measuring system corresponds to compiling the travel-times of a *sound* signal between a set of loudspeakers (transmitters) and microphones (receivers) distributed over the tomographic area, whose positions are known. These measured travel-times would then be used as the input data for inverse algorithms to reconstruct the airflow velocity and air temperature fields. Accordingly, the accuracy of the measuring system depends heavily on both the accuracy of the estimated travel times and the tomographic algorithms. Recent investigations on arrays of *acoustic* tomography are pointed towards optimization and reliability purposes. Many experimental layouts have been identified that could allow reducing the number of transceivers while keeping the accuracy of measurements. This paper gives a detailed review and discussion of recent advances in ATOM development applied to *outdoor* and indoor applications covering 90 articles published since 1990. It discusses the relevant challenges, opportunities, pros, and cons of existing approaches in relation to both travel time estimations and reconstruction algorithms, providing strategies to improve the sector of *acoustic* tomography. Accordingly, a comparative study of various reconstruction techniques is performed based on its systematic errors and theoretical strategies. This review inspires more successful explorations to improve the estimation of the air temperature and the airflow velocity using the ATOM technique.

Article

	Document title	Authors	Source	Year	Citations
242	**Multiple effects of visual-acoustic-thermal perceptions on the overall comfort of elderly adults in residential outdoor environments**	Du M.,Hong B.,Gu C.,Li Y., Wang Y.	Energy and Buildings , 283, 112813	2023	95

Hide abstract Find It @ UCL View at Publisher Related documents

We selected typical residential *outdoor* spaces and five common *sounds* (birdsong, dance music, traditional opera, conversation and traffic *sound*) with three A-weighted equivalent continuous *sound* pressure level (LAeq) (60, 70 and 80 dBA) as visual-*acoustic*-*thermal* conditions. 458 elderly adults completed a subjective questionnaire while environmental parameters (meteorological parameters, illumination intensity (LUX), visible green index (VGI) and LAeq) were simultaneously measured to determine effects of visual-*acoustic*-*thermal* perceptions on overall comfort. We found that: 1) When physiological equivalent temperature (PET) was above 43.80 °C, the elderly felt *thermally* uncomfortable. Older adults perceived traffic *sound* as *acoustic* uncomfortable when LAeq was higher than 66.1 dBA. A higher VGI decreased the sensitivity of respondents to LUX. 2) *Thermal* sensation vote (TSV) and *thermal* comfort vote (TCV) were susceptible to the *acoustic* and visual environments. The influence of the visual environment and PET on *acoustic* sensation vote (ASV) and *acoustic* comfort vote (ACV) were not significant. We found a significant correlation between PET and sunlight sensation vote (SSV). 3) *Acoustic* and *thermal* comfort had a one-vote veto tendency relative to overall comfort, but no absolute veto power. *Thermal* comfort was the most important factor affecting overall comfort in summer while *acoustic* comfort was the most important in spring. 4) A binary logistic regression to predict overall comfort of elderly adults had 84.7 % accuracy, indicating a good performance.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
243	**Outdoor environmental comfort evaluation for retail planning in a tropical business district using Integrated Environmental Modeller**	Lai P.-Y.,Koh W.S., Gopalan H.,... Tan E.,Tan J.	Plos One, 18(3 March), e0282106	2023	8

Hide abstract Find It @ UCL View at Publisher Related documents

This research proposes a simulation-based assessment of *outdoor thermal* and *acoustic* comfort for a planned business urban district in Singapore for retail planning using a customized OpenFOAM-centric multi-physics environmental simulation platform called the Integrated Environmental Modeller (IEM). IEM was employed to simulate the coupled impacts of solar radiation on wind and air temperature and wind and air temperature effects on traffic *noise* propagation in the district on the equinox and solstice day of the hottest period. Using IEM simulation results, we computed the *thermal* and *acoustic* comfort acceptability indicators derived from local field studies'

Document title	Authors	Source	Year	Citations

results. The spatial distribution of environmental comfort acceptability indicators in the worst-case scenario can be used to distinguish the zones exposed to *thermal* or *noise* influence. The *noise*-affected zones are near the main roads and overlap a part of the *thermal*-affected area. The *thermal*-affected area is almost everywhere in the studied sites in the worst-case scenario. Having *outdoor* retail spaces with both poor *thermal* and *acoustic* comfort is not recommended if the *thermal* and *acoustic* comfort cannot be improved simultaneously. For the high-level retail planning, a simplified parametric analysis considering solar irradiance blockage and wind speed enhancements, is provided. Considering the worst-case scenario, ≥50% *thermal* acceptability can be achieved by blocking 54%-68% solar irradiance among the pedestrian thoroughfares and the retail spaces. Coupled together, blocking the solar irradiance and enhancing the wind speed can further improve *thermal* comfort locally. These results can guide the retail mix (e.g., al fresco restaurants, pop-up kiosks etc.) near high footfall areas and provide reference for future plans combining landscape and infrastructure, (e.g., trees with shelter walkaways, green walls with *outdoor* ventilation fans etc.) taking into account the environmental acceptability of people working in or visiting the tropical urban district.

Article • *Open access*

244 **Implementation of learning content based on a competency approach in outdoor physical education classes** — Bula-Biteniece I., Radicuks R., Licis R., ... Smuka I., Lubinska I. — Journal of Physical Education and Sport, 23(3), 98 — 2023 — 5

Hide abstract | Find It @ UCL | View at Publisher | Related documents

Traditions for *outdoor* activities in Latvia have been influenced by both the pre-war and Soviet education system, as well as the new educational reforms, which are based on a competency approach. This article provides an overview of the teachers' and students' views on the implementation of Competency-based learning content in *outdoor* physical education classes. The qualitative study used a survey method. Students: the survey about the content of *outdoor* physical education classes was conducted in Riga secondary school No. 45 among 7th grade students (number of students-51), who participated in the implementation of established physical education class plans. Physical education teachers: selection criteria: 1) carries out *outdoor* physical education classes; 2) participated in the creation of curriculum based on the competency approach. Physical education teachers of 10 schools (5 in Riga, 5 in other cities of Latvia) were interviewed using a questionnaire with open-ended questions. The content analysis shows that most often the content of *outdoor* physical education classes is related to the classic physical education class tasks on the topics "Movement" and "Diverse activities to promote physical fitness", but less attention is paid to the topic "Human safety". In the opinion of both teachers and students, *outdoor* physical education classes are associated with the nature of adventure and an environment free from *noise* and crowding. Students prefer physical education classes that are dynamic and involving, with game elements, and the idea of physical education classes being conducted outdoors is attractive because it is related to the change of environment, also the nature of the adventure. The students indicate the weather conditions and the importance of *thermoregulation* as important factors that can affect the course of *outdoor* physical education classes. However, the teachers point out that more methodical materials are needed for a more successful *outdoor* physical education class.

Article

245 **Climate Change and Its Influence on the Energy Engineering Complex (Taking St. Petersburg as an Example)** — Mil'kov D.A., Yuferev Y.V., Tyutyunnikov A.I., Gorshkov A.S. — Thermal Engineering, 70(3) — 2023 — 1

Hide abstract | Find It @ UCL | View at Publisher | Related documents

Abstract: Observations of the Earth climate for the last 100–150 years have revealed that the atmospheric air temperature in the ground level shows quite a rapid growth. Megalopolises and other settlements behave as large sources producing *thermal* pollution of the Earth air basin. Transport facilities, heated buildings, and heat and electric networks release a significant amount of energy, which accumulates in the air medium of cities and then dissipates into the surrounding space. The most rapid change in the air temperature has been noted for the last 50 years. The greatest increase of the *outdoor* air temperature, e.g., in St. Petersburg, is observed in the cold period of the year. It has been determined that the average air temperature in the cold period of the year (from October to April) has increased by 1.51°C, whereas the average air temperature increased by 0.72°C for the same interval of time but in the warm period of the year (from May to September), i.e., half as much as in the cold period. It has been found from

Document title	Authors	Source	Year	Citations

studies of regulatory and reference documents that the standard climatic data specified in them lag significantly from the currently occurring climate changes. The article presents an analysis of the *outdoor* air temperatures in St. Petersburg for the period from 1992 to 2021 and an assessment of air temperatures in the coldest day and the coldest 5-day spell with reliabilities of 0.92 and 0.98 and also the absolute minimal air temperature for the considered 30-year interval of time. It is pointed out that the actual climatic indicators derived from statistical processing of daily data arrays are by 3–4°C higher than those specified in the relevant standards. As a result, when buildings are designed for the standard conditions specified by the existing regulations, the contractual loads turn out to be overestimated essentially in comparison with the actual ones, which is revealed in designing the configurations of settlement heat-supply systems. The consideration of actual climatic indicators opens up the possibility to decrease the contractual heat loads of consumers on a *sound* basis and, as a consequence, to achieve *thermal* energy saving without degrading the heat-supply quality.

Article

246 **Active Acoustic Sensing for 'Hearing' Temperature under Acoustic Interference** — Cai C., Pu H., Ye L., Jiang H., Luo J. — IEEE Transactions on Mobile Computing, 22(2) — 2023 — 20

Hide abstract Find It @ UCL View at Publisher Related documents

Though measuring ambient temperature is often deemed as an easy job, collecting large-scale temperature readings in real-Time is still a formidable task. The recent boom of network-ready (mobile) devices and the subsequent mobile crowdsourcing applications do offer an opportunity to accomplish this task, yet equipping commodity devices with ambient temperature sensing capability is highly non-Trivial and hence has never been achieved. In this paper, we propose *Acoustic Thermometer* (AcuTe+) as an interference-resilient ambient temperature sensor empowered by a single commodity smartphone. AcuTe+ utilizes on-board dual microphones to estimate air-borne *sound* propagation speed, thereby deriving ambient temperature. To accurately estimate *sound* propagation speed, we leverage the phase of chirp signals to circumvent the low sample rate on commodity hardware. In addition, we propose to use both structure-borne and air-borne propagations to address the multipath problem. Most importantly, we equip AcuTe+ with a mask-based desnoising algorithm to handle intensive *acoustic* interference. As a mobile, economical, highly accurate sensor, AcuTe+ may potentially enable many relevant applications, in particular large-scale indoor/*outdoor* temperature monitoring in real-Time. We have conducted extensive experiments on AcuTe+; the results demonstrate a median error of 0.6$^\circ°C even under severe *acoustic* interference (overall median 0.3$^\circ°C), and they also showcase the practical ability of AcuTe+ in real-Time distributed temperature sensing.

Article • *Open access*

247 **Beneficial Impacts on Environment and Society Through Smart Sustainable Maintenance of Public Real Estate** — Selicati V., Cardinale N. — International Journal of Heat and Technology, 41(1) — 2023 — 3

Hide abstract Find It @ UCL View at Publisher Related documents

Healthy, sustainable, energy-efficient buildings are more valuable since the green movement. Air quality, lighting, *acoustics*, and *thermal* conditions affect IEQ. Indoor air quality affects resident health. Chemicals or organisms contaminate the air. This work contributes to the goal of renovating and decarbonizing existing public administration buildings in the regional territory by integrating technologies and advanced building-plant systems management policies. This work focuses on building-plant energy diagnosis, its principles, regulations, and application procedures, especially IEQ and *thermo*-hygrometric comfort measurement and evaluation protocols. It follows a discussion of LCA for energy diagnosis and smart indoor air quality management and maintenance, such as quantifying human activity's impact on the indoor environment or adapting a modeling methodology to identify critical parameters to be monitored and controlled by building-plant systems' integrated sensor and actuator architecture. The goal is to find connections between indoor and *outdoor* air quality, occupant behavior in public buildings (especially schools and universities), and short- and long-term BRI. The study compares smart building diagnostic methods and procedures to European standards. This may improve Public Building Administration Social-LCA standards.

Article

	Document title	Authors	Source	Year	Citations
248	**The influence of outdoor thermal comfort on acoustic comfort of urban parks based on plant communities**	Mohammadzadeh N., Karimi A.,Brown R.D.	Building and Environment , 228, 109884	2023	85

Hide abstract Find It @ UCL View at Publisher Related documents

As human beings live in an environment where various factors interact and integrate, it seems vital to examine the effects of these factors on environmental perceptions. *Outdoor thermal* and *acoustic* environments are the main factors affecting the overall comfort of urban spaces. This research uses measurements and questionnaire surveys to explore the effects of *outdoor thermal* comfort dimensions on *acoustic* comfort, emphasizing the plant community in a city park (the El Goli Park of Tabriz, Iran). In this regard, the general comfort of the psychological equivalent temperature (PET) under different conditions of the *thermal-acoustic* environment has been evaluated and predicted. The results showed that a higher *thermal* sensation vote (TSV) and PET result in a significant decrease in soundscape quality (SQ), overall impression (OI), and the percentage of reported natural dominant *sound* (%RNDS), while an inverse trend has been reported in *thermal* comfort vote (TCV). On the other hand, by increasing TSV and PET, reported loudness (RL) has recorded an upward trend. However, this process was completely reversed in TCV. Furthermore, the analyzes showed that PET has a lower effectiveness compared to TSV and TCV and, it should be stated that the three indicators of *thermal* comfort did not show any significant relationship with landscape quality (LQ) and *sound* pressure level (SPL). These findings are important for managing and improving *outdoor acoustic* comfort and soundscapes, and it is expected that future studies will consider the acquired results in various green urban environments and provide a practical framework for landscape architects and urban planners.

Conference Paper • *Open access*

	Document title	Authors	Source	Year	Citations
249	**Integrating Simulation and Measurement Techniques to Model Outdoor Noise and Heat in Airport Neighbourhoods with Varying Urban Geometries**	Wuite G.,Peng Z.,Kim K., Lugten M.,Tenpierik M.	Building Simulation Conference Proceedings , 18	2023	0

Hide abstract Find It @ UCL View at Publisher Related documents

This study aims to evaluate the impact of different urban building geometries (six courtyards, two canyons, two slabs) on heat mitigation and aircraft *noise* attenuation, in order to support an evidence-based retrofit plan for future airport neighborhoods. Using'Pachyderm + ENVI-met simulations + field measurements', we found that the slanted-roof, low-rise courtyard exhibited optimal *acoustic-thermal* performance (SPL_{min} = 71.1 dB(A), σ_{UTCI} < 5 °C), while the mid-rise canyon demonstrated limited performance (SPL_{min} = 93.4 dB(A), σ_{UTCI} > 10 °C). These findings were observed under averaged boundary conditions of a 140 dB(A) aircraft *sound* source and a diurnal MRT range of 60 °C on a heatwave day in July 2022.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
250	**A Semi-Automatic Data Management Framework for Studying Thermal Comfort, Cognitive Performance, Physiological Performance, and Environmental Parameters in Semi-Outdoor Spaces**	Chen S.,Mihara K., Wong N.H.,Lee J.K.W., Tan C.L.	Sustainability Switzerland , 15(1), 183	2023	2

Hide abstract Find It @ UCL View at Publisher Related documents

Semi-*outdoor* space can be used as an alternative to short-term office activities to save office energy consumption and promote a healthy and nature-based working environment. This study evaluated the suitability of semi-*outdoor* space in four aspects including environmental measurements, physiological measurements, subjective measurements and cognitive performance tests. However, the manual processing and analysis of such multidimensional data can be time-consuming and error-prone. Hence, the objective of this study was to develop a semi-automatic method to manage and analyze the data from different instruments and platforms and two open-source applications (a stroop color and word test and a digit span test) for cognitive performance tests. These codes were critical to the success of the project, providing an effective framework for data extraction, data pre-processing, data analysis and performance tests. Eighty-nine people participated in the experiment of evaluation of *thermal* comfort, cognitive performance, physiological performance and environmental parameters in semi-*outdoor* spaces in a tropical setting. Each participant received cognitive tests to assess their selective attention, short-term memory,

	Document title	Authors	Source	Year	Citations

concentration and creativity quotient. Concurrently, qualitative measurements were conducted to assess *thermal* sensation, *thermal* comfort and *thermal* acceptability. The heart rate, skin temperature, and skin conductance of participants were measured throughout the experiments. Microclimatic variables such as illuminance, *noise* levels, dry-bulb air temperature, global temperature, relative humidity, air speed, and air direction were monitored simultaneously. To understand the effects of semi-*outdoor* spaces on participant performance, this study recorded participant performance in different environments through controlled experiments. Data related to participants in different settings include those shared (e.g., environmental measurement), and data unique to each participant (e.g., physiological performance). The results revealed that the subjects' cognitive and physiological performance did not change significantly after switching to the semi-*outdoor* space due to the availability of natural and mechanical ventilation, suggesting that short-term office activities in the semi-*outdoor* space are feasible in the tropics.

Conference Paper • *Open access*

☐ 251 **Cozie Apple: An iOS mobile and smartwatch application for environmental quality satisfaction and physiological data collection** — Tartarini F.,Frei M., Schiavon S.,Chua Y.X., Miller C. — Journal of Physics Conference Series , 2600(14), 142003 — 2023 — 34

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

Collecting feedback from people in indoor and *outdoor* environments is traditionally challenging and complex to achieve in a reliable, longitudinal, and non-intrusive way. This paper introduces Cozie Apple, an open-source mobile and smartwatch application for iOS devices. This platform allows people to complete a watch-based micro-survey and provide real-time feedback about environmental conditions via their Apple Watch. It leverages the inbuilt sensors of the smartwatch to collect physiological (e.g., heart rate, activity) and environmental (*sound* level) data. This paper outlines data collected from 48 research participants who used the platform to report perceptions of urban-scale environmental comfort (*noise* and *thermal*) and contextual factors such as who they were with and what activity they were doing. The results of 2,400 micro-surveys across various urban settings are illustrated in this paper, showing the variability of *noise*-related distractions, *thermal* comfort, and associated context. The results show that participants experienced at least a little *noise* distraction 58% of the time, with people talking being the most common reason (46%). This effort is novel due to its focus on spatial and temporal scalability and the collection of *noise*, distraction, and associated contextual information. These data set the stage for larger deployments, deeper analysis, and more helpful prediction models toward better understanding the occupants' needs and perceptions. These innovations could result in real-time control signals to building systems or nudges for people to change their behavior.

Conference Paper

☐ 252 **Multiple effects of visual-acoustic-thermal perceptions on the overall comfort of elderly adults in a humid subtropical city** — Du M.,Hong B. — Healthy Buildings 2023 Asia and Pacific Rim — 2023 — 0

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗

We selected typical residential *outdoor* spaces and five common *sounds* with three A-weighted equivalent continuous *sound* pressure level (LAeq) as visual-*acoustic*-*thermal* conditions. 458 elderly adults completed a subjective questionnaire while environmental parameters were simultaneously measured to determine effects of visual-*acoustic*-*thermal* perceptions on overall comfort. We found that: 1) *Thermal* sensation vote (TSV) and *thermal* comfort vote (TCV) were susceptible to the *acoustic* and visual environments. The influence of the visual environment and physiological equivalent temperature (PET) on *acoustic* sensation vote (ASV) and *acoustic* comfort vote (ACV) were not significant. 2) *Acoustic* and *thermal* comfort had a one-vote veto tendency relative to overall comfort, but no absolute veto power. *Thermal* comfort was the most important factor affecting overall comfort in summer while *acoustic* comfort was the most important in spring. 3) A binary logistic regression to predict overall comfort of elderly adults had 84.7% accuracy, indicating a good performance.

Book Chapter

☐ 253 **Aerogel-Inspired Materials Derived from Industrial Waste** — Duong H.M. — Springer Handbooks, Part F1485 — 2023 — 1

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

	Document title	Authors	Source	Year	Citations

In this chapter, we discuss fabrication methods, properties, and applications of eco-friendly macroporous aerogel-inspired materials made from recycled fibers of industrial waste feedstocks such as paper waste, fabric waste, plastic waste, and rubber tire waste, herein referred to as eco-aerogels. Properties afforded by such eco-aerogels are surveyed, including ultralow density, high porosity, large *sound* transmission loss, low *thermal* conductivity, high elasticity, and the ability to be made hydrophilic or superhydrophobic. The effects of fiber and crosslinker content on material microstructure are summarized. Applications including heat and *sound* insulation, oil spill and chemical remediation, *outdoor* apparel, *thermal* protection jackets, medical devices, personal care products, and packaging materials are reviewed.

Conference Paper

254 **A Low-Cost Thermal Imaging Device for Monitoring Electronic Systems Remotely** Humphreys J.,Secco E.L. Lecture Notes in Networks and Systems , 739 LNNS 2023 1

Hide abstract Find It @ UCL View at Publisher Related documents

This work aims at providing a low-cost solution to produce good quality *thermal* image that can be monitored remotely. The main application of such a system is to monitor electronic systems or installations as most industrial fires are caused by electronic or electrical systems. The paper presents the design and integration of the system, as well as a set of preliminary validation. The device was developed by combining and integrating a low-cost Raspberry Pi board and Pimoroni MLX90640 *thermal* camera with C++ and Python programming software. Multiple tests were then performed to validate the system's accuracy. The temperature of a daily life hot object was measured in time and compared vs the *thermal* measurements of a commercial IR gun (Etekcity Lasergrip 1080). Tests were performed indoor and *outdoor* to double check the effect of the environmental *noise*. The overall results show an average difference of ± 1.9 ℃.

Conference Paper

255 **Study on laser transmission characteristics of atmospheric turbulence disturbed by acoustic waves** Wang M.,Li Y. Proceedings of SPIE the International Society for Optical Engineering , 12554, 1255414 2023 0

Hide abstract Find It @ UCL View at Publisher Related documents

When laser propagates in the atmosphere, due to the influence of atmospheric turbulence, it will produce spot drift, light intensity flicker, beam spread and other phenomena, which restrict the development of laser engineering. *Acoustic* wave is a mechanical wave, when it propagates in the atmosphere, it will change the atmospheric density, affect the internal structure of atmospheric turbulence, and then affect the transmission of laser in atmospheric turbulence. In this paper, a coherent *acoustic* wave generator is designed, and the laser transmission experiment in the *outdoor* atmospheric turbulence environment disturbed by coherent *acoustic* waves is carried out. The variations of the beam drift fluctuation variance, scintillation index and beam diameter with coherent *acoustic* wave frequency and *acoustic* pressure level are studied. The results show that coherent *acoustic* waves will change the original structure of atmospheric turbulence and affect the variance of beam drift fluctuation, scintillation index and beam diameter when laser transmitted in atmospheric turbulence; The variance of beam drift fluctuation, scintillation index and beam diameter change with the change of *sound* source frequency and *acoustic* pressure level. This paper preliminarily explores the influence of *acoustic* wave disturbing atmospheric turbulence on laser transmission characteristics, and the research results provide a new idea for improving or interfering laser transmission characteristics in atmospheric turbulence.

Article

	Document title	Authors	Source	Year	Citations
256	**Research on Long Optical Path and Resonant Carbon Dioxide Gas Photoacoustic Sensor	长光程共振式二氧化碳气体光声传感器研究**	Li Z.-G.,Si G.-S.,Ning Z.-Q., ... Si B.-B.,Yang C.-P.	Guang Pu Xue Yu Guang Pu Fen Xi Spectroscopy and Spectral Analysis , 43(1)	2023	2

Hide abstract Find It @ UCL View at Publisher Related documents

Carbon dioxide (CO_2) is the raw material of plant photosynthesis and greenhouse gas. Its excessive emission will affect the ecological environment of animals and plants. Under the background of carbon peaking and carbon neutrality, it is of great significance to develop high-sensitivity CO_2detection devices. In order to monitor the change of CO_2concentration in the atmospheric environment, a long optical path and resonant CO_2gas photoacoustic sensor was designed, and a photoacoustic detection setup was built. Adistributed feedback laser (DFB) with a central wavelength of 2004 nm was used as the excitation light source. The laser entered into a spherical absorption cell made of diffuse reflective material, and multiple reflections occurred in the cell to increase the absorption path of the gas. To reduce the *thermal noise* generated by the absorption of light energy by the absorption cell, the outside of the cell was wrapped by two aluminum hemispheres with high *thermal* conductivity. An *acoustic* tube was coupled with the absorption cell. When the tube worked in the first-order longitudinal resonance mode, the photoacoustic signal was amplified and reached the maximum at the end of the tube. The CO_2relaxation rate was greatly accelerated, and the *thermal*, *acoustic* conversion efficiency was improved by saturated humidifying the sample, which further amplified the photoacoustic signal. The photoacoustic signal produced by the humidified sample was about 2. 1 times that of the dry sample. The photoacoustic detection setup was calibrated with a series of wet CO_2samples, and the results showed a good linear relationship between photoacoustic signals and concentrations. On this basis, the accuracy and stability of the setup were verified through the detection experiment of standard gas. Allan variance was used to evaluate the detection sensitivity of the setup under long-time operation. When the average time was 865 s, the detection sensitivity was - 0. 35 × 10^{-6}. Compared with the traditional T-type photoacoustic cell, the optical path was increased by - 20 times, and the photoacoustic signal was amplified by - 6 times. The setup was used to detect CO_2in the *outdoor* environment for 10 hours, and the average concentration of *outdoor* $CO_2$was - 381 × 10^{-6}. In conclusion, due to the combination of a long optical path, *acoustic* resonance and humidified samples, the photoacoustic signal of $CO_2$was effectively increased, which provided a relevant reference for the design of gas photoacoustic sensor and detection setup.

		Document title	Authors	Source	Year	Citations
257	Book Chapter	**Thermal-Acoustic and Indoor Comfort Performance of Nature-Based Materials for Green Buildings**	Santini C.,Fabiani C., Kousis I.,Pisello A.L., Cotana F.	Springer Tracts in Civil Engineering	2023	2

Hide abstract Find It @ UCL View at Publisher Related documents

Earth is an easy-to-find resource, and its application in construction involves reusing a natural material that can provide good energy efficiency and environmental comfort, thanks to its inertia and hygroscopicity. This chapter analyses the *thermo-acoustic* performance of earth-based concretes produced using the Shot-earth technique and traditional moulding with the addition of different amounts of fibers in the mix for structural health monitoring (SHM) purposes (smart-earth). The two types of samples differ in the method of fabrication, which leads to different *thermal*, *acoustic*, and surface porosity performance, although the quantities of each component are the same. In addition to this type of compound, suitable for structural applications, a fibers-functionalized mix was also developed and tested as an innovative plaster for *outdoor* applications. Each component's most critical *thermal acoustic* and hygroscopic performance was evaluated at a small scale and in the field to identify their potential in building applications.

		Document title	Authors	Source	Year	Citations
258	Book Chapter	**Climate Parameters, Heat Islands, and the Role of Vegetation in the City**	Xhexhi K.	Urban Book Series	2023	11

Hide abstract Find It @ UCL View at Publisher Related documents

Climate has a strong influence on urban planning and also plays a fundamental role in soil composition affecting the character of plants and animals. The climate is a combination of different meteorological factors that characterized a specific region over a specific time. The movement of the Sun and Earth inclination toward it is the most important

Document title	Authors	Source	Year	Citations

factors which determine the characteristics of the climate. The global movement of the air from equator toward poles and vice versa influences also drastically the climate change. In this section are also included breezes. Also air relative humidity plays an important role shaping the climate. It plays a very important role in cloud formation. The built environment can influence the urban climate of the city through height, form, and applied building materials. Building form and shape, building materials, openings in the façade, and their orientation often are determined by climate parameters. Urban density, directions, and street network and their shape are related to the climate change, apart from topography. Density of urban structures produces important effects on the microclimatic conditions in external spaces. The bioclimatic behavior of each type of *outdoor* spaces depends on the relationship with its surrounding elements. The city of Tirana is included in Mediterranean climate characterized by hot dry summers and mild cold wet winters with high daily *thermal* excursions. All of the countries bordering the Mediterranean Sea benefit from the Mediterranean climate. Specifically spotted, Tirana is part of Mediterranean coastal zone (<22 °C in temperature) with an annual temperature of 15.3 °C. In the inner city can be found areas with high pollution, which normally contain a high percentage of business and industry, settlements, and traffic, so-called “sealed” areas. They constitute “hyperthermic areas” or “heat islands”. The term “heat island” describes built-up areas which are hotter than nearby rural areas. Heat islands can affect communities by increasing summertime peak energy demand, air-conditioning costs, air pollution, greenhouse gas emissions, heat-related illness and mortality, and water quality. The aim of the bioclimatic design is also the reduction of the heat-island phenomenon. The use of materials with different energy absorption capabilities and integration of green spaces into urban life is important. The vegetation effects on microclimate and in the built environment are present. Trees are necessary and indispensable to human life. They provide shelter for us, provide food, and their fibers cloth us. Implementation of the green belts in the city is one of the strategies in order to reduce heat-island effect. Vegetation also can provide a deep feeling of attachment to certain places and trees which are part of it. The use of trees avoids monotony and ensures diversification. They play a crucial role creating microclimates which influence the *thermal* comfort of people reducing heating or cooling loads. Vegetation also creates cooler microclimates around the building through evapotranspiration, reduces wind speed, regulates humidity, and reduces temperature. It is observed that tree species have a greater influence on humidity than on temperature. Trees can also absorb and reduce the reverberation time of the *sound*, behaving as *sound* buffers. The benefits of tree plantation in the inner city are enormous and they can’t be neglected.

Book Chapter

☐ 259 **A Methodology to Optimize Thermal Conditions of Built Forms for Humans and Birds in a Birds Sanctuary** — Sowmiya R.,Vijayalaxmi J. — Lecture Notes in Civil Engineering , 316 — 2023 — 0

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

Ecological sustainability should be a holistic approach, where we consider all the biotic and abiotic factors of an ecosystem. Due to climate change, changes in vegetation patterns, a rapid increase of urbanization, depletion of resources, etc., in many ecological cases, we have crossed the line of conservation and now we face the phase of rejuvenation or revival of an eco-sensitive space. For the benefit of flora and fauna, environment much research is done, but the consideration of other species is very less. Sanctuaries being home to 80% of the migratory and native birds has to be rejuvenated by making them a vital space for avian habitat. This paper deals with the impact of buildings in a bird sanctuary on the indoor *thermal* performance of humans and the impact on birds (breeding temperature) due to human intervention. The site chosen for this study is the Chitrangudi Birds Sanctuary in Tamil Nadu, India. To achieve the optimal *thermal* conditions for the birds in and around a bird sanctuary, many stages of analysis and strategies benefiting both humans and birds have to be considered. The strategies derived after analysis for the site considering human adaptive *thermal* comfort and bird breeding temperature of each species are—roof, WWR%, shading device, the shape of the building, and jaali. Quantification of different passive strategies that balance both indoor comfort for humans and *outdoor* comfort for birds is carried out. The various conditions for which iterations are carried out are Orientation and form—10 conditions, aspect ratio:11 ratio each for two perimeter conditions (22 conditions of aspect ratio), roof structure and material—15 conditions of various pitch angles and overhang, WWR and Height—20 Opening condition of WWR, building height—7 height conditions, 5 Sill Heights and 5 Window heights, level of openings—8 conditions, perforated screens—6 configurations, and shading devices—8 conditions for optimizing indoor air temperature as well as lighting levels. Future studies can include *noise*. This study establishes a methodology for optimizing indoor comfort for humans with minimum disturbance to the requirement

	Document title	Authors	Source	Year	Citations

of avian microclimate in a bird sanctuary. This methodology can be followed in other sanctuaries to ensure safe human interventions to assist flora and fauna.

Conference Review

260 **2023 IEEE 17th International Conference on the Experience of Designing and Application of CAD Systems, CADSM 2023 - Proceedings** — [No Authors Found] — 2023 IEEE 17th International Conference on the Experience of Designing and Application of CAD Systems Cadsm 2023 Proceedings — 2023 — 0

Hide abstract Find It @ UCL View at Publisher

The proceedings contain 45 papers. The topics discussed include: designing the topology of microelectromechanical systems by machine learning methods; development of a remote-control system for a mobile vibration-driven robot; models of recurrent distributions statistical averaging for electronic components with fluctuations; inverse-dynamic neural controller simulation; a method of creating virtual pixels in matrix; the computer modeling of the *thermal* agent hydrodynamics through the alcohol distillery stillage stationary layer; method for estimating the topological structure of self-organized networks; a new hybrid method for predicting recommendations for collaborative recommender systems; software-defined transmitter to support automatic dependent surveillance-broadcast; deductive matrix synthesis for fault simulation; investigation of response from the micro objects of complex shape irradiated by *acoustic* wave; *outdoor* positioning for industrial workplace; a software complex for researching algorithms for working with graphs; and hydrogen adsorption in porous silicon: simulation and control method.

Conference Paper

261 **Understanding Indoor Environmental Quality (IEQ) of Naturally Ventilated Educational Buildings—A Systematic Literature Review** — Ali A.M.,Acharya S. — Smart Innovation Systems and Technologies, 343 — 2023 — 1

Hide abstract Find It @ UCL View at Publisher Related documents

Most people in developed and developing countries spend an average of 80–90% of their time indoors. Due to this, the importance of a building's Indoor Environmental Quality (IEQ) is of utmost importance. Several studies have shown that IEQ impacts building occupants' health, comfort, safety, well-being, efficiency, and productivity. In the urban context of developing countries, most buildings are naturally ventilated and are easily susceptible to poor IEQ compared to air-conditioned buildings due to their connection with the *outdoor* environment. In addition, most developing countries are in a hot and humid climate, making such spaces even more vulnerable. This issue becomes even more critical in educational buildings, as children spend more than 25% of their time in the classroom, and their lungs and other organs are developing. Therefore, a systematic literature review of naturally ventilated educational buildings has been conducted to understand the IEQ performance of such buildings and the importance of the four major IEQ parameters. It was found that the *thermal* environment causes the most discomfort, followed by the *acoustic* environment, and the visual environment causes the least discomfort. But, there are some confusions with IAQ. Therefore, further investigation is required.

Article

262 **Thermal characteristics of vehicle wake induced by the interaction between hydrodynamic wake and cold skin** — Luo F.,Shuai C.,Du Y.,Ma J. — Ocean Engineering, 267, 113272 — 2023 — 20

Hide abstract Find It @ UCL View at Publisher Related documents

The wake temperature signal generated by the motion of an underwater vehicle can be captured by infrared detectors, threatening the concealment of the vehicle. To clarify the formation mechanism and physical phenomena of *thermal* wake, detached-eddy simulations are conducted to establish a *thermal* wake calculation model, and the volume-of-fluid model is employed to capture the *thermal* wake characteristics of the water–air interface. Based on

Document title	Authors	Source	Year	Citations

the interface heat diffusion and transfer, a high-resolution cold skin boundary layer is established to study the interaction between hydrodynamic wake and temperature boundary layer. The numerical results show that the surface *thermal* wake presents a long regular strip-shaped characteristic with eddy current structure, which is basically consistent with the results of *outdoor* wake infrared test in phenomenon and overall intensity. The wake *thermal* signature originates from the modulation of the cold skin microstructure at the water surface by the wake vorticity. By applying the proposed evaluation method for the *thermal* characteristics of water surface wakes, the influence of navigation parameters and cold skin parameters on the *thermal* characteristics of water surface wakes is systematically discussed, and the *thermal* characteristics of wakes and their influencing factors are summarized.

Conference Paper • *Open access*

263 **An environmental quality assessment of office buildings: The impact of glass façade on internal and external users** | Muraj I.,Ostojic S.,Versic Z. | Iop Conference Series Earth and Environmental Science , 1196(1), 012060 | 2023 | 4

Hide abstract | Find It @ UCL | View at Publisher | Related documents

People are spending an increasing amount of time inside buildings. Working spaces have a very strong influence on peoples' activities and well-being. Therefore, the quality of indoor environment is important for all occupants, as it can contribute to illness and lack of productivity. An increasing number of office buildings with glass facades impact not only internal occupants, but also external users who spend time around those buildings. This paper aims to assess the interaction of internal and external users and the glass façades of office buildings in Zagreb, Croatia. The focus of this research is not only physical parameters of glass facade: materials, proportions, colours, *thermal* and *acoustic* characteristics, airtightness, and its impact on occupants. Also, the relationship between inside and outside is essential for spatial awareness. Having access to views of the outside and providing natural light are important for occupants' well-being. Solar shading system, anti-glare protection, the tint of the glass, transparency and translucency of glass façade also significantly affect external users and their perception of the building. The graduate course 'Architecture and Technology 1' at the Faculty of Architecture, University of Zagreb, aims to educate master students on assessing the environmental performance of existing buildings. Analysing examples of built office buildings with glass facades and the results of student survey of internal and external users of those buildings, this paper discusses working conditions, indoor and *outdoor* atmosphere, technical solutions, as well as architectural approaches in accordance with implementation of principles of sustainable development. The results of this research should provide useful information on users' satisfaction, well-being, and requirements for future application in glass facade design for office buildings.

Article

264 **Prediction of enhanced bipolar disorder in autistic children using denoising convolutional neural network** | Manjunath K.M., Veeramani V. | Network Modeling Analysis in Health Informatics and Bioinformatics , 11(1), 36 | 2022 | 0

Hide abstract | Find It @ UCL | View at Publisher | Related documents

Bipolar disorder (BD) causes depression, anxiety, irritability, hyperactivity, and other behavioral changes in autistic children. An accurate BD examination helps the doctors to prescribe the correct treatment and dosage level. Patients with BD have previously undergone the Aberrant Behavior Checklist (ABC) in clinics for examination and which takes only a short amount of time. Continuous monitoring of autistic children is a major problem for physicians when assessing autistic children’s examinations. In this paper, autistic child BD assessment is performed through the *thermal* radiometric pixel of facial regions such as the face, eye, mouth, cheek, and forehead *thermal* images. *Thermal* images are obtained from continuous monitoring of *thermal* cameras such as the indoor and *outdoor* environments. The facial *thermal* regions have a crumbling effect (CE) and hypomelanotic disorder (HD) in the skin leads to *noise* in *thermal* pixel values, resulting in inaccurate measurement. Because of these CE and HD effects, *thermal* images have both scaled and unscaled *noises*, as well as white *noise*. *Thermal* facial region images are applied with Non-Decimated Wavelet Transform (NDWT) and Denoising Convolutional Neural Network (DnCNN) filters. In comparison

	Document title	Authors	Source	Year	Citations

to ABC, the suggested NDWT and DnCNN gives an accurate measurement for assessment of BD, emotions, and repetitive behaviors in autistic children with about 95% of accuracy in identification and examination.

Article • *Open access*

265 **A Case Study of Vignetting Nonuniformity in UAV-Based Uncooled Thermal Cameras** — Yuan W.,Hua W. — Drones, 6(12), 394 — 2022 — 18

Hide abstract | Find It @ UCL | View at Publisher | Related documents

Uncooled *thermal* cameras have been employed as common UAV payloads for aerial temperature surveillance in recent years. Due to the lack of internal cooling systems, such cameras often suffer from *thermal*-drift-induced nonuniformity or vignetting despite having built-in mechanisms to minimize the *noise*. The current study examined a UAV-based uncooled *thermal* camera vignetting regarding camera warmup time, ambient temperature, and wind speed and direction, and proposed a simple calibration-based vignetting migration method. The experiments suggested that the camera needed to undergo a warmup period to achieve stabilized performance. The required warmup duration ranged from 20 to 40 min depending on ambient temperature. Camera vignetting severity increased with camera warmup time, decreasing ambient temperature, and wind presence, while wind speed and direction did not make a difference to camera vignetting during the experiments. Utilizing a single image of a customized calibration target, we were able to mitigate vignetting of *outdoor* images captured in a 30 min duration by approximately 70% to 80% in terms of the intra-image pixel standard deviation (IISD) and 75% in terms of the pixel-wise mean (PWMN) range. The results indicated that *outdoor* environmental conditions such as air temperature and wind speed during short UAV flights might only minimally influence the *thermal* camera vignetting severity and pattern. Nonetheless, frequent external shutter-based corrections and considering the camera nonlinear temperature response in future studies have the potential to further improve vignetting correction efficacy for large scene temperature ranges.

Article • *Open access*

266 **Heat Stress During Arba'een Foot-Pilgrimage (World's Largest Gathering) Projected to Reach "Dangerous" Levels Due To Climate Change** — Choi Y.-W.,Eltahir E.A.B. — Geophysical Research Letters, 49(19), e2022GL099755 — 2022 — 9

Hide abstract | Find It @ UCL | View at Publisher | Related documents

Arba'een, a Muslim pilgrimage, is one of the largest annual mass gatherings in the world, with a date fixed according to the lunar calendar. Most pilgrims start their long walking journey from Basra/Najaf toward Karbala (about 70–500 km) and are significantly affected by *outdoor* weather conditions during this period. Here, based on simulations performed using carefully-selected climate models, we project that heat stress during the pilgrimage is likely to reach a "dangerous" level, defined according to the US National Weather Service criteria, by the end of this century. Moreover, a significant increase in consecutive occurrence of hot days and hot nights is expected within the coming decades, which may cause a high incidence of heat-related disorders as the human body may not recover from the daytime heat loads. Our study suggests that *sound* adaptation measures and stringent mitigation actions must be established to ensure a safe pilgrimage in the future.

Article • *Open access*

267 **Effects of Acoustic Perception on Outdoor Thermal Comfort in Campus Open Spaces in China's Cold Region** — Chen H.,Hong B.,Qu H.,Geng Y.,Su M. — Buildings, 12(10), 1518 — 2022 — 36

Hide abstract | Find It @ UCL | View at Publisher | Related documents

To explore the effects of *acoustic* perception on *outdoor thermal* comfort, *acoustic* perception in five typical open spaces in a campus in China's cold region with common soundscapes was evaluated by using meteorological measures, *sound* level devices, and a questionnaire survey. Eight adjectives were used to evaluate an individual's *acoustic* perception, and the physiological equivalent temperature (PET) was selected as a *thermal* index. The correlations between *acoustic* perception and *thermal* comfort in these typical open spaces were further analyzed. We demonstrated that: (1) *Acoustic* sensation and comfort varied significantly among *sound* types (STPs). Respondents reported pleasant perceptions of broadcasting music, running water and birdsong, and wind (a gentle breeze) and

	Document title	Authors	Source	Year	Citations

insects. (2) Although there was no significant difference in the *thermal* sensation vote (TSV), we found large differences in the *thermal* comfort vote (TCV) among STPs. (3) The respondents' neutral PET (NPET) varied among STPs. In autumn, the ranked order of NPET was machines > running water and birdsong > crowds > broadcasting music > wind (a gentle breeze) and insects. In winter, the order of the NPET was machines > wind (a gentle breeze) and insects > crowds > broadcasting music > running water and birdsong. (4) When people perceived "*acoustic* comfort", their TSV improved, and vice versa.

Article

268 **Thermal-Inertial SLAM for the Environments with Challenging Illumination** — Jiang J.,Chen X.,Dai W.,Gao Z.,Zhang Y. — IEEE Robotics and Automation Letters, 7(4) — 2022 — 45

Hide abstract Find It @ UCL View at Publisher Related documents

In recent years, longwave infrared (LWIR) cameras have become potential in visual simultaneous localization and mapping (SLAM) research since the delivered *thermal* images can provide information beyond the visible spectrum and are robust to environment illumination. However, due to modality differences, SLAM methods designed for visible cameras cannot be directly applied to *thermal* data. In this paper, we propose a *thermal*-inertial SLAM method for all-day autonomous systems. To overcome the challenge of the *thermal* data association, the proposed method represents several improvements, including singular-value-decomposition-based (SVD-based) image processing and *ThermalRAFT* tracking methods. Based on the characteristics of the *thermal* images, the SVD-based image processing method can exploit the fixed *noise* pattern of *thermal* images and enhance the image quality to improve the performance of subsequent steps, including *thermal* feature extraction and loop detection. To achieve real-time and robust feature tracking, we develop *ThermalRAFT*, an efficient optical flow network with iterative optimization. Moreover, the system introduces a bag-of-words-based loop detection method to maintain global consistency in long-term operation. The experimental results demonstrate that the proposed method can provide competitive performance in indoor and *outdoor* environments and is robust under challenging illumination conditions.

Article • *Open access*

269 **Multi-domain human-oriented approach to evaluate human comfort in outdoor environments** — Cureau R.J.,Pigliautile I.,Kousis I.,Pisello A.L. — International Journal of Biometeorology, 66(10) — 2022 — 34

Hide abstract Find It @ UCL View at Publisher Related documents

Human comfort outdoors is widely investigated, but most studies explore the comfort domains singularly. This paper aimed to evaluate human comfort in parks, verifying the importance of using a multi-domain (simultaneously evaluating *thermal*, visual, *acoustic*, and air quality) and multi-disciplinary (combining environmental and social fields) approach. A walk through a pre-defined path from one park to another was repeated twice per day on four consecutive days in June, with three participants per walk. The two investigated parks are in central Italy and were chosen because they differ in their design and spatial characteristics. Environmental data were recorded with an innovative wearable device during the whole walk, and surveys were used to assess people's perceptions of the parks. Despite observed differences in collected physical parameters, the survey's responses were similar, and different comfort domains showed dependence on each other in the two parks. Logistic regression models were developed for each park, and they revealed that the qualitative information predicted the overall comfort level more accurately than the environmental data. In detail, the models based on environmental data resulted in R^2 equal to 0.126 and 0.111 in Parks 1 and 2, respectively, whereas using the survey answers increased it up to 0.820 (Park 1) and 0.806 (Park 2). This study contributes to addressing the gap in multi-domain comfort studies outdoors and confirms the importance of using multi-disciplinary and multi-domain approaches for a complete comfort analysis, supporting holistic human-biometeorology-oriented models and forecasting opportunities that can promote improvements in urban environmental quality and liveability.

Article • *Open access*

270 **Experimental and numerical assessment of permeability functions in closed cavity façades** — Lori G.,Van Den Brande K.,Van Den Bossche N.,De Bleecker H.,Belis J. — Glass Structures and Engineering, 7(3) — 2022 — 1

	Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

This paper is the second in a series on the behavior of closed cavity façades (CCF) under *thermo*-mechanical loading excitations. CCF are a novel trend for sustainable high quality double skin façade solutions, characterized by a minimal maintenance demand. The air cavity is designed according to stringent air tightness requirements, provided with a small dry air flow that preserves the relative humidity, minimizing the risk for condensation and dust ingress. This extremely reduced permeability, with respect to typical double skin design, gives a scenario in conflict with the general prescriptions for the structural calculations of double skins under wind and climatic loading. This paper is the second of a series of three documents that aim to identify the structural shortcomings in the current codes and to propose efficient calculation methods and modifications to the current calculation strategies. This will overcome a critical design paradox in double skin and in particular CCF aim to reach the highest sustainability performance, by means of outstanding *thermal* and *acoustic* insulation efficiency and providing the cavity with a clean and dry environment, suitable for a significant upgrade of the façade life expectancy. On the other hand, an overly conservative calculation approach is generally against the optimization of the sustainability performance indices, determining an excessive use of materials and then impacting the overall life cycle cost under several perspectives. In particular, the objective of this second paper is to describe in a measurable way the permeability behavior of a CCF, defining statistical variability and nonlinearity effects measured during dedicated experimental testing. The permeability functions represent a fundamental input for the proposed assessment tool introduced by the first paper and it will be seen that an accurate quality control during the manufacturing can ensure the robustness of an optimized calculation approach, based on the load sharing mechanism between the skins. In the third paper it will be shown that the proposed tool, fed by the experimental permeability functions, appears adequate to predict the façade structural behavior under the superimposition of *thermal* loading, wind loading and dry air flow effects, when compared with measurements collected during *outdoor* testing on a façade specimen.

Article • *Open access*

271 **Actual energy performance and indoor climate in Finnish NZEB daycare and school buildings** Ahmed K.,Hasu T., Kurnitski J. Journal of Building Engineering , 56, 104759 2022 23

Hide abstract Find It @ UCL View at Publisher Related documents

The study addressed the improvement of building energy performance along with a good *thermal* comfort and indoor air quality (IAQ) in newly built Finnish daycare and school buildings. Indoor Environmental Quality (IEQ), calculated and actual energy performance, investment and life cycle cost (LCC) were monitored. Ventilation airflow rates, temperature, and CO_2 were measured onsite in five daycare and school buildings during summer and winter, and an occupant questionnaire survey was conducted in parallel with onsite measurements. Energy performance was evaluated based on an energy performance certificate (EPC) and actual metered consumption to figure out possible performance gaps. Simulations were conducted to calibrate models and analyse improvements in ventilation system operation, which showed energy-saving potential while ensuring good *thermal* comfort and IAQ. The *thermal* comfort and IEQ of all buildings were excellent and were in line with the results of the questionaries survey except in 'School 3', which had specific odour and *noise* problems. The measured energy use of all buildings except 'School 3' was increased by a factor of 2.1–2.6 compared to the EPC value, caused by poor control of a ventilation system and the presence of some non-regulated uses such as hot kitchens, washing machines, and dryers. LCC considerably depended on actual energy use stressing the importance of accurate energy prediction. It was shown that the reduction of energy use by a factor of 1.9–3.1 is possible without compromising indoor climate by adjustment of correct operation hours, improving ventilation system control, and limiting excessive *outdoor* airflows in some cases.

Conference Paper • *Open access*

272 **STUDY ON THE DYNAMIC NATURAL VENTILATION RATES OF OCCUPIED RESIDENCE** Liu S.,Jian Y.,Zhu W., ... Bian M.,Liu Z. E3s Web of Conferences , 356, 03058 2022 0

Hide abstract Find It @ UCL View at Publisher Related documents

The residential natural ventilation rates have a significant impact on indoor *thermal* comfort and air quality and building energy consumption. The characteristics of the indoor-*outdoor* temperature difference and wind pressure change over time, as well as the occupants' window opening behavior and the use of HVAC systems, resulting in the

	Document title	Authors	Source	Year	Citations

residential air change rate being dynamic with time. Many previous studies of residential ventilation measured the steady-state air change rates, which does not reflect the actual dynamic characteristics. In this study, a field measurement was conducted in a bedroom of one natural ventilation residential building in Beijing with continuously monitoring the CO_2 concentration, indoor air temperature, and *outdoor* meteorological parameters for one year. Using the CO_2 released by occupants as a tracer gas, the extended Kalman filter based on the Transient Mass Balance Equation (TMBE) was adopted to calculate the dynamic air change rate. This method can effectively filter the CO_2 concentration measurement *noise*. The trend of air change rate with each influencing factor was analyzed. This study is expected to lay the foundation for future studies of dynamic air change rates in residential buildings.

Article

273 **Pre-dimensioning of ventilation openings in double façades | Vordimensionierung von Lüftungsöffnungen in Doppelfassaden** — Friedrich M.,Schweers K., Wellershoff F. — Bauphysik, 44(4) — 2022 — 0

Hide abstract Find It @ UCL View at Publisher Related documents

Pre-dimensioning of ventilation openings in double façades. Double façades improve *sound* insulation while natural ventilation and offer the shading in the space between the façades, protection from higher wind loads. With regard to natural ventilation, however, it should be noted that the air flows must pass through two façade layers and the air exchange rate cannot be planned with the existing assumptions for single-skin façades. The potentially higher temperatures in the intermediate façades space also require special consideration when planning summer heat protection. Climate data were recorded, and air exchange measurements were taken on 10 buildings in Hamburg with corridor double façades and box window double façades. Based on these data, a calculation approach of the *outdoor* air exchange was created and applied in transient *thermal* building simulations. A method was created with which the required size of the ventilation openings can be determined quickly and without software application in early design phases.

Article • *Open access*

274 **Post-Occupancy Evaluation of Green Technologies for a High-Rise Building Based on User Experience** — Ye C.,Yao L.,Meng Y., Zhang Y.,He G. — Sustainability Switzerland , 14(15), 9538 — 2022 — 14

Hide abstract Find It @ UCL View at Publisher Related documents

Post-occupancy evaluations of buildings help us understand the effectiveness of building designs. Most of such evaluations focus on the overall building performance and on the indoor air environment, and few focus on individual technologies. In this paper, we report a user experience-based evaluation on multiple green technologies applied in a high-rise building constructed 17 years ago. These technologies include a circular building design, innovative floor plans, a corridor-type, double-skin facade with cavity shading, and three-dimensional greening design. Data were obtained through a questionnaire for the building occupants and on-site measurements of environmental variables. The results show that the occupants were most satisfied with the three-dimensional greening, natural lighting and ventilation in the public spaces, and the *sound* and *thermal* environment of the rooms with the double facade. They were not satisfied with room ventilation, solar shading, or lighting adjustment provided by the double facade. In addition, the aesthetic appearance of the circular building was not recognized by the occupants, who appeared to pay more attention to its space efficiency than the external appearance. This was true especially for older occupants. The users expressed a strong demand for *outdoor* leisure spaces and green vegetation. However, the practicality and functionality of three-dimensional greening, along with the interference of the glass curtain wall on vegetation growth, calls for more attention to the design. The female occupants were found to be more sensitive to the *thermal* and *acoustic* environment. Based on these results, we provide recommendations on the maintenance of the building and the future design of these measures.

Article

275 **Impact of improved indoor environment on recovery from COVID-19 infections: a review of literature** — Afful A.E., Osei Assibey Antwi A.D.D., Ayarkwa J.,Acquah G.K.K. — Facilities, 40(11-12) — 2022 — 9

Hide abstract Find It @ UCL View at Publisher Related documents

	Document title	Authors	Source	Year	Citations

Purpose: This study aims to explore the impact of the indoor environment on recovery from COVID-19 infections. Extant literature on the impact of the four key themes of the indoor environment (indoor air quality, indoor *thermal* quality, daylighting and visual comfort, and *acoustic* comfort) on COVID-19 infection and recovery rates were reviewed. Design/methodology/approach: Data collection for this study was based on extant literature within the *Scopus* database and scoped to a time frame of 2020–2021 because the topical issue of indoor environmental quality (IEQ) and its impact on COVID-19 arose in the wake of the pandemic. In total, 224 documents were systematically desk reviewed from various journals. Findings: The study identified that air pollutants such as $PM_{2.5}$ and PM_{10} as well as air-conditioned places, low ambient temperatures, poor ventilation and no views of the *outdoor* environment were deteriorating factors for COVID-19 patients. On the other hand, proper ventilation, the use of air cleaners, views of the *outdoor* environment and allowance for ample daylighting were improvement factors for COVID-19 patients. The inter-relationship of the various concepts was presented in an ontology chart. Practical implications: As COVID-19 still exists and keeps evolving, this study provides suggestions to industry professionals, especially health-care Facility Managers, to create a post-pandemic environment focusing on the IEQ and finding long-term and reliable solutions for the well-being of occupants. Adaptability is crucial. New, creative technology solutions are being introduced daily, but it is up to the facility managers and health-care professionals to analyse and specify the most cost- and outcome-effective technologies for their facility. Originality/value: The study brought to light the pivotal role of the indoor environment on the health and well-being of occupants, particularly in the contraction, spread, prevention and control of infectious diseases such as COVID-19.

Article

☐ 276 **Overlaps in space utilisation patterns and IEQ conditions observed in Activity-Based Working supportive office** — Gocer O.,Candido C., Gocer K.,... Alizadeh T., Sarkar S. — Building and Environment , 220, 109273 — 2022 — 10

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

One of the potential positive aspects of spaces designed to support Activity-Based Working (ABW) concept is to allow people to choose to work from zones that suit their Indoor Environmental Quality (IEQ) preferences. The vast majority of studies reporting findings relevant to IEQ performance of ABW spaces rely on spot measurements and/or subjective surveys and as such it is difficult to map longitudinal actual utilisation patterns and IEQ performance. This study contributes to this topic by analyzing a longitudinal dataset of space utilisation of an ABW over a twelve-month period. Analysis aims to identify overlaps between space utilisation patterns and expected IEQ performance in order to understand which types of zones occupants seem to prefer to work from. Occupants' demographics (age and gender) were also analyzed with respect to the use of space. Further, specific physical markers were investigated, namely proximity to window, air-conditioning outlets, and high-low *noise* locations. The impact of physical markers on workplace choice was classified and predicted during a day using a decision-tree diagram developed on the basis of the CHAID algorithm. Overall, workers tended to choose to work within the inner zones (after 11 a.m.) during winter. Findings show that female workers moved away from air-conditioning outlets. Additionally, female workers worked from workstations situated in high and medium *noise* zones. Results illustrated the potential to use utilisation datasets to understand how workers move away or towards zones in response to expected IEQ performance - *thermal* comfort (cold/warm), *acoustics* (noisy/quiet), and access to the *outdoor* environment/view (window) proximity.

Article • *Open access*

☐ 277 **Design of Multimodal Sensor Module for Outdoor Robot Surveillance System** — Uhm T.,Park J.,Lee J., ... Ki G.,Choi Y. — Electronics Switzerland , 11(14), 2214 — 2022 — 10

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

Recent studies on surveillance systems have employed various sensors to recognize and understand *outdoor* environments. In a complex *outdoor* environment, useful sensor data obtained under all weather conditions, during the night and day, can be utilized for application to robots in a real environment. Autonomous surveillance systems require a sensor system that can acquire various types of sensor data and can be easily mounted on fixed and mobile agents. In this study, we propose a method for modularizing multiple vision and *sound* sensors into one system, extracting data synchronized with 3D LiDAR sensors, and matching them to obtain data from various *outdoor* environments. The proposed multimodal sensor module can acquire six types of images: RGB, *thermal*, night vision,

	Document title	Authors	Source	Year	Citations

depth, fast RGB, and IR. Using the proposed module with a 3D LiDAR sensor, multimodal sensor data were obtained from fixed and mobile agents and tested for more than four years. To further prove its usefulness, this module was used as a monitoring system for six months to monitor anomalies occurring at a given site. In the future, we expect that the data obtained from multimodal sensor systems can be used for various applications in *outdoor* environments.

Article • *Open access*

278 **Compact Decentral Façade-Integrated Air-to-Air Heat Pumps for Serial Renovation of Multi-Apartment Buildings** — Ochs F.,Monteleone W.,Dermentzis G.,Siegele D.,Speer C. — Energies, 15(13), 4679 — 2022 — 7

Hide abstract Find It @ UCL View at Publisher Related documents

To address the huge market of renovation of multi-apartment buildings, minimal-invasive decentral serial-renovation solutions are required. One major challenge in the design of decentral heat pumps is to find the optimal balance between, on one hand, compactness and pleasant design, and on the other hand, efficiency and minimal *sound* emissions. A comprehensive holistic design and optimization process for the development of decentral heat pumps, from the component level, to the system level, and up to the building level, is developed. A novel façade-integrated speedcontrolled exhaust air to supply air heat pump combined with a mechanical ventilation system with heat recovery and recirculation air was developed and simulated in a reference flat. Compared to a traditional supply air heat pump without recirculation, it shows only slight performance improvement, but allows significantly better *thermal* comfort and control, independently from the hygienic air flow rate and from the heating and cooling loads. Detailed measurement and simulation results are presented for several functional models with heating power of around 1 kW up to 2.5 kW. The design was optimized by means of CFD simulations to allow for low pressure drop, homogeneous flow, and low *sound* emissions. Moreover, mock-ups of innovative façade-integrated heat pump *outdoor* units are presented.

Article • *Open access*

279 **Holistic Operational Signatures for an energy-efficient district heating substation in buildings** — Hong Y.,Yoon S. — Energy, 250, 123798 — 2022 — 23

Hide abstract Find It @ UCL View at Publisher Related documents

District heating (DH) is recognized as a sustainable energy infrastructure in cities and buildings. DH substation operation is essential to bridging the gap between plants and *thermal* clients in achieving the intended DH values and visions. Therefore, in this study, an operation analysis method was developed for building-side DH substations. Specifically, Holistic Operational Signature (HOS) was suggested with multiple signature elements (x-OS) and representations to consider comprehensive operational correlations. A HOS-based analytics is proposed to provide more insights and *sound* reasoning about the DH substation design, operation, control, and heating efficiency potentials, compared with the existing Energy Signature. In a case study for winter, the proposed HOS method was applied to a target DH substation serving actual residential buildings using the real operational datasets. The six HOSs (HOSs 1–6) were derived with their six elements to identify the representative heating consumption levels and patterns, partial heating load ratios, and operation and control efficiency. Excessive water flow rates were identified based on the partial heating load elements (less than 50% in most cases) of all signatures. Inefficient signatures (HOSs 1–4) that do not follow the conventional control principle according to *outdoor* air temperatures were captured, especially for high heating demand patterns, which accounted for 72.5% of the total in winter. According to the signatures, the demand-based control can decrease the heating supply temperature setpoint by about 1.4 °C and 0.6 °C, respectively for HOS2 and the winter. It is recommended that the valve operation should be improved to lower the hunting levels (within ±1.5 °C) of heating water temperature control, especially in the afternoon with low heating demands.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
280	**Influence of atmospheric conditions on measured infrasound from wind turbines**	D'Amico S., Van Renterghem T., Botteldooren D.	Journal of Wind Engineering and Industrial Aerodynamics, 225, 105021	2022	5

Hide abstract Find It @ UCL View at Publisher Related documents

Scientific interest towards low frequencies and infrasound from wind turbines has renewed over the last decade as the size of wind turbines and the number of wind farms have continuously increased. However, few comprehensive studies are found assessing such *noise* emitted by wind turbines in response to meteorological parameters, including turbulent characteristics of the atmospheric boundary layer. The current work presents a detailed long-term measurement campaign combining infrasound recordings close to wind turbines and high-resolution meteorological parameters at three heights. An artificial neural network (ANN) model, developed based on the measurement data, was able to adequately predict both blade passing frequency levels and broadband infrasonic *noise*. The model allowed discriminating between the broadband inflow *noise* emitted by the wind turbines and the remaining microphone-induced wind *noise* under the wind-shielding dome for a wide range of atmospheric conditions. At the same time, the ANN allowed deriving physically plausible relationships between meteorological and turbulence parameters and wind turbine emitted infrasound components. Dominant predictors for the infrasound production were mean wind speed, turbulence intensity and turbulent vertical heat flux.

Article

281	**Environmental satisfaction, mood and cognitive performance in semi-outdoor space in the tropics**	Mihara K.,Chen S., Hasama T.,... Lee J.K.W., Wong N.H.	Building and Environment, 216, 109051	2022	35

Hide abstract Find It @ UCL View at Publisher Related documents

The use of semi-*outdoor* space for the well-being of occupants is promoted in the tropics. Although a semi-*outdoor* space is often used for break space, environmental satisfaction and work performance are unclear in a semi-*outdoor* space as work activity space compared with air-conditioned (AC) room. This study focuses on the use of semi-*outdoor* space for short-term work activities such as office work and meetings. Experiments were carried out to evaluate environmental satisfaction, mood and cognitive performance in the cool and warm seasons in an AC room and a semi-*outdoor* space in Singapore. Eighty-nine participants answered questionnaires and conducted cognitive performance tests of Stroop test, visual backward digit span test, typing test and mind map test for 70 min per case in the AC room and semi-*outdoor* space with or without the operation of a ceiling fan. We analyzed the relationships among environmental satisfaction, mood, work performance by Structural Equation Modeling. The conditions which 80% of participants accepted in the semi-*outdoor* space were *thermal* environment ranged from 23.3 °C to 30.1 °C SET*, lighting levels of more than 260 lx and *noise* levels of less than 48 dBA. No differences in cognitive performance were found between the AC room and semi-*outdoor* space ($p > 0.05$). Mood influenced work performance ($p < 0.001$) while environmental satisfaction did not ($p = 0.28$). Self-reported performance, especially stress, influenced creativity ($p = 0.047$). These findings lead to the possibility of short-term work activities in the semi-*outdoor* space without a decrease in work performance.

Article • *Open access*

282	**Intercomparison of Vaisala RS92 and RS41 Radiosonde Temperature Sensors under Controlled Laboratory Conditions**	Rosoldi M.,Coppa G., Merlone A.,Musacchio C., Madonna F.	Atmosphere, 13(5), 773	2022	10

Hide abstract Find It @ UCL View at Publisher Related documents

Radiosoundings are essential for weather and climate applications, as well as for calibration and validation of remote sensing observations. Vaisala RS92 radiosondes have been widely used on a global scale until 2016; although in the fall of 2013, Vaisala introduced the RS41 model to progressively replace the RS92. To ensure the highest quality and homogeneity of measurements following the transition from RS92 to RS41, intercomparisons of the two radiosonde models are needed. A methodology was introduced to simultaneously test and compare the two radiosonde models inside climatic chambers, in terms of *noise*, calibration accuracy, and bias in temperature measurements. A pair of RS41 and RS92 radiosondes has been tested at ambient pressure under very different temperature and humidity

	Document title	Authors	Source	Year	Citations

conditions, reproducing the atmospheric conditions that a radiosonde can meet at the ground before launch. The radiosondes have also been tested before and after fast (within ≈ 10 s) temperature changes of about ±20 °C, simulating a scenario similar to steep *thermal* changes that radiosondes can meet when passing from indoor to *outdoor* environment during the pre-launch phase. The results show that the temperature sensor of RS41 is less affected by *noise* and more accurate than that of RS92, with *noise* values less than 0.06 °C for RS41 and less than 0.1 °C for RS92. The deviation from the reference value, referred to as calibration error, is within ±0.1 °C for RS41 and the related uncertainty (hereafter with coverage factor k = 1) is less than 0.06 °C, while RS92 is affected by a cold bias in the calibration, which ranges from 0.1 °C up to a few tenths of a degree, with a calibration uncertainty less than 0.1 °C. The temperature bias between RS41 and RS92 is within ±0.1 °C, while its uncertainty is less than 0.1 °C. The fast and steep *thermal* changes that radiosondes can meet during the pre-launch phase might lead to a *noise* increase in temperature sensors during radiosoundings, up to 0.1 °C for RS41 and up to 0.3 °C for RS92, with a similar increase in their calibration uncertainty, as well as an increase in the uncertainty of their bias up to 0.3 °C.

Article • *Open access*

283 **Direct heat flux sensing for window shading control in passive cooling systems** — Danis J.,Mishra S., Rempel A.R. — Energy and Buildings , 261, 111950 — 2022 — 7

Hide abstract Find It @ UCL View at Publisher Related documents

Greenhouse gas emissions from space cooling are expected to double globally by 2050, giving urgency to the development of low-carbon cooling methods. Solar heat gain through windows is a leading contributor to cooling loads, and accordingly, operable shading strategies have been established that respond to numerous environmental parameters, including solar radiation, illumination, time of day, and indoor and *outdoor* air temperature. While these have shown excellent performance, optimal setpoints are typically specific to climates, seasons, and spaces, preventing development of controls with consistent effectiveness across varying conditions. Here we investigate an alternative parameter, window surface heat flux, in which a universal setpoint of 0 W/m^2 identifies transitions between window heat gain and loss. Examination of heat flux signals acquired with low-cost sensors first revealed the associated *noise* to be consistent but too great for control application. Representative *noise* characteristics were then used to construct noisy signals for simulations, allowing evaluation of *noise* mitigation by digital filtering methods and signal persistence requirements. Strikingly, shading controlled by the resulting output reduced window heat gain by 54–78% in six contrasting climates, comparable to or exceeding the performance of established approaches, indicating that heat flux sensing is now an outstanding candidate for shading control in passive cooling systems.

Article

284 **Measuring Method for Long-Distance Dynamic Indication Accuracy of Laser Irradiation System | 激光照射系统远距离动态指示精度的测量方法** — Wenbo J.,Bingkun H., Genming C. — Zhongguo Jiguang Chinese Journal of Lasers , 49(7), 0704003 — 2022 — 1

Hide abstract Find It @ UCL View at Publisher Related documents

Objective The indication accuracy of the laser irradiation system is an important factor influencing the strike accuracy of laser-guided weapons. When the irradiation target of the laser irradiation system is a moving target, the collected images possess a large perspective distortion, which affects the calculation of the real indication accuracy. Only a single camera cannot solve this problem well. Compared with the single-camera vertical target imaging measurement method, the measurement method that uses visible light and near-infrared cameras to simultaneously capture the vertical target has more advantages. The visible light and near-infrared cameras are used to detect the moving target and collect the laser spot, respectively. This method has higher imaging quality than the method of simultaneous imaging of the single-camera target and the spot. Note that to achieve good registration accuracy for a long focal length telephoto optical system, these conditions, including large effective registration target surface, controlled position accuracy of the positioning hole, and cooperation of the registration target plate must be considered. Also, the imaging of the registered target is susceptible to interference from the external environment. Simultaneously, due to the continuous movement of the target, the images collected by the dual-camera imaging system at different times have different perspective distortions. Furthermore, the accuracy of the perspective distortion correction affects the final measurement result of the indication accuracy. Thus, our proposed dual-

Document title	Authors	Source	Year	Citations

imaging system measurement method without reference distance but with dual-perspective correction is useful for the measurement of the far-field dynamic indication accuracy of the laser irradiation system. Methods Before calculating the dynamic indication accuracy, we solve the essential problem that the images of the dual cameras do not overlap due to the mechanical structure and difference of cameras, that is, the image registration of the dual-imaging system. The problem of large-angle perspective distortion at different moments caused by movement is to correct the perspective distortion of the imaging system. Here visible light and near-infrared dual cameras of the coaxial optical system are used for telephoto imaging. Also, we design and construct the collimator star point array reticle and registration model of the visible light and near-infrared imaging system, respectively, to complete the indoor registration of the dual-imaging system. For the *outdoor* experiment, a theoretical model of dual-perspective correction without reference distance is established using the cooperative target on the moving target board to realize the perspective distortion correction of the visible light target board and near-infrared laser spot images, and extract the cross target and laser spot from the corrected image. Next, the center position is calculated to obtain the far-field dynamic indication accuracy of the laser irradiation system. Results and Discussions At 1 km imaging distance for the dual-imaging system and 10 experiments, the registration and corresponding pixel standard deviations are 0.94 mm and 0.08 pixel, respectively. In the *outdoor* dynamic measurement accuracy calibration experiment of the measurement system and combined with the geometric imaging formula, the horizontal and vertical image spacing of the cooperative target point without perspective distortion on the target board image 1 km away are 2. 18 and 0. 92 mm, respectively. Continuous 340 frames of images are used for dynamic measurement accuracy statistical calculation, as shown in Fig. 7(e). Results show that the horizontal and vertical measurement errors of the measurement system are mainly floating, and the maximum measurement error value is less than 28.28 mm, corresponding to the image. The deviation is 2 pixel. The average measurement error and standard deviation of the continuous 340 frame image pairs are 2. 45 and 9. 90 mm, respectively. The system can be effectively used in the *outdoor* evaluation of the indicating performance of the laser irradiation system. We note that the process of evaluating the *outdoor* dynamic indication accuracy of the laser irradiation system is the same as that of the *outdoor* dynamic measurement accuracy calibration experiment of the measurement system. However, a slight difference is that the cooperative self-inspection lamp becomes the actual laser irradiation system spot, that is, the detection target of the near-infrared camera is replaced and becomes the laser spot. Fig. 8(b) shows the near-infrared laser spot image corrected by the dual perspective projection, and its signal-to-*noise* ratio is good. Fig. 8(e) shows the result of fusing images of the two cameras. Here, the laser spot is located at the lower left of the target board. Fig. 8(f) shows the dynamic accuracy test results of 376 consecutive frames of images (note: the direction of the y pixel coordinate axis of the image is from top to bottom). The measurement result data show that the dynamic indication error of the laser irradiator is floating with an average indication error and standard deviation of 354.65 and 58.28 mm, respectively. Conclusions In this paper, we propose a laser far-field dynamic indication accuracy measurement method without reference distance using dual-perspective correction. First, we simplify the image registration process of the dual-imaging system and improve the image registration accuracy of the dual-camera imaging system for the registration standard deviation of 0.08 pixel. The actual registration standard deviation for a corresponding imaging distance at 1 km is 0.94 mm. Furthermore, we calibrate the dynamic measurement accuracy of the measurement system using a cooperative self-test lamp on the target plate by dual-perspective correction theoretical model without reference distance. The average measurement error of the measurement system is 2. 45 mm with a dynamic measurement standard deviation of 9.90 mm. Finally, we perform practical dynamic indication accuracy tests of laser irradiation systems, to further verify the effectiveness of the proposed measurement method.

Article • *Open access*

☐ 285	**Exploring the Multisensory Interaction between Luminous, Thermal and Auditory Environments through the Spatial Promenade Experience: A Case Study of a University Campus in an Oasis Settlement**	Berkouk D.,Bouzir T.A.K., Boucherit S., ... Matallah M.E.,Mazouz S.	Sustainability Switzerland , 14(7), 4013	2022	29

Hide abstract Find It @ UCL View at Publisher Related documents

This paper aimed to develop a multisensory approach in a university campus, based on quantitative and qualitative approaches, investigating sense walk experiences (*thermo*-visual *sound* walk) under interactions of luminous, *thermal*, and auditory environments. The study was conducted in October 2021, in Chetma university campus in Biskra city, southern Algeria, which remains a famous oasis settlement of arid regions over the country. A comparative and

	Document title	Authors	Source	Year	Citations

correlation analysis was performed between the physical dimensions collected through a walking experience in three campus routes (*outdoor*, semi-*outdoor* and indoor). In addition, a multisensory survey of the walking experience on perceptual dimensions was evaluated in parallel to the empirical contribution. The paper shows that walkers' *thermal* levels were balanced between neural and slightly hot in different spatial aspects. The glare was almost unperceived regarding the luminous conditions in the study site. The auditory experience reveals that the conducted points were generally quiet and well placed for educational requirements. Findings also show a strong relationship between the physical dimensions of the luminous and auditory environment. Furthermore, the findings suggest that the *thermal* and luminous environments are more perceptible than the auditory environment for the walkers of the *outdoor* and indoor routes. In contrast, the semi-*outdoor* route is often perceptible by the perceptual dimensions of the luminous and auditory environments. The findings on sensorial thresholds and spatial adaption are essential for the educational practices' architectural and urban strategies for the Saharan cities and oasis settlements.

Article

☐ 286 **Assessing the Superhydrophobic Performance of Laser Micropatterned Aluminium Overhead Line Conductor Material** — Lian C., Emersic C., Rajab F.H., ... Lowndes R., Li L. — IEEE Transactions on Power Delivery, 37(2) — 2022 — 19

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

Overhead electric supply lines are subject to water and ice accumulation, causing *noise* and weight increase, and leading to faults or failures. In this research, micro-textures were laser-ablated on aluminium substrates representative of overhead line conductors to achieve superhydrophobic properties. Samples were then subjected to a range of tests to measure the resilience and durability of such surfaces to conditions experienced by overhead line conductors. Tests to examine *thermal* ageing, *thermal* cycling, UV exposure, long term ambient *outdoor* environment exposure, and corona exposure testing were conducted and the evolution of surface superhydrophobicity was quantified. A high degree of resilience was observed for most of the testing. The feasibility of manufacturing upscale is discussed in addition to optimising test methods.

Article • *Open access*

☐ 287 **Performance of Dual-Hop Relaying for OWC System Over Foggy Channel With Pointing Errors and Atmospheric Turbulence** — Rahman Z., Shah T., Zafaruddin S.M., Chaubey V. — IEEE Transactions on Vehicular Technology, 71(4) — 2022 — 21

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

Optical wireless communication (OWC) over atmospheric turbulence and pointing errors is a well-studied topic. Still, there is limited research on signal fading due to random fog in an *outdoor* environment for terrestrial wireless communications. In this paper, we analyze the performance of a decode-and-forward (DF) relaying under the combined effect of random fog, pointing errors, and atmospheric turbulence with a negligible line-of-sight (LOS) direct link. We consider a generalized model for the end-to-end channel with independent and not identically distributed (i.ni.d.) pointing errors, random fog with Gamma distributed attenuation coefficient, double generalized gamma (DGG) atmospheric turbulence, and asymmetrical distance between the source and destination. We develop density and distribution functions of signal-to-*noise* ratio (SNR) under the combined effect of random fog, pointing errors, and atmospheric turbulence (FPT) channel and distribution function for the combined channel with random fog and pointing errors (FP). Using the derived statistical results, we present analytical expressions of the outage probability, average SNR, ergodic rate, and average bit error rate (BER) for both FP and FPT channels in terms of OWC system parameters. We also develop simplified and asymptotic performance analysis to provide insight on the system behavior analytically under various practically relevant scenarios. We demonstrate the mutual effects of channel impairments and pointing errors on the OWC performance, and show that the relaying system provides significant performance improvement compared with the direct transmissions, especially when pointing errors and fog becomes more pronounced.

Article • *Open access*

☐ 288 **A Sustainable Approach to Build Insulated External Timber Frame Walls for Passive Houses Using Natural and Waste Materials** — Georgescu S.-V., Sova D., Campean M., Cosereanu C. — Forests, 13(4), 522 — 2022 — 11

Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

This paper presents structures of timber-framed walls designed for passive houses, using natural and waste resources as insulation materials, such as wool, wood fibers, ground paper, reeds (Phragmites communis), and Acrylonitrile Butadiene Styrene (ABS) wastes. The insulation systems of stud walls composed of wool–ABS composite boards and five types of fillers (wool, ABS, wood fibers, ground paper, and reeds) were investigated to reach U-value requirements for passive houses. The wall structures were designed at a thickness of 175 mm, including gypsum board for internal wall lining and oriented strand board (OSB) for the exterior one. The testing protocol of *thermal* insulation properties of wall structures simulated conditions for indoor and *outdoor* temperatures during the winter and summer seasons using HFM-Lambda laboratory equipment. In situ measurements of U-values were determined for the experimental wall structures during winter time, when the temperature differences between outside and inside exceeded 10 °C. The results recorded for the U-values between 0.20 W/m^{2}K and 0.35 W/m^{2}K indicate that the proposed structures are energy-efficient walls for passive houses placed in the temperate-continental areas. The vapour flow rate calculation does not indicate the presence of condensation in the 175 mm thick wall structures, which proves that the selected *thermal* insulation materials are not prone to degradation due to condensation. The research is aligned to the international trend in civil engineering, oriented to the design and construction of low-energy buildings on the one hand and the use of environmentally friendly or recycled materials on the other.

Article • *Open access*

289 **Associations between indoor soundscapes, building services and window opening behaviour during the COVID-19 lockdown**

Torresin S.,Albatici R., Aletta F.,... Oberman T., Kang J.

Building Services Engineering Research and Technology , 43(2)

2022 28

Hide abstract Find It @ UCL View at Publisher Related documents

Results of an online survey conducted during the COVID-19 lockdown among 848 home workers living in London (United Kingdom) and in Italy are reported with a focus on (1) the impacts of building services on the perception of the *acoustic* environment while working and relaxing at home and (2) the factors associated with window opening behaviour. The analyses showed no significant difference in soundscape appropriateness for relaxation depending on the heating, ventilation and cooling system typologies, and in soundscape appropriateness for working from home (WFH) based on the ventilation strategy. Higher soundscape appropriateness for WFH was associated with houses equipped only with radiant floors for heating in Italy and with air-cooling systems in London. In London, air systems resulted in higher perceived dominance of *noise* from building services compared to other systems. Overall, rooms with less dominant *sounds* from building services were evaluated as more appropriate for working and relaxing. The dominance of sky or buildings from the window view, *outdoor* noisiness, *noise* sensitivity, age and gender were not significantly associated with participants' window opening behaviour while WFH. Differently, participants viewing more vegetation from windows in Italy were more likely (odds ratio: 1.279) to keep the window open while WFH.

Article

290 **The NEXT.ROOM: Design principles and systems trials of a novel test room aimed at deepening our knowledge on human comfort**

Vittori F.,Chiatti C., Pigliautile I.,Pisello A.L.

Building and Environment , 211, 108744

2022 38

Hide abstract Find It @ UCL View at Publisher Related documents

Occupants' interaction with building systems, and more generally occupants' behaviour, is strongly related to human perception of the surroundings. Progresses in human comfort theories rely upon data collection during tailored experiments, stressing the influence of single or specific combinations of environmental stimuli on diversified human responses. Laboratory experiments are typically performed in test rooms, i.e., rooms that are equipped to provide continuously monitored and controlled environmental boundaries. The NEXT.ROOM is a novel test room built up for investigating the multidisciplinary complexity of human perception, by controlling and monitoring environmental factors belonging to the four domains of comfort (*thermal*, visual, air quality, and *acoustics*) through centralized and human-centred systems. The study presents rationales behind its design and the selection of human-centred dedicated devices. Trials of the installed systems provide operational ranges for designing human comfort experiments to be performed within the NEXT.ROOM. Limits in fast-dynamics and range of internal temperature are

	Document title	Authors	Source	Year	Citations

also highlighted, since the adopted systems are mainly associated to common residential uses. The internal position of the test room also limits daylight access but the window designed in-line with the external laboratory window still guarantees the *outdoor* view from inside the NEXT.ROOM. A further innovation is the introduction of Virtual Reality tools for testing different internal layouts or even *outdoor* scenarios under highly controlled and carefully monitored environmental conditions.

Article

291 **The multisensory interaction between auditory, visual, and thermal to the overall comfort in public open space: A study in a tropical climate**

Nitidara N.P.A., Sarwono J., Suprijanto S., Soelami F.X.N.

Sustainable Cities and Society , 78, 103622

2022

95

Hide abstract Find It @ UCL View at Publisher Related documents

Human comfort, one of the most critical factors in public space utilization, is influenced by a multisensory interaction between stimuli in the built environment, namely auditory, *thermal*, and visual. However, the combined interaction between these stimuli and how it affects the perception of comfort and thus the use of space, especially for conditions in public open spaces in a tropical-climate country, is not fully understood. This study utilizes the Structural Equation Modeling method to model the interaction between the auditory, *thermal*, and visual factors in public open spaces by observing the physical parameters of the environment and the perceptions of space users. The results indicate a multisensory relationship between the visual and *thermal* factors induced by the sun exposures, in which Visual Sensation increases with an increase in *Thermal* Sensation. Another multisensory relationship exists between auditory and *thermal* factors, where Auditory Sensation increases with increased *Thermal* Sensation and *noise* level, which explains the dynamic in public open spaces. Additionally, changes in the auditory factor have the most significant influence on comfort in public open space, followed by *thermal* and visual factors. The findings of this study provide valuable information to support urban designers in creating comfortable urban public spaces.

Article • *Open access*

292 **How to Stay Cool: Early Acoustic and Thermal Experience Alters Individual Behavioural Thermoregulation in the Heat**

Udino E., Mariette M.M.

Frontiers in Ecology and Evolution , 10, 818278

2022

9

Hide abstract Find It @ UCL View at Publisher Related documents

Climate change is pushing organisms closer to their physiological limits. Animals can reduce heat exposure – and the associated risks of lethal hyperthermia and dehydration – by retreating into *thermal* refuges. Refuge use nonetheless reduces foraging and reproductive activities, and thereby potentially fitness. Behavioural responses to heat thus define the selection pressures to which individuals are exposed. However, whether and why such behavioural responses vary between individuals remains largely unknown. Here, we tested whether early-life experience generates inter-individual differences in behavioural responses to heat at adulthood. In the arid-adapted zebra finch, parents incubating at high temperatures emit “heat-calls,” which adaptively alter offspring growth. We experimentally manipulated individual early life *acoustic* and *thermal* experience. At adulthood, across two summers, we then repeatedly recorded individual panting behaviour, microsite use, activity (N = 2,402 observations for 184 birds), and (for a small subset, N = 23 birds) body temperature, over a gradient of air temperatures (26–38°C), in *outdoor* aviaries. We found consistent inter-individual variation in behavioural *thermoregulation*, and show for the first time in endotherms that early-life experience contributes to such variation. Birds exposed prenatally to heat-calls started panting at lower temperatures than controls but panted less at high temperatures. It is possible that this corresponds to a heat-regulation strategy to improve water saving at high temperature extremes, and/or, allow maintaining high activity levels, since heat-call birds were also more active across the temperature gradient. In addition, microsite use varied with the interaction between early *acoustic* and *thermal* experiences, control-call birds from cooler nests using the cooler microsite more than their hot-nest counterparts, whereas the opposite pattern was observed in heat-call birds. Overall, our study demonstrates that a prenatal *acoustic* signal of heat alters how individuals adjust behaviourally to *thermal* challenges at adulthood. This suggests that there is scope for selection pressures to act differently across individuals, and potentially strengthen the long-term fitness impact of early-life effects.

Article

	Document title	Authors	Source	Year	Citations
293	**Outdoor environment assessment tool for existing neighbourhoods based on the multi-criteria decision-making method**	Zhao K.,Jiang Z.,Li D.,Ge J.	Building and Environment , 209, 108687	2022	10

Hide abstract Find It @ UCL View at Publisher Related documents

Numerous existing neighbourhoods, such as many of those built-in 1980–2000 in China, have poor physical environmental conditions. Supplemented by field investigations, this paper puts forward an *outdoor* environment evaluation tool designed for existing neighbourhoods based on the multi-criteria decision-making (MCDM) method. Evaluation indicators of the *acoustic*, luminous, and *thermal* environments and air quality were chosen and classified according to the characteristics of existing neighbourhoods. The weight of each indicator was determined through opinions from both residents and experts. To achieve a comprehensive score, the calculation was carried out based on a multi-level system for neighbourhoods with different conditions/requirements. The proposed methodology was then applied to measure the *outdoor* environment performance of four existing neighbourhoods in Hangzhou, China. Results showed that, for all neighbourhoods, the *outdoor* environments were shown to be inadequate regarding *acoustic* and luminous environments, with improvement potential for the *thermal* environment and air quality. Corresponding suggestions for renovation and upgrading were put forward. The proposed tool can inform renovation potential of existing neighbourhoods from both residents' and experts' opinions, diagnosis of *outdoor* physical environments in four aspects, and development of renovation strategies under different environmental conditions/requirements.

Article

	Document title	Authors	Source	Year	Citations
294	**Combined effects of visual-acoustic-thermal comfort in campus open spaces: A pilot study in China's cold region**	Geng Y.,Hong B.,Du M., Yuan T.,Wang Y.	Building and Environment , 209, 108658	2022	108

Hide abstract Find It @ UCL View at Publisher Related documents

To investigate the combined effects of *thermal*, *acoustic* and visual comfort in open spaces, this study selected five typical spaces on a university campus with five common *sounds* (broadcasting music, running water & birdsong, wind (a gentle breeze) & insects, crowds, and machines) in a cold region of China. 418 volunteers were asked to complete a subjective sensory questionnaire in a randomly combined audio-visual environment while meteorological measures were taken on-site. Four primary outcomes emerged. First, broadcasting music improved individual subjective *thermal* comfort votes (TCV). Under the moderate heat stress ($0.5 \leq$ *thermal* sensation vote [TSV] < 2.5), a comfortable *acoustic* environment helped reduce individuals' subjective TSV and increase TCV. Second, the subjective *acoustic* comfort vote (ACV) of broadcasting music, wind & insect *sound* and machine *noise* decreased with the increase of Universal *Thermal* Climate Index (UTCI); respondents perceived crowds, machines and wind & insects louder when TCV = 2 and –2. Third, under no *thermal* stress ($-0.5 \leq$ TSV < 0.5)/moderate heat stress ($0.5 \leq$ TSV < 2.5), TSV increased with increasing illumination intensity (LUX). Under the strong heat stress ($2.5 \leq$ TSV ≤ 3), there was no significant difference in TCV among subjective visual comfort vote (VCV) levels. Finally, when the LUX class was neutral/slightly bright, respondents felt darker in *thermo*-neutrality. Additionally, VCV and TCV were positively correlated under each LUX class.

Article

	Document title	Authors	Source	Year	Citations
295	**Combined effects of the thermal-acoustic environment on subjective evaluations in outdoor public spaces**	Yin Y.,Zhang D.,Zhen M., ... Luo W.,Feng W.	Sustainable Cities and Society , 77, 103522	2022	53

Hide abstract Find It @ UCL View at Publisher Related documents

The public space environment is a complex and comprehensive system composed of multiple factors that are interconnected and can both promote and restrict each other. *Thermal* environment and *acoustic* environment are the main factors that affect the overall comfort of urban open spaces. Therefore, this study selects a university campus as a research location for an *outdoor* public space to explore the change characteristics of human comfort in the *thermal-acoustic* environment with the object of providing a reference for improving the comfort of the *outdoor* public space environment. The study results show that the effects of *thermal-acoustic* environment on the subjective comfort are quite different in autumn and winter; the *acoustic* environment has a greater impact on subjective comfort in high and low temperature environments, while the effect of *acoustic* environment on subjective comfort in

	Document title	Authors	Source	Year	Citations

a *thermally* neutral environment is not obvious; the overall comfort evaluation is the highest in a 60–70 dB(A) *acoustic* environment, and an excessively noisy or quiet *acoustic* environment increases the discomfort; both the *acoustic* environment and the *thermal* environment are important factors affecting the subjective comfort of the human body, but the *thermal* environment is more important than the *acoustic* environment.

Article • *Open access*

296 **Green CURIOCITY: A study protocol for a European birth cohort study analysing childhood heat-related health impacts and protective effects of urban natural environments**

Van Den Bosch M., Basagana X., Mudu P., ... Vrijheid M., Nieuwenhuijsen M.J.

BMJ Open, 12(1), e052537

2022

4

Hide abstract Find It @ UCL View at Publisher Related documents

Introduction The European climate is getting warmer and the impact on childhood health and development is insufficiently understood. Equally, how heat-related health risks can be reduced through nature-based solutions, such as exposure to urban natural environments, is unknown. Green CURe In *Outdoor* CITY spaces (Green CURIOCITY) will analyse how heat exposure during pregnancy affects birth outcomes and how long-term heat exposure may influence children's neurodevelopment. We will also investigate if adverse effects can be mitigated by urban natural environments. A final goal is to visualise intraurban patterns of heat vulnerability and assist planning towards healthier cities. Methods and analysis We will use existing data from the Human Early-Life Exposure cohort, which includes information on birth outcomes and neurodevelopment from six European birth cohorts. The cohort is linked to data on prenatal heat exposure and impact on birth outcomes will be analysed with logistic regression models, adjusting for air pollution and *noise* and sociobehavioural covariates. Similarly, impact of cumulative and immediate heat exposure on neurodevelopmental outcomes at age 5 will be assessed. For both analyses, the potentially moderating impact of natural environments will be quantified. For visualisation, Geographical information systems data will be combined to develop vulnerability maps, demonstrating urban € hot spots' where the risk of negative impacts of heat is aggravated due to sociodemographic and land use patterns. Finally, geospatial and meteorological data will be used for informing GreenUr, an existing software prototype developed by the WHO Regional Office for Europe to quantify health impacts and augment policy tools for urban green space planning. Ethics and dissemination The protocol was approved by the Comité Ético de Investigación Clínica Parc de Salut MAR, Spain. Findings will be published in peer-reviewed journals and presented at policy events. Through stakeholder engagement, the results will also reach user groups and practitioners.

Article

297 **Indoor noise A-level assessment related to the environmental noise spectrum on the building facade**

Zaporozhets O., Fiks B., Jagniatinskis A., ... Karpenko S., Mickaitis M.

Applied Acoustics, 185, 108380

2022

14

Hide abstract Find It @ UCL View at Publisher Related documents

There is a growing body of research reporting that unwanted indoor *noise* can negatively affect the well-being of the occupants. According to the World Health Organisation (WHO), indoor guideline values in dwellings are 30 dBA ($L_{Aeq,night,indoor}$), which in turn should also include appropriate airborne and impact *sound* insulation of façades. Such low *noise* values in dwellings are difficult to measure, which is why indirect methods of indoor *noise* assessment (including the accurate calculation and measurement procedures for *sound* insulation of dwelling façades) are an important issue worth investigating. Therefore, in this paper, a new methodology of how indoor *noise* A-level may be assessed by applying the *outdoor noise* monitoring and standardized *sound* insulation measurement of the façade is proposed. It also provides an overview of the existing difficulties in performing accurate calculations of the *sound* insulation on the façades. The proposed calculation procedure for indoor equivalent *noise* A-level estimation is also evaluated. The highest accuracy was achieved by adding to the calculation a specific spectrum adaptation term $C_{spec,100-3150}$ for the *noise* emitted façade in-situ instead of a standard spectrum adaptation term $C_{tr,100-3150}$, defined by the unified spectra corresponding to road traffic, and suggested by the ISO 717–1 for the calculations. The specific spectrum adaptation term was calculated based on the averaged spectrum of the specific *noise* events (different from road traffic *noise*) that contribute to the *outdoor* environment at the locations where measurements were carried out.

	Document title	Authors	Source	Year	Citations

When assessed in such a way, the equivalent *sound* levels of indoor *noise*, which are produced by different types of *outdoor noise* sources, show a good agreement with the directly measured levels indoors.

Article

298 **Outdoor environmental investigation of old communities during summer in hot summer and cold winter regions**

Li D.,Zhao K.,Ge J.,Ren M.

Indoor and Built Environment , 31(1)

2022

8

Hide abstract Find It @ UCL View at Publisher Related documents

The *outdoor* physical environments of many old communities are unable to meet modern living requirements and require improvement urgently. This is especially true for communities in regions with extremely hot summers and cold winters, which may cause additional difficulties. This study used measurements and questionnaire data to investigate *outdoor* environment (the *thermal*, *sound*, light environments and air quality) and resident sensations in four such communities. Results revealed some key factors affecting *outdoor* environmental comfort, including summer shading, winter sunlight, air temperature, air quality, the *sound* environment and odour. The results also revealed that the main problems were *noise*, insufficient illumination and high air temperatures in summer, all of which had seriously affected comfort during *outdoor* activities. Specifically, *noise* values during the most unfavourable periods were between 57.4 and 80.6 dB(A), while average air temperatures were between 32.3°C and 35.8°C, and average illuminances were below 2.9 lx. As such, this paper proposes improvement measures such as *sound* barriers and greening. Further, we propose a method for prioritizing these factors for such improvements based on a combination of subjective and objective factors. This study provides data and technical references for the reconstruction of old communities in regions with hot summers and cold winters.

Conference Paper

299 **Acceptable environmental conditions for working in semi-outdoor space in the tropics**

Mihara K.,Chen S., Hasama T.,... Lee J.K.W., Wong N.H.

17th International Conference on Indoor Air Quality and Climate Indoor AIR 2022

2022

0

Hide abstract Find It @ UCL View at Publisher Related documents

The aim of this study was to ascertain the acceptable conditions of environmental factors in semi-*outdoor* space in the tropics. Experiments were conducted in a semi-*outdoor* space in Singapore. A total of 89 participants were surveyed in a semi-*outdoor* space with or without the operation of a ceiling fan. Eighty percent of participants found the accepted *thermal* environment ranging from 23.3 to 30.1 °C SET*. Eighty percent of participants were satisfied with the lighting levels of more than 260 lx and *noise* levels of less than 48 dBA. These findings promote the use of the semi-*outdoor* space during breaks and short-term work activities.

Article • *Open access*

300 **S-shaped dependence of the sound pressure level in outdoor propagation on the effective sound speed gradient**

Hohenwarter D., Mursch-Radlgruber E., Kirisits C.

Acta Acustica, 6, 13

2022

3

Hide abstract Find It @ UCL View at Publisher Related documents

The effective *sound* speed gradient is used to describe the meteorological conditions during *sound* measurements at roads and railways. Meteorological parameters were assessed up to a height of 10 m. The *sound* level differences between a reference point close to the passing vehicles and at distances of 100 500 m from motorways and railway tracks were determined. These differences were found to correlate well with the effective *sound* speed gradient determined from the measured temperature and wind speed gradients which follow the day/night cycle as a result of the reversing air temperature gradient, incoming solar radiation and wind conditions. The correlation with *sound* level differences can be approximated by an S-shaped function which is constant for large positive or negative gradients of the effective *sound* speed. These threshold values are a consequence of the local meteorological and attenuation conditions. The analysis shows that large effective *sound* speed gradients are mainly a result of the wind conditions whereas gradients without a substantial wind speed contribution are low and a result of the temperature gradient. In the distance range between 150 m and 250 m, the meteorological influences cause a level increase of 3 4

	Document title	Authors	Source	Year	Citations

dB(A) for downward refracted *sound* rays (favourable *sound* propagation) compared to a situation without meteorological influence (effective *sound* speed gradient of zero). In the same distance range, meteorological conditions cause a maximum *sound* level attenuation of 5 10 dB(A) for upward curved *sound* rays (unfavourable *sound* propagation).

Conference Paper

301 **Multi-Objective Optimization Of Thermo-Acoustic Comfort Of School Buildings** — Colarossi D.,Di Loreto S., Tagliolini E.,Principi P., Serpilli F. — Building Simulation Applications , 2022-June — 2022 — 3

Hide abstract Find It @ UCL View at Publisher Related documents

The reduction of the environmental impact of the building sector is one of the top priorities in the “climate change challenge”. As the primary energy consumption of the building decreases, a high level of indoor comfort must be maintained. Both *thermal* parameters, lighting, *acoustic* level, and indoor air quality affect indoor comfort. These aspects are fundamental, especially in school buildings, where a good level of indoor comfort can help student to stay focused. This paper proposes a methodology for a combined optimization of the energetic and the *acoustic* performance of a school building. A case study, located in the center of Italy, was analyzed. Firstly, the *thermal* and *acoustic* performance was determined. Then a list of interventions was hypothesized and simulated, involving both the building envelope, the lighting and *thermal* plants. Normalized *acoustic* insulation of partitions between adjacent rooms, *acoustic* insulation of the façade and reverberation time were evaluated. The *outdoor* and ambient *noise* levels were based on the main characteristics of the façade (type and stratifications of opaque and transparent components, ventilation system, etc.). Results show that the optimal combination of interventions reduces the CO2 emissions of 88.55 % and the global energy performance index of 85.2 %. The indoor *sound* pressure level due to traffic *noise* is reduced by 19 dB after *acoustic* insulation of the façade, while further treatments to indoor surfaces should be implemented to reduce internal reverberation time and to improve speech intelligibility. The combined optimization shows that the highest reduction of the global impact (89.2 %) is obtained by weighting 80 %/20 % the *acoustic*/*thermal* performance.

Conference Paper

302 **New Sensing Systems for Securing Virtual Walls at Outdoor Based on True Differential Digital TMOS** — Blank T.,Brouk I., Bar-Lev S.,... Bruno G., Nemirovsky Y. — Proceedings IEEE International Symposium on Circuits and Systems , 2022-May — 2022 — 3

Hide abstract Find It @ UCL View at Publisher Related documents

This paper presents a non-imaging differential digital passive Infra-Red (PIR) remote sensing system using CMOS-SOI-MEMS transistors as the *thermal* sensor. A large 660mu {mathrm m}, {mathrm x}, 660mu{mathrm m} pixel area is developed by the 8x8 mosaic matrix of 60mu {mathrm m} ,{mathrm x}, 60mu{mathrm m} sub-pixels connected in-parallel. The mosaic sensors, which are manufactured by nano-fabrication methods in CMOS FABs, exhibit enhanced performance and robust manufacturing in wafer level processing and vacuum packaging. Since the sub-pixels are *thermally* isolated, the *thermal* time constant of the large pixel is determined by that of the sub-pixel, which is sim {mathrm 80} msec when packaged in vacuum of sim 1{mathrm Pa}. For *outdoor* operation, two identical large pixels are differentially measured. The pixels view the detected scene with an optics that may be based either on mirror optics or Fresnel plastic lenses. The optics defines a narrow field of view (± 3 degrees) as required for curtain sensors. Furthermore, the optics forms 'cockeyed' vision which enables differential measurement that cancels the environmental' *noise*' and allows *outdoor* operation. The overall measured performance for detecting human targets at extended ranges and hot spots detection are reported. This sensor outperforms *thermopiles* and pyroelectric sensors at *outdoor* operation.

Conference Paper • *Open access*

303 **Vehicular Visible Light Communications Noise Analysis and Autoencoder Based Denoising** — Turan B.,Kar E.,Coleri S. — 2022 Joint European Conference on Networks and — 2022 — 8

	Document title	Authors	Source	Year	Citations
			Communications and 6g Summit Eucnc 6g Summit 2022		

Hide abstract Find It @ UCL View at Publisher Related documents

Vehicular visible light communications (V-VLC) is a promising intelligent transportation systems (ITS) technology for vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communications with the utilization of light emitting diodes (LEDs). The main degrading factor for the performance of V-VLC systems is *noise*. Unlike traditional radio frequency (RF) based systems, V-VLC systems include many *noise* sources: solar radiation, background lighting from vehicle, street, parking garage and tunnel lights. Traditional V-VLC system *noise* modeling is based on the additive white Gaussian *noise* assumption in the form of shot and *thermal noise*. In this paper, to investigate both time correlated and white *noise* components of the V-VLC channel, we propose a *noise* analysis based on Allan variance (AVAR), which provides a time-series analysis method to identify *noise* from the data. We also propose a generalized Wiener process based V-VLC channel *noise* synthesis methodology to generate different *noise* components. We further propose convolutional autoencoder (CAE) based denoising scheme to reduce V-VLC signal *noise*, which achieves reconstruction root mean square error (RMSE) of 0.0442 and 0.0474 for indoor and *outdoor* channels, respectively.

Conference Paper • *Open access*

☐ 304	**Reducing Object Detection Uncertainty from RGB and Thermal Data for UAV Outdoor Surveillance**	Sandino J.,Caccetta P.A., Sanderson C.,Maire F., Gonzalez F.	IEEE Aerospace Conference Proceedings , 2022-March	2022	11

Hide abstract Find It @ UCL View at Publisher Related documents

Recent advances in Unmanned Aerial Vehicles (UAVs) have resulted in their quick adoption for wide a range of civilian applications, including precision agriculture, biosecurity, disaster monitoring and surveillance. UAVs offer low-cost platforms with flexible hardware configurations, as well as an increasing number of autonomous capabilities, including take-off, landing, object tracking and obstacle avoidance. However, little attention has been paid to how UAVs deal with object detection uncertainties caused by false readings from vision-based detectors, data *noise*, vibrations, and occlusion. In most situations, the relevance and understanding of these detections are delegated to human operators, as many UAVs have limited cognition power to interact autonomously with the environment. This paper presents a framework for autonomous navigation under uncertainty in *outdoor* scenarios for small UAVs using a probabilistic-based motion planner. The framework is evaluated with real flight tests using a sub 2 kg quadrotor UAV and illustrated in victim finding Search and Rescue (SAR) case study in a forest/bushland. The navigation problem is modelled using a Partially Observable Markov Decision Process (POMDP), and solved in real time onboard the small UAV using Augmented Belief Trees (ABT) and the TAPIR toolkit. Results from experiments using colour and *thermal* imagery show that the proposed motion planner provides accurate victim localisation coordinates, as the UAV has the flexibility to interact with the environment and obtain clearer visualisations of any potential victims compared to the baseline motion planner. Incorporating this system allows optimised UAV surveillance operations by diminishing false positive readings from vision-based object detectors.

Article • *Open access*

☐ 305	**Window Opening Behavior of Residential Buildings during the Transitional Season in China's Xi'an**	Yang X.,Liu J.,Meng Q., ... Zhang L.,Lian Y.	Complexity, 2022, 4593101	2022	14

Hide abstract Find It @ UCL View at Publisher Related documents

Window opening behavior in residential buildings has important theoretical significance and practical value for improving energy conservation, indoor *thermal* comfort, and indoor air quality. Climate and cultural differences may lead to different window opening behavior by residents. Currently, research on residential window opening behavior in northwest China has focused on indoor air quality, and few probabilistic models of residential window behaviors have been established. Therefore, in this study, we focused on an analysis of factors influencing window opening behavior and the establishment of a predictive model for window opening behavior. Four typical residential buildings in different locations and building types in Xi'an were selected. The indoor and *outdoor* environments and window opening states were measured. Subsequently, a multivariate analysis of variance was used to determine the factors

	Document title	Authors	Source	Year	Citations

that had a significant effect on window opening behavior. Single- and multiparameter logistic regression models for window opening behavior were established. Of all the measured factors, we found that indoor temperature and CO2 concentration, *outdoor* temperature, and relative humidity had significant effects on window opening behavior, and indoor relative humidity and *noise* did not. Meanwhile, the temperature was positively correlated with the window opening probability, whereas indoor CO2 concentration and *outdoor* relative humidity were negatively correlated. The prediction accuracy of the multiparameter model was promising, at almost 75%, and the model can provide theoretical support for modelling residential buildings in Xi'an.

Article • *Open access*

306 **TiO2nano-coated thin film PV glazing with superior thermal resistance, self-cleaning, electricity generation and adaptive optical control**

Cuce E.,Cuce P.M.,Riffat S. | International Journal of Low Carbon Technologies , 17 | 2022 | 18

Hide abstract Find It @ UCL View at Publisher Related documents

A unique nano-coated photovoltaic (PV) glazing technology with superior multifunctional features, *thermally* resistive PV glazing (TRPVG), is introduced, and for three different configurations of TRPVG (TRPVG-Air12, TRPVG-Ar12, TRPVG-Ar16), UVC/UVA absorption, *noise* reduction, *thermal* insulation, electricity generation, visible light and solar radiation control are evaluated through an extensive experimental methodology. Energy production and *acoustic* tests are conducted in a simulation house, whereas the rest of the experiments are carried out under real operating conditions. The results reveal that each sample is capable of blocking 100% of incoming UVC and UVA light. Visible light control of TRPVG-Ar12 (Glass 1) is found to be 94.4%, whereas it is 88.9% for TRPVG-Air12 (Glass 2) and 93.6% for TRPVG-Ar16 (Glass 3). Solar radiation blockage of Glasses 1-3 is found to be 93.5%, 90.9% and 94.8%, respectively. Average temperature difference between front and rear glazing is determined to be 21.3°C, 19.9°C and 21.7°C for Glasses 1, 2 and 3, respectively. A total of 25 independent *acoustic* tests are performed for Glass 3, and the sample is observed to reduce 33% of *outdoor noise* in dBA. Solar simulator tests reveal that Glass 3 can generate 102.6 W of electricity per square metre of PV module area.

Conference Paper

307 **Edge and Fog Computing for IoT: A Case Study for Citizen Well-Being**

Bianconi L.,Lechiara Y., Bixio L.,... Viti F.,Bozzano R. | Lecture Notes of the Institute for Computer Sciences Social Informatics and Telecommunications Engineering Lnicst , 442 LNICST | 2022 | 2

Hide abstract Find It @ UCL View at Publisher Related documents

Citizen well-being during indoor and *outdoor* activities is a prerogative in the paradigm of smart city. Air quality, together with *thermo*-hygrometric, light and *noise* comfort are some of the most relevant parameters that affect the perception of the state of well-being. An internet of things (IoT) network based on LoRaWAN end-nodes has been set up with the aim to provide an approach to improve the well-being of citizens living in an urban district in Savona (Liguria region, Italy) under the umbrella of the PickUP project. The main focus of the project was the design of innovative methods and tools for energy and environmental management and the reduction of consumption in heterogeneous districts. The proposed IoT architecture comprises a Fog node constituted of a Raspberry Pi 4 to analyze in near-real-time the heterogeneous data and to provide feedback to the proper actuators in case of need of changes in the ventilation and heating of the occupied room. Moreover, the infrastructure provides insights about the air quality in *outdoor* areas to local authorities, in order to implement mitigation action or strategies. Results demonstrate the usefulness of the proposed architecture with a precise focus on social impact, but also reveal economic implications linked to an intelligent use of energy resources and to the development of energy efficiency strategies.

Conference Paper

	Document title	Authors	Source	Year	Citations
308	**Computational Analysis of Aerothermal Characteristics of Skygardens and Comparative Assessment of Different Attenuators**	Mohammadi M.,Tien P.W.,Calautit J.	Building Simulation Conference Proceedings	2022	0

Hide abstract Find It @ UCL View at Publisher Related documents

Buildings consisting of vegetative measures, such as green roofs and skygardens hold multiple benefits including improvements in the local microclimate, attenuating *noise* and air pollution, and physiological wellbeing. Nevertheless, skygardens located on high levels, may be exposed to high wind speeds, endangering the safety of occupants. Despite its popularity, very few studies have investigated the aero-*thermal* characteristics of such *outdoor* spaces and the influence of surrounding vegetation. This study utilises Computational Fluid Dynamics (CFD) to analyse 3 types of wind buffers on the wind characteristics of a skygarden located on a high-rise building. The Realizable k-e setup was employed where the vegetation was modelled as a porous zone. Results indicate that not all types of barriers can generate comfortable environment within the region, and taller barriers provide more conducive environment. Porous barriers can attenuate wind speeds in the depth as well as the height of the space, while also producing some local cooling.

	Conference Review				
309	**2022 IEEE 28th International Symposium for Design and Technology in Electronic Packaging, SIITME 2022 - Conference Proceedings**	[No Authors Found]	2022 IEEE 28th International Symposium for Design and Technology in Electronic Packaging Siitme 2022 Conference Proceedings	2022	0

Hide abstract Find It @ UCL View at Publisher

The proceedings contain 53 papers. The topics discussed include: solutions for *acoustic noise* caused by multilayer ceramic capacitors; study regarding the configuration setup used inside an EMC laboratory for NFC communication measurement; color detector for visual impaired users; conceptual model of a system for controlling the process of egg-hatching in incubators; analysis of different types of wireless communication modules with a real-time spectrum analyzers; electromagnetic and electric field radiation inside electric vehicles and classic vehicles; performance analysis of *thermoelectric* cooler — *thermoelectric* generator system for heat recovery applications; real-time embedded framework debugger; mapping the environment at range: implications for camera calibration; gesture interaction with 3d printed hybrid compression and inertial tracking device based on plastic embedded antenna for virtual reality integration; printed circuit board assembly modeling for predictive reliability assessment; and preliminary study for *outdoor* testing: effect of moisture and temperature on the stability of UV dye sensitized solar cells.

	Article • *Open access*				
310	**Impact of COVID-19 protocols on IEQ and students' perception within educational buildings in Southern Spain**	Aguilar A.J.,de la Hoz-Torres M.L.,Oltra-Nieto L.,Ruiz D.P.,Martinez-Aires M.D.	Building Research and Information , 50(7)	2022	16

Hide abstract Find It @ UCL View at Publisher Related documents

The COVID-19 pandemic has highlighted the fact that air quality is essential in buildings. In the case of educational buildings, teaching activities were moved to an online format during the 2020/2021 academic year, with only activities such as exams remaining face-to-face. This strategy required the development of protocols to ensure that classrooms were safe spaces. This study assesses the impact of these protocols in the indoor environmental conditions of educational buildings in Southern Spain. For this purpose, a measurement campaign was carried out at the Fuentenueva Campus of the University of Granada. The results show that the protocols have guaranteed effective ventilation. However, other indoor environmental variables have also been affected, including the satisfaction of

Document title	Authors	Source	Year	Citations

users during exams due to temperature, relative humidity (RH) and *noise*. The highest levels of satisfaction were related to indoor lighting, while the highest levels of dissatisfaction were related to the indoor *thermal* environment. Among the main causes of dissatisfaction were draughts and *outdoor noise*, directly related to natural ventilation protocols during the pandemic. Based on these findings, current pandemic protocols should be revised and redesigned to minimize the impact on student satisfaction and perceived learning performance from the identified environmental sources in this research.

Conference Paper

311 **A Multiband Uncooled Infrared Camera for Measuring Volcanic SO2Gas Concentration and Temperature Distributions** — Jitsufuchi T. — International Geoscience and Remote Sensing Symposium IGARSS, 2022-July — 2022 — 2

Hide abstract Find It @ UCL View at Publisher Related documents

Measuring the surface phenomena of a volcano is useful for predicting its short-term activity. To this end, methods for remotely sensing SO2 gas emitted from volcanoes have been developed. One such method is an imaging sensor for observing infrared radiation in the longwave infrared (LWIR) region. This method allows for day and night observation of infrared radiation. In this study, we develop a multiband LWIR camera named SPIC-UC/4VGA prototype for the quantitative imaging of volcanic surface phenomena, such as surface temperature and gas distribution. The SPIC-UC/4VGA prototype consists of four uncooled infrared cameras equipped with an internal optical filter (Cameras 1, 2, 3, and 4). All four cameras employ focal plane array (FPA)-type amorphous silicon microbolometer elements. Cameras 1-3 measure the 7500-14000, 9000-14000, and 7950-9300 nm regions, which match the SO2 absorption bands, and Camera 4 measures the 11785-12785 nm region. These four cameras can acquire frame-synchronized data at 30 fps. The SPIC-UC/4VGA prototype is realized as an environment-resistant type in consideration of *outdoor* use. Using a shutterless operation, the performance evaluations indicate that all cameras achieve an absolute temperature accuracy of ± 2 K. For SO2 detection, Camera 3 can achieve an NETD (*noise* equivalent temperature difference) of approximately 0.39 K. The evaluation of SO2 measurement accuracy by simulation using the same S02 gas concentration distribution conditions as those under which our original airborne hyperspectral sensor (ARTS) actual observations were made on April 8, 2008 reveals that the developed Camera 3 can detect SO2 gas concentration distributions under background conditions at a temperature of 50°C with errors of ± 1 ppmv. These results indicate that the developed SPIC-UC/4VGA prototype can be used for volcanic SO2 gas quantitative detection.

Article

312 **Examination and optimization of classroom indoor environments in China's hot summer and cold winter regions** — Ren M.,Zhao K.,Huang Z., Ge J. — Science and Technology for the Built Environment, 28(10) — 2022 — 3

Hide abstract Find It @ UCL View at Publisher Related documents

The school's indoor environment significantly affects students' health and learning performance. Although related studies have been carried out in many regions, only a few have examined indoor classroom environments and proposed the energy-saving optimization of heating and ventilation in winter in China's hot summer and cold winter areas. This study aims to comprehensively investigate classroom indoor environmental quality in China's hot summer and cold winter areas, and propose targeted optimization strategies. It first examined the indoor environment, including indoor air, *acoustic*, visual, and *thermal* quality, of two typical middle schools. The indoor CO_2 concentration in winter was found to be unsatisfactory. Subsequently, how the indoor layout, *outdoor* wind and occupancy density influence indoor CO_2 concentration was examined through simulation, which indicated that high occupancy density is the main cause. Finally, this study proposed several ventilation optimization strategies and performed an energy analysis to maintain sufficient air quality and a comfortable *thermal* environment in high-density classrooms. The results show that compared with the intermittent mode, the fixed ventilation and heating mode could maintain a more comfortable indoor environment and achieve a 30% energy saving. A reference for the improvement of classroom indoor environment in hot summer and cold winter areas was provided.

	Document title	Authors	Source	Year	Citations

Article • *Open access*

313 **The Effect of Different Specifications of Passive Spaces on Residents' Satisfaction in Adjoining Spaces within a Hot Dry Climate**

Alzahrani M.,Majid R.A., Karban A.S.,Majrashi A.

International Journal on Advanced Science Engineering and Information Technology , 12(2)

2022

4

Hide abstract Find It @ UCL View at Publisher Related documents

Passive spaces are a passive design strategy that aims to reduce energy consumption and increase user satisfaction in buildings. One example of passive space is the air shaft. The air shaft is a vertical void within the building from the ground level to the roof level, and it provides the building with natural ventilation and daylight, especially in deep-plan buildings. However, the function of the air shaft is questioned due to its impacts on residents' needs. This study assesses the effects of air shaft specifications on residents' satisfaction with the indoor environment quality of air shafts and adjoining spaces. Survey questionnaires were distributed to residents of apartment buildings. The results proved that air shafts have a significant negative impact on residents' satisfaction. The findings of cross-tabulation analysis illustrate a significant relationship between the air shafts' specifications and the residents' answers. The analysis also showed that the air shafts that are closed from the bottom and include A\C *outdoor* units have a more negative impact on the *thermal* environment and air quality. Regarding the air shaft areas, the small areas have a high negative response regarding bad smell, the view, visual and *acoustic* privacy, and *thermal* environment. From the indoor environment quality perspective, this study emphasizes the need to consider the impact of air shaft design on a building's performance and residents' satisfaction. The results of this study are expected to contribute to the development of future passive spaces design.

Article

314 **The influence of perceived aesthetic and acoustic quality on outdoor thermal comfort in urban environment**

Lau K.K.-L.,Choi C.Y.

Building and Environment , 206, 108333

2021

111

Hide abstract Find It @ UCL View at Publisher Related documents

Thermal comfort in *outdoor* space is essential for human health and human wellbeing. The comfortable *outdoor* space enhances urban livability and sustainability. Currently, the influence of environmental quality on human *thermal* comfort is not conclusive. Research on the interrelation between perceived environmental quality and subjective human *thermal* comfort is needed to have a concrete argument. This paper examines the relationship between perceived aesthetics, perceived *acoustics*, and the *outdoor thermal* comfort in Hong Kong during the hot summer, by conducting questionnaires and on-site meteorological measurement. *Thermal* sensation vote (TSV) showed a strong, negative association with the perceived aesthetics vote and *acoustics* vote as calculated for 1 °C UTCI bin. It was also revealed that the groups with satisfactory of perceived *acoustic* and aesthetics have a significantly higher comfort vote than that of unsatisfactory groups. Findings suggest that humans in a perceptually quiet and beautiful *outdoor* environment have a significantly higher *thermal* tolerance, and lower *thermal* sensitivity. This work provides valuable data on the effects of perceived environmental quality on *outdoor thermal* comfort in subtropical hot summers in high density urban settings. These findings help the urban development in the *outdoor* urban environment in our changing climate. Urban planner and designer can create a more satisfactory aesthetic and *acoustic* environment to improve the *thermal* tolerance and adaptation of individuals in the *outdoor* urban environment.

Review • *Open access*

315 **How housing conditions determine the welfare of pigs**

Ludwiczak A., Skrzypczak E., Skladanowska-Baryza J., ... Slosarz P.,Racewicz P.

Animals, 11(12), 3484

2021

32

Hide abstract Find It @ UCL View at Publisher Related documents

Humans who care for pigs prefer an environment that not only allows the pigs to express their natural behaviors but also limits the development of aggression and stereotypes. Most of the behavioral and health problems encountered

	Document title	Authors	Source	Year	Citations

by pigs in barren, conventional conditions are solved by alternative housing systems. However, it is not known whether these systems are advantageous in terms of the performance of pigs. In this work, we review the effects of housing systems on pigs' behaviors and performance, which are among the major indicators of the welfare of these animals. Research results point out that *outdoor* systems are more ideal for sows and fatteners than buildings. Nonetheless, *outdoor* housing is associated with two major effects in both groups: increased activity due to environmental exploration and higher space allowance, and increased incidence of injuries compared to indoor systems. Sows are more active when housed in groups, but they experience an increased frequency of injuries. According to the literature, group-housed sows give birth to healthy piglets with good daily weight gains. The difference in the fattening and slaughter performance of pigs raised indoors vs. outdoors remains unclear, and the results reported so far are inconsistent. *Outdoor* systems seem to be associated with a higher incidence of osteochondrosis and lesions of elbow and hock joints, whereas indoor systems cause a greater degree of body soiling in pigs. Based on the reviewed literature, it may be concluded that *outdoor* housing helps to solve behavioral issues in pigs but leads to other problems in pig production.

Article

316 **Experimental Study on Phase Change Material Used in Detachable Arms Protective Clothing | 相变材料用于可拆卸护臂热防护服的实验研究** Chang M.,Wang Y., Zhang Y.,Chen Z.,Huang L. Journal of Refrigeration , 42(6) 2021 1

Hide abstract Find It @ UCL View at Publisher Related documents

Working under high-temperature conditions is becoming increasingly common in summer; therefore, the working conditions of delivery workers, cleaners, security guards, and other personnel should be ensured, which is in line with China's people-oriented development strategy. To overcome the shortcomings of traditional cooling clothing, such as a poor cooling effect, low battery life, high energy consumption, and high *noise*, *thermal* protective clothing with detachable arms for human operation in a high-temperature summer environment is designed using the characteristics of the phase change material $Na_2SO_4{\cdot}10H_2O$, which absorbs heat at high temperatures. On this basis, a simulation experiment with water as the heating source and a real human test are conducted. The test results show that in the simulation test with hot water, under the working conditions of 30 ℃, 41 ℃, and 45 ℃, the temperatures of the four measuring points (the front chest, side abdomen, back, and abdomen) increase to the highest level in 60 min, and the average temperature of the four measuring points is finally reduced to approximately 26 ℃. The protective clothing exhibits good cooling performance at 30 ℃, and poor cooling performance at 45 ℃. In the real human test, the average temperature of the four measuring points in the protective clothing can finally decrease to approximately 26 ℃ within 3 h under the working conditions of 37. 1 ℃, 39 ℃, and 41 ℃. Under the high-temperature condition of 60 ℃, the average temperature of the four measuring points can finally decrease to approximately 31 ℃ within 3 h, and the average temperature of the inside and outside of the protective arm finally decrease to approximately 32 ℃, which are both lower than the critical temperature of the human body burning at 45 ℃. Therefore, the protective clothing exhibits a good cooling effect and can meet the *thermal* comfort requirements of *outdoor* workers in summer.

Article • *Open access*

317 **Passive window ventilation openings in every-day use** Hoffmann C.,Geissler A., Hauri C.,Huber H. Building and Environment , 206, 108259 2021 4

Hide abstract Find It @ UCL View at Publisher Related documents

The integration of passive window ventilation openings (PWVO, small air inlets integrated in the window frame) with additional exhaust fans in the kitchen and bathroom(s) can ensure a user-independent basic air change rate in dwellings. The project reported on herein is focused on how well buildings with PWVO work in real life. Altogether 28 multi-family houses are investigated by site visits, a survey among the inhabitants and measurements in eight flats. The survey is conducted in the winter 2017/18 and addresses user ventilation behavior, *thermal* and *acoustic* comfort, IAQ and user satisfaction. The 270 completed questionnaires returned allow tentative inferences for future ventilation concepts. For instance, it is found that in dwellings heated with radiators, air draught caused by PWVO is reported less frequently then in dwellings with a floor heating. The measurements in the winter 2018/19 are twofold: short-term measurements with a focus on the volume flow rates, the airtightness and the relative pressures in the flats. Long-

	Document title	Authors	Source	Year	Citations

term measurements comprise CO_2 concentrations, interior and exterior air temperatures and humidities and operation modes of the extract fans. In six out of eight flats the measured *outdoor* air flow rate covers only 35–80 % of the amount recommended by the Swiss building standards. The airtightness of all flats varies between q_{a50}-values of 0.5 and 1.3 $m^3/(h\ m^2)$ (±10 %). CO_2 measurements show that in the sleeping rooms the mean is between 650 and 3'440 (±70–160) ppm.

Conference Paper • *Open access*

318 **The influence of perceived environmental quality on thermal comfort in an outdoor urban environment during hot summer** — Lau K.K.-L.,Choi C.Y. — Journal of Physics Conference Series, 2042(1), 012047 — 2021 — 4

Hide abstract Find It @ UCL View at Publisher Related documents

Thermal comfort in *outdoor* spaces is essential for human health and human wellbeing. A comfortable *outdoor* space enhances urban livability and sustainability. Previous studies on *outdoor* human *thermal* comfort highlighted that apart from the microclimate conditions, the psychological and physiological factors play an important role in human *thermal* comfort. The influence of environmental quality on human *thermal* comfort is being examined in this paper. A survey with a total of 1842 *thermal* comfort responses was conducted during a hot summer in Hong Kong. Perceived aesthetic and *acoustic* quality votes are strongly associated with *Thermal* Sensation Votes (TSV). *Thermal* Comfort Votes (TCV) in the satisfied aesthetic group and the satisfied *acoustic* group are significantly higher than that in the not satisfied group. A sensation of comfort was confirmed by 39.8% and 38.4% of participants in the satisfied aesthetic group and the satisfied *acoustic* group, while only 22.2% and 23.9% of the members of the not satisfied group felt comfortable. The study suggested that the perceived environmental qualities are highly associated with *thermal* sensation and *thermal* comfort, and a beautiful and quiet environment can improve the *thermal* comfort and *thermal* tolerance.

Article • *Open access*

319 **Machine learning-based prediction of surface checks and bending properties in weathered thermally modified timber** — van Blokland J.,Nasir V., Cool J.,Avramidis S., Adamopoulos S. — Construction and Building Materials, 307, 124996 — 2021 — 26

Hide abstract Find It @ UCL View at Publisher Related documents

Machine learning (ML)-based models, decision tree and ANFIS, were used to predict the degree of surface checking and bending properties of 30-month weathered *thermally* modified timber. The results showed that the investigated initial board properties did not allow accurate predictions of surface checks. ML regression and clustering analysis confirmed important variables for accurate predictions of bending properties were dynamic stiffness, *acoustic* velocity, density and lowest local bending modulus. ML models performed better than conventional regression models used for timber grading, and a prediction accuracy of 80–90% for bending stiffness and 50–70% for bending strength could be achieved.

Article

320 **Performance of externally bonded fiber-reinforced polymer retrofits in the 2018 Cook Inlet Earthquake in Anchorage, Alaska** — Tatar J.,Sattar S., Goodwin D.,... Dukes J., Segura C. — Earthquake Spectra, 37(4) — 2021 — 5

Hide abstract Find It @ UCL View at Publisher Related documents

As part of the effort to improve the seismic performance of buildings in Alaska (AK), many of the deficient structures in Anchorage, AK, were retrofitted—some with externally bonded fiber-reinforced polymer (EBFRP) composite systems. The 2018 magnitude 7.1 Cook Inlet earthquake that impacted the same region offered an opportunity to evaluate the performance of EBFRP retrofits in a relatively high-intensity earthquake. This study summarizes the following findings of this field investigation: (1) the performance of EBFRP-retrofitted structures in the Cook Inlet earthquake and (2) the observations concerning the condition of FRP retrofits from over a decade of exposure in a subarctic environment. A deployment team from the National Institute of Standards and Technology (NIST) in collaboration with the University of Delaware (UD) Center for Composite Materials conducted post-earthquake inspections of EBFRP retrofits in multiple buildings to assess their performance during the earthquake and condition with respect to weathering. EBFRP debonding was documented with infrared *thermography* and *acoustic* sounding

	Document title	Authors	Source	Year	Citations

and the bond quality between EBFRP and concrete was assessed using pull-off tests. Visual inspections showed no major signs of earthquake damage in the EBFRP-retrofitted components. However, evaluation of debonding and pull-off test results suggested that *outdoor* conditions may have led to bond deterioration between EBFRP and concrete from installation defects that grew over time, freeze–thaw expansion from moisture present at the FRP/concrete interface, differences in *thermal* expansion of the materials, or a combination thereof. The carbon fiber–reinforced polymer (CFRP) bond to concrete was found to be more vulnerable to *outdoor* exposure than the glass fiber–reinforced polymer (GFRP) bond. Earthquake effects on FRP/concrete bond could not be assessed due to the lack of baseline data.

Article

321 **Improving local pedestrian-level wind environment based on probabilistic assessment using Gaussian process regression** — Wu Y.,Zhan Q.,Quan S.J. — Building and Environment, 205, 108172 — 2021 — 27

Hide abstract Find It @ UCL View at Publisher Related documents

Many wind comfort assessment standards have been proposed to evaluate wind comfort and safety in spaces with intended use for pedestrians, such as pathways, building entrance areas, amenity spaces, and *outdoor* sitting spaces. However, the optimization of the wind environment presents a unique combination of complexities in assessment, simulation, and design. Conventionally, designers rely heavily on wind tunnel experiments or computational fluid dynamics simulations based on which only limited proposals are examined to select the one with the best wind performances. A new design framework for proactively improving the wind conditions at various measuring positions is demonstrated in this paper. Gaussian process regression models are combined with the Latin hypercube sampling method to approximate the relationships between design variables and wind velocities at discrete locations. Gaussian process regression is a non-parametric model well-suited for modeling the non-linear and stochastic wind behaviors. Moreover, it can provide the prediction mean and variance, and the latter is incorporated into a prediction quality function to quantify the modeling uncertainty. In the case study, with the objectives of maximizing the winter and summer wind comfort and minimizing the modeling errors, the robust and near-optimal designs of a target building in an infill development project are explored using the evolutionary search algorithm. We believe that this framework can be migrated to other urban design practice to better predict and simultaneously optimize the microclimate performances at locations of interest, such as solar access, indoor ventilation efficiency, and *thermal* comfort.

Article

322 **Thermal and acoustic performance of cement fibreboard and bamboo buildings** — Singh S.,Chourasia A., Bisht R.S.,Yadav S. — Indian Journal of Engineering and Materials Sciences, 28(5) — 2021 — 4

Hide abstract Find It @ UCL View at Publisher Related documents

The present study evaluates the *thermal* and *acoustic* performance of building constructed from aerated concrete-cement fibreboards and bamboo composite. The *thermal*, humidity, ambient, and indoor temperature parameters are analyzed for a 24-hours cycle from April to May and December to January. The average temperature plot shows the maximum indoor temperature does not exceed 28-32°C for April to May and 12-16°C for December to January for both the buildings. *Thermal* admittance is observed as 7.3 W/m^{2}K and 12 W/m^{2}K for cement fibreboard and bamboo composite buildings respectively. *Outdoor* to indoor *noise* reduction (OINR) values at frequencies ranging from 80 Hz to 4000 Hz are noted and analysed. The maximum *noise* isolation provided by cement fibreboard and bamboo composite walls are found to be ~40 dBA and 35 dBA at 500 Hz. With proper ventilation and avoidance of direct sunlight in the room, a comfortable atmosphere can be achieved for these buildings.

Article

	Document title	Authors	Source	Year	Citations
323	**ROSI: A Robotic System for Harsh Outdoor Industrial Inspection - System Design and Applications**	Rocha F.,Garcia G., Pereira R.F.S.,... Costa R.R., Lizarralde F.	Journal of Intelligent and Robotic Systems Theory and Applications , 103(2), 30	2021	40

Hide abstract Find It @ UCL View at Publisher Related documents

Belt Conveyors are essential for transporting dry bulk material in different industries. Such structures require permanent inspections, traditionally executed by human operators based on cognition. To improve working conditions and process standardization, we propose a novel procedure to inspect conveyor structures with a ground robot composed by a mobile platform, a robotic arm, and a sensor-set. Based on field experience, we introduce ROSI, a new robotic device designed for long-term operations in harsh *outdoor* environments. The mobile robot has a hybrid locomotion system, using wheels to reduce energy consumption while covering long distances, and also flippers with tracks to improve mobility during obstacle negotiation. A mechanical passive switch allows decoupling tracks' traction, reducing components wear and energy consumption without raising mechanical complexity. Aiming the robot-assisted operation, control strategies help to (i) command both the mobile platform and a robotic manipulator considering the system whole-body model, (ii) adjust the contact force for touching the conveyor structure during vibration inspection, and (iii) climb stairs while automatically adjusting the flippers. Machine Learning algorithms detect conveyors' dirt build-ups, roller failures, and bearing faults by processing visual, *thermal* and *sound* data as inspection functionalities. The algorithms training and validation use a dataset collected from running conveyors at Vale, presenting detection accuracy superior to 90%. Field test results in a mining site demonstrate the robot capabilities to stand for the harsh operating conditions while executing all the required inspection tasks, stating ROSI as a disruptive solution for Belt Conveyor inspections and other general industrial operations.

Article

324	**Heterogeneous photooxidation of 6:2 polyfluoroalkyl phosphoric acid diester on dust mineral components under simulated sunlight and the influence of relative humidity and oxygen**	Yao Y.,Meng Y.,Sun H.	Chemosphere, 281, 130713	2021	15

Hide abstract Find It @ UCL View at Publisher Related documents

Polyfluoroalkyl phosphoric acid diesters (diPAPs) have been widely and increasingly detected in various environmental mediums. The degradation of diPAPs brings perfluoroalkyl carboxylic acids (PFCAs) concerned of adverse health effects. DiPAPs mainly occur in particulate matter in ambient air and their photo-degradation behaviors have not been investigated. In this study, heterogeneous photo-degradation of 6:2 diPAP was studied on four model mineral components in ambient dust. 6:2 diPAP was found to undergo a fast degradation on titanium dioxide (TiO_2) particles as well as on artificial mineral dust containing TiO_2 (2.67% in mass) to produce C5–C7 PFCAs and other intermediates. Based on monitored intermediates and further degradation tests on important intermediates, *thermodynamic* calculation of energy barrier and Gibbs Free Energy was used to explain the observed degradation patterns and accordingly the degradation pathways of diPAPs were proposed. The increase in relative humidity promotes the production of hydroxyl radicals, which enhances the hydrolysis of 6:2 mono- and di-PAPs and the yield of C5 and C7 PFCAs. Oxygen is critical for radical formation, which bypasses the production of fluorotelomer carboxylic acid. Results of this study for the first time demonstrate that diPAP may account for additional PFCA sources in both indoor and *outdoor* environments and the heterogeneous degradation pathways were different from those of volatile fluorotelomer alcohols.

Article

	Document title	Authors	Source	Year	Citations
325	**Heat recovery ventilation system combined with boiler room preheating to improve the indoor environment of rabbit house in severe cold areas | 热回收通风系统结合锅炉房预热改善严寒地区兔舍室内环境**	Liu P.,Guo Y.,Ni J.,... Tian J., Wu Z.	Nongye Gongcheng Xuebao Transactions of the Chinese Society of Agricultural Engineering , 37(18)	2021	0

Hide abstract Find It @ UCL View at Publisher Related documents

Heat recovery ventilation (HRV) is type of energy saving ventilation. It has been widely used in animal house to alleviate contradiction between *thermal* comfort and ventilation in recent years. However, air heat exchanger freeze of HRV caused by severe cold weather limited HRV application in northeast China. It is necessary to find an economic way to ensure HRV being operated without freezing. Establish a more reasonable integration of heating and ventilation system in severely cold area. An experiment was conducted in rabbit house during winter in Arun Banner, northeast China. The rabbit house was heated by a coal boiler in the next room. Hot water produced by boiler was supplied to heating radiator inside the rabbit house. The boiler room was warm because of some extra heat release by boiler. To prevent freezing, the boiler room was used to preheat cold fresh air before it entered the heat exchanger. The HRV was installed on the ground of the rabbit house. It was an integration of HRV and boiler heating system. Temperature, relative humidity and concentration of noxious was measured with automatic recording sensor instrument. Data was collected to evaluate the decrease in air pollutants and air temperature of the rabbit house. Preheating effect of the boiler room and the HRV were tested when the system was in operation. The performance of the HRV and prevention of freezing in heat exchanger was evaluated in typical cold weather in northeast China. The economic benefit of the system was analyzed. The results showed that when the HRV and boiler system began to operate, concentrations of ammonia (NH_3) and carbon dioxide (CO_2) in rabbit house dropped by 46% and 64%, respectively. Moreover, indoor air temperature only dropped by an average of 1.8 ℃ when the *outdoor* air temperature varied from -15.8 to -11.8 ℃. It means that the system can significantly improve indoor air quality with barely a slight decline of indoor air temperature. Respiratory and skin disease decreased by 16% and 4%, respectively. It is great potential for better health and production of rabbits. The boiler room heated fresh air by 8.1 ℃, and the HRV heated fresh air by 6.4 ℃ on average. No frozen of the heat exchanger was observed. Heated load of the boiler room and the heat recovery load of the HRV were 6.8 and 5.6 kW, respectively. Sensible heat recovery efficiency of the HRV was 75.6%. Coefficient of performance of the HRV was 6.2. The system was operated efficiently. The method of freezing prevention ensure HRV in good condition among the whole cold season in northeast China. It is an economic option because the system can operated without extra energy consumption. The return of investment is 3 years in this work condition. The integration of HRV and boiler allowed HRV and boiler heating to operate with high efficiency and solve the contradiction between heating and ventilation in severely cold weather in the rabbit house. This finding can provide a *sound* reference for the design of animal house heating and ventilation system in severely cold climate northeast China.

	Book Chapter				
326	**Advancements in thermal comfort modeling using modern sensing and computational technologies**	Choi J.-H.	Research Methods in Building Science and Technology	2021	3

Hide abstract Find It @ UCL View at Publisher Related documents

Despite numerous technical efforts to enhance *thermal* comfort in multiple building typologies, *thermal* stress has been reported as the most serious problem among building users for several decades. Many advanced *thermal* comfort prediction models have been investigated and applied, but there are still significant limitations to meet the individual occupants' *thermal* satisfaction while minimizing their *thermal* stress. One main reason is the limited technical parameters that are considered in *thermal* comfort prediction models. In addition, while many indoor and *outdoor* environmental attributes dynamically change almost every second, the current *thermal* comfort prediction models often assume many of these attributes to be constant over time. Human bio-signals, such as skin temperature, heart rate, heart rate variability, and electro-dermal activity, have been investigated as critical components affected by the ambient *thermal* condition while dynamically changing depending on the user's environmental perception. Considering that *thermal* stress has frequently been reported as the most significant stressor compared to visual, air, and *acoustic* environmental stimuli, these physiological signals have been reported

	Document title	Authors	Source	Year	Citations

as reliable indicators for effective *thermal* comfort/stress estimation in recent advanced studies. The present chapter discusses the current technical limitations in *thermal* comfort prediction and proposed methods to overcome these limitations with the aid of human physiological responses as a function of the user's bio-signals. Also, the principles of integrative approach with human factors are introduced as a futuristic solution in *thermal* comfort/stress estimation, which may help to completely replace the current equation-based *thermal* comfort models with data-driven approaches. Additional technical solutions are also discussed as potential research agenda to investigate in the domain of *thermal* comfort research.

Article • *Open access*

327 **Experimental and computational model for a neonatal incubator with thermoelectric conditioning system** — Casado A.R., Larrode-Diaz M., Zacarias F.F., Molina R.H. — Energies, 14(17), 5278 — 2021 — 5

Hide abstract Find It @ UCL View at Publisher Related documents

This work describes the design, construction and testing of a *thermo*-electric conditioning device installed in a neonatal incubator with the aim of improving the precision in the regulation of the interior air temperature, reducing *noise* and interior vibration, and improving the life of the neonate. A simplified one-dimensional *thermal* model has been developed, made up of resistances and *thermal* capacities that simulate the *thermal* behaviour of all the elements of the system from end to end. All the equations of the model are obtained in a nodal way, allowing the mathematical relationship between the input and output to be known. This model makes it possible to improve temperature control, avoiding the deviations that occur in the traditional model controlled by sensors at both ends. The computational model allows to predict the variation of temperatures in transient and permanent regime. This model allows the design and sizing of the *thermoelectric* system for different *outdoor* environmental conditions and the selection of the number of Peltier modules needed to satisfy the heating demand of other incubators with different geometry and capacity. The results of the computational model show good agreement with the experimental tests, despite being a simplified 1D nodal model. The results obtained show a coefficient of operation (COP) of 1.38, achieving higher performance than the current traditional electrical resistance system (COP = 1). In addition, a CFD study has been carried out to check the air patterns, to see the temperature uniformity and to estimate the number of air changes per hour (HVAC) inside the incubator.

Review • *Open access*

328 **A review of cultural background and thermal perceptions in urban environments** — Naheed S., Shooshtarian S. — Sustainability Switzerland, 13(16), 9080 — 2021 — 19

Hide abstract Find It @ UCL View at Publisher Related documents

Thermal comfort is among the chief indicators of the sustainability of *outdoor* spaces. However, the complex nature of comfort represents the interaction of several determinants that leads to a perception of the *thermal* environment. Recently, researchers have paid particular attention to non-physical factors to understand the mechanisms involved in *thermal* perceptions in urban environments. The extant literature has contended that culture and cultural background are determinants to individuals' *thermal* perceptions. Therefore, this study aimed to review how the link between *outdoor thermal* comfort (OTC) and cultural background is investigated. This is, to the best of our knowledge, the first review study on the subject. The study used a systematic literature review approach based on secondary data available in relevant and contemporary literature. The findings first showed the scarcity of research on cultural background and OTC; however, all studies identified corroborated the significant impact of cultural background on *thermal* perceptions. Notably, the cultural background was found to be the source of variation in *thermal* perceptions, tolerance to, and preference for certain *thermal* conditions, *thermal* comfort requirements and expectations, choice of clothing, and environmental attitudes. The findings provide a *sound* basis for future researchers to address the research gaps identified. The study also raises policy makers' and designers' awareness of urban environment users' genuine needs and requirements.

Article

	Document title	Authors	Source	Year	Citations
329	**Enhancing the performance of heat pumps by immersing the external unit in underground water storage tanks**	Al-Nimr M.A., Al-Waked R.F., Al-Zu'bi O.I.	Journal of Building Engineering, 40, 102732	2021	6

Hide abstract Find It @ UCL View at Publisher Related documents

Renewable energy sources have a considerable potential to contribute to sustainable development and have the potential to reduce the reliance on fossil fuels. Direct-Expansion Water Source Heat Pump (DX-WSHP) could be used in air conditioning applications as an energy-efficient environmental-friendly system. In this study, DX-WSHP potential for air conditioning of a multi-level residential building was numerically examined. A closed-loop of copper pipes filled with R134a, as a refrigerant, was immersed in an underground water tank that is used as a heat sink/source. Cooling/Heating loads of the residential unit under three different climate conditions were estimated at different floor levels. The proposed system takes advantage of a higher heat transfer rate in water and better *thermal* performance when compared with traditional Air Source Heat Pumps (ASHP). A mathematical model was built to examine the performance of the system using the Engineering Equations Solver (EES) software. Results have shown that the DX-WSHP has outperformed the ASHP at a lower (higher) condensing (evaporating) temperature in the cooling (heating) mode with higher values of COP varies from 30% to 118%. Furthermore, there was a significant reduction in economic and environmental impacts, CO_2 emissions, and electrical energy consumption of the DX-WSHP system for the three different climate conditions. The reduction in CO_2 emissions varied from 8% to 40%. Additional advantages of the proposed system were found: energy-saving and overcoming the problems concerning heat rejection to the surrounding air and the *outdoor* fan *noise*.

Article • *Open access*

	Document title	Authors	Source	Year	Citations
330	**Light-induced off-axis cavity-enhanced thermoelastic spectroscopy in the near-infrared for trace gas sensing**	Zheng K., Zheng C., Lien H.U., ... Wang Y., Tittel F.K.	Optics Express, 29(15)	2021	21

Hide abstract Find It @ UCL View at Publisher Related documents

A trace gas sensing technique of light-induced off-axis cavity-enhanced *thermoelastic* spectroscopy (OA-CETES) in the near-infrared was demonstrated by combing a high-finesse off-axis integrated cavity and a high Q-factor resonant quartz tuning fork (QTF). Sensor parameters of the cavity and QTF were optimized numerically and experimentally. As a proof-of-principle, we employed the OA-CETES for water vapor (H_2O) detection using a QTF (Q-factor ~12000 in atmospheric pressure) and a 10cm-long Fabry-Perot cavity (finesse ~ 482). By probing a H_2O line at 7306.75 cm^{-1}, the developed OA-CETES sensor achieved a minimum detection limit (MDL) of 8.7 parts per million (ppm) for a 300 ms integration time and a normalized *noise* equivalent absorption (NNEA) coefficient of $4.12 \times 10^{-9} cm^{-1} W Hz^{-1/2}$. Continuous monitoring of indoor and *outdoor* atmospheric H_2O concentration levels was performed for verifying the sensing applicability. The realization of the proposed OA-CETES technique with compact QTF and long effective path cavity allows a class of optical sensors with low cost, high sensitivity and potential for long-distance and multi-point sensing.

Conference Review

	Document title	Authors	Source	Year	Citations
331	**ISSCS 2021 - International Symposium on Signals, Circuits and Systems**	[No Authors Found]	Isscs 2021 International Symposium on Signals Circuits and Systems	2021	0

	Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher

The proceedings contain 73 papers. The topics discussed include: a CMOS charge-domain global shutter pixel characterization system; indirect characterization of indoor RF EMF exposure to nearby mobile phone base stations; a study on human reaction time in a mixed reality environment; application design principles for road usersâ€™ safety; UAV embedded system â€“ a selection process; *acoustic* monitoring of *outdoor* areas by a sensor consisting of four microphones; on change detection in the state of the human faces by *thermal* images and Renyi information measures; hateful meme detection with multimodal deep neural networks; presentation of a development kit for IoT applications and a potential solution for communications security; and deep autoencoder-based detection of electricity stealth cyberattacks in AMI networks.

Conference Paper

☐ 332 **Using thermal vision for extended VINS-mono to localize vehicles in large-scale outdoor road environments** — He M., Rajkumar R.R. — IEEE Intelligent Vehicles Symposium Proceedings, 2021-July — 2021 — 5

Hide abstract Find It @ UCL View at Publisher Related documents

A monocular VIO (Visual-Inertial Odometry) system provides a compact, low-cost, and easily-deployed configuration for relative localization. However, using *thermal* vision for VIO is much less studied than using a visible-spectrum camera. A *thermal*-vision camera works under all lighting conditions and has been used to detect pedestrians, cars and animals at nighttime to provide ADAS (Advanced Driver Assistance System) functions. Common problems in directly using *thermal* images in conventional VIO methods are: (1) lower signal-to-*noise* ratio and fewer reliable feature points in the texture-less *thermal* images, (2) periodic recalibration hampers *thermal* image capturing and feature-point tracking, and (3) the 'jello' effect of the rolling shutter readout architecture is sensitive to aggressive vehicle maneuvers. Extended VINS-Mono, proposed in our previous work [1], aims at providing relative and absolute localization of a vehicle in large-scale *outdoor* road environments by introducing (1) absolute localization methods to enable VINS-Mono to output local and global state estimates simultaneously, (2) vehicle speed readings for fast (re-)initialization and reliable scale estimates, and (3) DNN-based object detection methods to remove nonstationary objects from the visible scene. In this paper, we show that Extended VINS-Mono can use *thermal* images to provide relative and absolute localization even when light conditions are very poor. We conducted several experiments on a 25 Km-trip journey through highways, tunnels, urban areas and suburban areas in Pittsburgh, USA during daytime and nighttime to evaluate the performance of Extended *Thermal* VINS-Mono, including (re-)initialization, accuracy, rate, and latency. Our evaluation confirms that using *thermal* vision for localization satisfies localization requirements in large-scale *outdoor* road environments when the visible-spectrum camera performs poorly.

Article

☐ 333 **Extracting boundary conditions for sound propagation calculations using augmented reality | Ermittlung von Randbedingungen für Schallausbreitungsberechnungen mittels Augmented Reality** — Sporr A., Blank-Landeshammer B., Kasess C.H., ... Waubke H., Reichl C. — Elektrotechnik Und Informationstechnik, 138(3) — 2021 — 2

Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

The number of air-to-water heat pumps has increased over the last years, resulting in a rising number of installed *outdoor* units. These objects constantly emit *noise* into the environment during the heat pumps' runtime. To avoid conflicts with neighbors or even a relocation of the already placed heat pump, an app is developed to estimate the emitted *sound* of the virtual heat pump on site. This publication presents adapted *sound* calculation methods suitable for *outdoor* use and on mobile devices. Based on this, boundary conditions are specified that must be met by the *sound* data of the respective *sound* source and the objects recorded on site using augmented reality in real time. Both the determination of the *sound* data and the identification of objects using AR are explained in detail. Furthermore, different virtual objects that alter the *sound* propagation can be placed in the real environment and configured with *sound*-relevant properties. This allows the estimation of the usefulness of *sound*-reducing measures already before realization. An example of the identification of individual objects, the placement of virtual walls and heat pumps, and the final visualization of the *sound* propagation in different ways on the mobile device is presented.

Article • *Open access*

☐ 334 **Outdoor thermal comfort: Analyzing the impact of urban configurations on the thermal performance of street canyons in the humid subtropical climate of Sydney** — Abdollahzadeh N., Biloria N. — Frontiers of Architectural Research, 10(2) — 2021 — 139

Hide abstract Find It @ UCL View at Publisher Related documents

The quality of *outdoor* space is becoming increasingly important with the growing rate of urbanization. Visual, *acoustic*, and *thermal* balance degradation are all negative impacts associated with *outdoor* comfort in dense urban fabrics. Urban morphology thus needs assessment and optimization to ensure favorable *outdoor thermal* comfort (OTC). This study aims to evaluate the *thermal* performance of streets in residential zones of Liverpool, NSW, Australia, and tries to improve their comfort index (Physiological Equivalent Temperature) to reveal optimum urban configurations. This evaluation is done by investigating the following urban design factors affecting OTC using computational simulation techniques: street orientation, aspect ratio, building typology, and surface coverage. Our findings reveal that street canyon orientation is the most influential factor (46.42%), followed by aspect ratio (30.59%). Among the influential meteorological parameters (air temperature, wind speed, humidity and solar radiation), wind velocity had the most significant impact on the *thermal* comfort of the *outdoor* spaces in this coastal region, which typically experiences intense airflow. The results of our analysis can be utilized by multiple stakeholders, allowing them to understand and extract the most vital design factors which contextually influence the *thermal* comfort of *outdoor* spaces. *Outdoor thermal* comfort has a direct effect on the health and wellbeing of occupants of *outdoor* spaces.

Article

☐ 335 **Combining suspended radiant ceiling with diffuse ventilation – Numerical performance analysis of low-energy office space in a temperate climate** — Krusaa M.R.,Hviid C.A. — Journal of Building Engineering, 38, 102161 — 2021 — 18

Hide abstract Find It @ UCL View at Publisher Related documents

The development of low energy buildings present new opportunities to develop HVAC products that works well with renewable energy sources. In this paper, we present a novel comfort system that combines a full radiant ceiling with diffuse ventilation into one flexible ceiling solution, HVACeiling. The purpose of the paper is to investigate how the combination changes heating and cooling performance and how the *thermal* storage cycle of the exposed slab in the closed plenum affects energy consumption and *thermal* comfort. The heating and cooling performance was investigated numerically in IDA ICE with daily based occurrence temperatures and compared to a reference with a radiant ceiling and mixing ventilation. The HVACeiling was a perforated gypsum board with pipes embedded that is used for diffuse ventilation and *acoustical* absorption as well as a radiant ceiling. Water was circulated day and night at fixed supply temperatures of 20–22 °C without any other controls. The convective surface heat transfer coefficients of the plenum was modelled in a preliminary CFD study and compared to literature and experiments. It was found that for daily based occurrence scenarios, with *outdoor* temperature in the range 23–28 °C, the heat exchange from the ceiling was increased 5–16% compared to the reference. The studies also disclosed improved *thermal* comfort, up

	Document title	Authors	Source	Year	Citations

to 1.5 °C lower peak operative temperature in the occupied zone down from 27.2 °C. This implies that *thermal* comfort is easier achievable with the HVACeiling, even with a small temperature offset that is well suited for renewable sources.

Article

336 **The role of social vs. physical environmental nuisances in affecting stress among Jewish and Muslim women in Israel** — Saadi D.,Tirosh E., Schnell I. — Science of the Total Environment, 768, 144474 — 2021 — 3

Hide abstract Find It @ UCL View at Publisher Related documents

We calculate the effects of selected social and physical environmental nuisances on the autonomic nervous system balance among Jewish and Muslim mothers. Seventy-two Jewish and Muslim young healthy mothers from an Arab and Jewish neighboring city were tested in seven urban environments. Three social mediators (social discomfort, participation in household's decision-making and freedom of movement) and three physical mediators (*thermal* load, carbon monoxide and *noise*) were considered. Mean differences between Jewish and Muslim heart rate variability (HRV) and the effects of the mediating variables were calculated by ANOVA and stepwise multiple regressions. The results suggest that HRV was predominantly affected by social discomfort. Jewish and Muslim mothers experienced similar levels of social discomfort albeit; their responses differed by type of environment. Jewish mothers experienced stronger social discomfort in *outdoor* environments while Muslim mothers felt stronger social discomfort at home. Crossing ethnic boundaries was associated with increase in stress in both groups. Social discomfort played a stronger role in predicting stress in inter-ethnic environments. In conclusion, the study shows that social discomfort is a predominant environmental factor in predicting stress and related risk to health as reflected in the ANS balance across ethnicities. In addition ethnic specific factors like women freedom of movement played a minor role in effecting stress. At the same time the physical environmental aspects played essential role in effecting stress.

Article • *Open access*

337 **Long-term outdoor observation of geomagnetic fields using tunneling magnetoresistance-based magnetometer and detection of long-period oscillations of geomagnetic pulsations: Prospects for earthquake disaster prevention** — Isogami S.,Oishi Y., Furukawa C.,Ohnishi N., Okubo K. — Applied Physics Express, 14(5), abf66a — 2021 — 1

Hide abstract Find It @ UCL View at Publisher Related documents

We investigated the appropriateness of a tunneling magnetoresistance (TMR)-based magnetometer for detecting geomagnetic fields with longperiod oscillations through observation of the geomagnetic pulsation (Pi). Pi was not sufficiently observed in the raw outputs due to *thermal noise*; a bandwidth-limitation treatment helped extract Pi without data stacking. The detectable intensity was reduced to 1 nT with a period of 30 s, but detection was impossible for 100 s. The frequency characteristics revealed that reducing the cutoff frequency of a TMR magnetometer is a promising approach for detecting field oscillations occurring over long periods, such as 100 s. ? 2021 The Japan Society of Applied Physics.

Conference Review

338 **20th Sustainable, Environment and Architecture** — [No Authors Found] — Iop Conference Series Earth and Environmental Science, 738(1) — 2021 — 0

Hide abstract Find It @ UCL View at Publisher

The proceedings contain 80 papers. The topics discussed include: *thermal* environment control of buildings using installation of plants and metal panels on brick walls; *thermal* comfort of the *outdoor* transition space in the dean's office building; *thermal* comfort analysis of residential home in coastal city based on physiological equivalent temperature (PET) index and operative temperature zone; *thermal* and visual comfort analysis using CBE tools and Velux simulation of classroom in R.21 in Gedung Kuliah Bersama V University of Bengkulu; design of sustainability of

	Document title	Authors	Source	Year	Citations

solar panel integration in a green building complex of Wonogiri regent office; the influence of the *sound* absorber application to the *acoustics* conditions and the *thermal* transfer value of the building envelope in open plan office; optimization of *thermal* comfort in Rifain digital printing: vertical garden with water wall system concept; energy efficiency in church building based on Sefaira energy use intensity standard; and estimation of energy consumption efficiency in office rooms cooling systems to create *thermal* comfort for the user.

Article

339 **Machine learning-based prediction of internal checks in weathered thermally modified timber** — van Blokland J.,Nasir V., Cool J.,Avramidis S., Adamopoulos S. — Construction and Building Materials , 281, 122193 — 2021 — 20

Hide abstract Find It @ UCL View at Publisher Related documents

This study investigated possibilities to predict the presence of internal checks in *thermally* modified Norway spruce timber after 2.5 years of weathering based on the initial properties of the boards. Machine-learning classification enabled sorting the input parameters based on their relative importance for accurate predictions. The parameters of *thermally* modified timber with the highest relative importance were annual ring width followed by initial moisture content, density and dynamic stiffness. Whereas after kiln drying these were, density, annual ring width, initial moisture content and *acoustic* velocity. The results showed that predictions are possible, and an accuracy of 67% was achieved by using annual ring width combined with density and initial moisture content, or *acoustic* velocity that can be determined after either kiln drying or *thermal* treatment.

Article • *Open access*

340 **Noise Reduction for Improvement of Ultrasonic Monitoring Using Coda Wave Interferometry on a Real Bridge** — Wang X.,Chakraborty J., Niederleithinger E. — Journal of Nondestructive Evaluation , 40(1), 14 — 2021 — 28

Hide abstract Find It @ UCL View at Publisher Related documents

Reinforced concrete bridges are iconic parts of modern infrastructure. They are designed for a minimum service life of 100 years. However, environmental factors and/or inappropriate use might cause overload and accelerate the deterioration of bridges. In extreme cases, bridges could collapse when necessary maintenance lacks. Thus, the permanent monitoring for structure health assessment has been proposed, which is the aim of structural health monitoring (SHM). Studies in laboratories have shown that ultrasonic (US) coda wave interferometry (CWI) using diffuse waves has high sensitivity and reliability to detect subtle changes in concrete structures. The creation of micro-cracks might be recognized at an early stage. Moreover, large-volume structures can be monitored with a relatively small number of US transducers. However, it is still a challenge to implement the CWI method in real SHM practical applications in an *outdoor* environment because of the complex external factors, such as various *noise* sources that interfere with the recorded signals. In this paper, monitoring data from a 36-m long bridge girder in Gliwice, Poland, instrumented with embedded US transducers, *thermistors*, and vibrating wire strain gauges, is presented. *Noise* estimation and reduction methods are discussed, and the influence of traffic, as well as temperature variation, are studied. As a result, the relative velocity variation of US waves following the temperature change with a very high precision of 10^{-4}% is shown, and a good bridge health condition is inferred. The influence of lightweight real traffic is negligible. The study verified the feasibility of the implementation of the CWI method on real bridge structures.

Article

341 **Cooling operation analysis of multienergy systems in a nearly zero energy building** — Li H.,Zhang S.,Yu Z.,Wu J., Li B. — Energy and Buildings , 234, 110683 — 2021 — 9

Hide abstract Find It @ UCL View at Publisher Related documents

A 4025 m^2 office building was built in Beijing, China, in 2014 as a living laboratory and prototype to explore the technical energy systems of nearly zero energy buildings (NZEBs) in China. Not only the building itself but also the heating, ventilation and air conditioning (HVAC) systems were designed with flexibility to explore their performance. Solar-assisted air conditioning (SAAC) and ground source heat pump (GSHP) systems with different terminal units were designed as HVAC systems for either experimental, comparison, or measurement purposes. In this paper, the

Document title	Authors	Source	Year	Citations

cooling operation of multienergy systems is discussed on the basis of actual measurement data. HVAC systems have three possible types of operation configuration types, namely, SAAC priority, GSHP only and multimode. Operation scenarios for these configurations are presented first, and the small-scale simulation was carried out to analyze the performance of SAAC priority and GSHP#1 only mode. It indicates that SAAC could save about 47% of energy than that of GSHP#1. Then, the actual operation performance in 2018 was analyzed by reference to the operation data. Through this study, it is found that 1) multienergy systems of SAAC and GSHP operated well in this NZEBs during the summer season; 2) a 10 RT absorption chiller can work well for an NZEB building of approximately 1000 m^2 when *outdoor* enthalpy is low and the solar radiation is stronger than 400 W/m^2 as a result of high building *thermal* performance and air tightness; 3) SAAC can save about 40 to 50% of energy compared to that of GSHP system when providing services for the same terminal system under the same *outdoor* weather conditions and 4) The GSHP system is a good complement to the SAAC system which works well in June, September and early July when *outdoor* enthalpy is low. This study indicates that a combination of GSHP and SAAC systems is highly promising for NZEBs in cold climate zones such as Beijing, where the solar radiation is *sound*.

Article

☐ 342 **Thermal performance experiment of heat storage wall with flat micro heat pipe array in solar greenhouse | 日光温室平板微热管阵列蓄热墙体热性能试验**

Guan Y.,Wang T.,Wei M., ... Hu W.,Duan S.

Nongye Gongcheng Xuebao Transactions of the Chinese Society of Agricultural Engineering , 37(3)

2021

6

Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

Composite hetero-structure wall is widely used for the north wall of solar greenhouse in China. A *thermal* stable thick layer in this type of wall can greatly weaken the heat storage capacity of the greenhouse wall. This study aims to improve the temperature of the *thermal* stable layer, and thereby promoting the heat storage capacity of north walls in a solar greenhouse. A new type of wall was proposed for the solar greenhouse with a flat Micro Heat Pipe Array (MHPA) wall or heat storage wall. The wall was made by bending the flat micro heat pipe array into a zigzag-shape while mixing the cement mortar to form an MHPA block. Then, the MHPA block and ordinary blocks were built into the MHPA wall. The evaporation section of the flat micro heat pipe array was placed on the inner surface of the MHPA wall, and the condensation section was placed inside the MHPA wall to achieve the purpose of enhancement heat transfer. Two small-scale greenhouse comparison test-benches were set up on site with the MHPA wall (named as MHPA greenhouse) and the ordinary block wall (named as ordinary greenhouse). The evaluation parameters included the temperature of the greenhouse wall, the daily heat storage/release of the wall, and the greenhouse environmental temperature. A systematic investigation was made to compare the characteristics of heat storage/release of the MHPA wall, and the effect of improving the greenhouse *thermal* environment on representative days. The results showed that the average internal surface temperature of the MHPA wall on typical sunny days (from 17: 00 to 9: 00 the next day) was 1.6-2.3℃ higher than that of an ordinary wall. From indoor to *outdoor*, the average temperature inside the MHPA wall for the 0-370 mm area along the wall thickness direction increased by 2.7-4.0℃. During the heat storage period (from 9: 00 to 17: 00), the average environmental temperature of the MHPA greenhouse was 0.2-0.5℃ lower than that of an ordinary greenhouse, and the average ground surface temperature was 1.8-1.9℃ lower. During the heat release period, the average environmental temperature of the MHPA greenhouse was 1.2-1.5℃ higher than that of an ordinary greenhouse, and the average ground surface temperature was 0.6-1.0℃ higher. The MHPA wall can play the role of "peak cutting and valley filling" in the *thermal* environment of a greenhouse. The total daily heat storage of the MHPA wall was 8.92%-14.35% higher than that of an ordinary wall, and the total daily heat release was 2.24%-8.07% higher. In the heat storage and release rates of a wall on December 27, the maximum and average heat storage rates of the MHPA wall were 9.63% and 10.99% higher than those of the ordinary wall, and the maximum and average heat release rates were 9.23% and 5.60% higher than those of the ordinary wall, respectively. Moreover, the heat release rate of the MHPA wall was higher than that of an ordinary wall between 23: 00 and 7: 00 the next day, with an average increase of 11.53%. Therefore, the introduction of the MHPA wall into the wall of a solar greenhouse can effectively improve the heat storage and release capacity of the wall materials for a better *thermal* environment. These findings can provide a *sound* reference for the application of a flat micro-heat pipe array to the heat storage wall in a solar greenhouse.

Book Chapter

☐ 343 **Influence of internal thermal insulation on the sound insulation of walls** Zeitler B. Energy Efficient Retrofit of Buildings by Interior Insulation Materials Methods and Tools 2021 0

Hide abstract Find It @ UCL View at Publisher Related documents

Adding any sort of lining to a building element, e.g., to a wall or floor, influences the *sound* transmission through that building element. In this chapter, the change of *sound* transmission or *sound* insulation caused by adding an internal *thermal* insulation lining to exterior wall is discussed. First, a short review of the physical phenomena related to the *sound* insulation of a wall with and without internal *thermal* insulation is given. This should help understand the wall's ability to insulate indoor spaces from *outdoor noise*. Following, the measured *sound* insulation, described by the standardized weighted *sound* reduction index R_w, is presented for three types of linings on a masonry wall. Of the three types, the detached stud linings perform best and give the highest improvement when added to a wall (up to 20 dB), linings attached to the wall with furring strips show a lesser improvement (up to 4 dB), and the lowest *acoustically* performing linings are the panelized composite linings that even result in a worsening of *sound* insulation when they are attached over the entire surface. The role, the spectra of the *outdoor noise* sources, e.g., traffic *noise*, can play on the indoor levels is also touched upon. The *sound* insulation of the exterior wall with lining is put into perspective with the other elements of the exterior wall, such as windows and ventilators. The internal *thermal* linings do not only influence the *sound* transmission from outdoors to indoors but also the *sound* transmission from indoors

	Document title	Authors	Source	Year	Citations

to indoors. An example suggests how and why this should also be taken into consideration when selecting appropriate interior *thermal* insulation.

Conference Paper

344 **Propagation loss of low frequency horn loudspeakers: Is "throw" a real phenomenon?** Hipperson J. | 150th Audio Engineering Society Convention AES 2021 | 2021 | 0

Hide abstract Find It @ UCL View at Publisher Related documents

Horn loading is frequently used in *sound* reinforcement to increase efficiency and directivity of high and mid frequency transducers. Low frequency horn loudspeakers are less common due to their large size. Increases in available amplifier power and *thermal* dissipation in transducers have led to widespread use of high power, low efficiency dual 18" bass reflex loudspeakers. However, some manufacturers and enthusiasts continue to develop and use low frequency horn loudspeakers for their high efficiency and subjective audio quality. In the fields of live event production and *noise* control, there is sometimes a perception, or "urban myth" that horn low frequency loudspeakers project or "throw" *sound* a further distance than direct radiating low frequency loudspeakers. This is either considered to be beneficial, or problematic depending on the context. Considering the relevant *acoustic* theory, it is not immediately apparent why this should be the case, providing the loudspeakers are level matched and of similar physical dimensions. Unfortunately, there is very little investigation of low frequency horns in previous literature to aid in providing a definitive answer. Measurements in this paper demonstrate that horn and direct radiating low frequency loudspeakers and arrays closely follow the theory, and the difference in propagation loss is within measurement uncertainty. The implication for *noise* control of *outdoor* events, is that bass loudspeaker size and type are not especially relevant factors, and focus should instead be on system/array design and site layout.

Book Chapter

345 **Ventilative Cooling and Air Pollutants** Carrilho da Graca G., Martins N.R. | Polito Springer Series | 2021 | 4

Hide abstract Find It @ UCL View at Publisher Related documents

The majority of office and other non-domestic buildings use mechanical cooling and ventilation, even when an optimized natural ventilation (NV) system could meet cooling and fresh air requirements. However, in most large cities, the *outdoor* environment is contaminated with a combination of *noise*, fine particles, heat and toxic gases. This contaminated environment has a detrimental impact on naturally ventilated buildings due to their lack of filtration and *outdoor noise* attenuation systems. This chapter presents a numerical analysis of the effect of fine particle pollution (PM2.5) on the NV potential of office buildings in California, Europe and Asia. Several years of measured weather and PM2.5 concentration data were used to perform dynamic *thermal* and airflow simulation analysis. Detailed simulation results show that a hybrid NV system can reduce the air-conditioning and ventilation electricity consumption of a well-designed office building by up to 83% (which can be increased to up to 93% by the availability of personal comfort systems), in comparison to an office using, during all working hours, a mechanical cooling and ventilation system equipped with a high-efficiency particle filter. Unfortunately, in this hybrid approach, high levels of *outdoor* PM2.5 penetrate the indoor environment, increasing occupant cumulative exposure by up to six times. To overcome this problem, two exposure control approaches were tested. Using NV only during moments of low *outdoor* PM2.5 concentrations limits the exposure increase to up to three times but at the cost of reducing energy savings. Equipping NV openings with an electrostatic filter would result in a similar exposure reduction, but at a very low energy cost, taking full advantage of NV's saving potential.

Conference Paper

346 **Research on Heat Dissipation Performance and Long-term Reliability of the Flapping Wing Cooling Technology Applied to the 5G Communications Equipment** Guo Y., Zhang X., Lan D., Zhu Y. | 2021 27th International Workshop on Thermal Investigations of Ics and Systems Therminic 2021 | 2021 | 3

Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

To further improve the heat dissipation efficiency of the 5G communications equipment, this study innovatively applies the flapping wing cooling technology to *outdoor* 5G base stations. A series of experimental and numerical studies have been conducted to explore effects of the flapping wing cooling device on heat dissipation performance and long-term reliability of 5G communications devices. Specifically, (1) Taguchi method was used to determine geometric parameters (length, width and spacing) of the flapping wing cooling device's blades, where airflow and *noise* were objective functions of the optimization method; (2) additionally, Computational Fluid Dynamics simulation and *thermal* tests were conducted to demonstrate advantages of forced convection compared to natural convection. The results indicate that: (1) when blades of the flapping wing cooling device are 15 mm in width, 70 mm in length, 14 mm in spacing, the airflow reaches the maximum of 10.7 CFM, with the *noise* level meeting the permissible *noise* standard of the 5G *outdoor* devices; (2) the flapping wing cooling device reduces the temperature of electronic components of the 5G AAU by about 10 °C on average. The temperature drop of electronic components near the fan outlet is 6 °C higher than that of electronic components 0.5 m away from the fan outlet. In terms of reliability, Finite Element Method simulation was conducted to research the failure probability of the flapping wing cooling device under vibration fatigue; besides, based on a series of environmental durability tests, the reliability of the flapping wing cooling device for long-term usage outdoors was explored. The results show that: (1) the maximal Mises stress of blades is lower than fatigue limit of its own materials, and vibration fatigue life meets the requirement of 10 years; (2) the flapping wing cooling device has passed high temperature aging test, temperature cycling test, high temperature and high humidity aging test, salt spray test, demonstrating the feasibility of long-term *outdoor* operation. In conclusion, the flapping wing cooling device can effectively reduce the size and weight of the 5G communications devices by improving the heat dissipation efficiency, which is a great breakthrough in enhancing the competitiveness of products, and what is more, it is reliable to operate outdoors for a long time.

Article

347 **Investigation of Multifield Characteristics in Braking of Lightweight Discs on a Self-Developed Platform** | Pei Y.-C.,Wang S.,Xia Z.-R.,Zhang F. | Tribology Transactions, 64(2) | 2021 | 1

Hide abstract Find It @ UCL View at Publisher Related documents

Disc brakes are widely used in various kinds of light vehicles. Currently, it has become a tendency to reduce the mass and volume of brake systems because of the increasing lightweight requirement for vehicles, especially for passenger vehicles in populated cities and for *outdoor* sport. However, there is often conflict between light weight and reliability. In this article, the braking characteristics of a thin disc brake were investigated experimentally, using a self-developed inertia braking platform. The platform was designed to conveniently measure multifield characteristics in real time. The goal of the experiment is to verify the special performance of a brake system with a very thin solid disc. Measurements of transient temperature, lateral vibration, and *noise* in the dry friction contacts of a steel solid thin disc and two resin circular pads were performed. A frequency–time analysis method was used for the lateral vibration of the disc and braking *noise*. The results suggest that one of the reasons for lateral vibration is the mounting error of the thin disc–shaft system, which should receive extra attention for safety reasons. In addition, the braking *noise* is structural borne.

Book Chapter

348 **Ventilative Cooling and Control Systems** | Breesch H.,Merema B. | Polito Springer Series | 2021 | 3

Hide abstract Find It @ UCL View at Publisher Related documents

Robust control systems are essential to guarantee an efficient usage of ventilative cooling. The most important challenges of controlling natural ventilative cooling are the *outdoor* environmental conditions like wind, rain, *noise* and pollen; security and interaction with and satisfaction of the user. The main control strategies and components of natural ventilative cooling, including actuators and sensors are described. From case studies, the following lessons are learned. The main driver for control of natural ventilative cooling systems is *thermal* comfort and *outdoor* weather conditions. Optimization and commissioning of ventilative cooling control is critical to maximize the cooling potential as well as to prevent overcooling. Conclusions on the interaction of the user with the ventilation cooling control are twofold. On the one hand, automatic control is preferred to manual control of ventilative cooling. On the other hand, the user should be able to overrule this automatic control. Suggestions for design are formulated. The users and their

	Document title	Authors	Source	Year	Citations

expectations of controllability play a central role. Moreover, the future maintenance of the actuators have to be taken into account in the design phase.

Conference Paper

349 **The NACHOS CubeSat-based hyperspectral imager: Laboratory and field performance characterization**

Post K.W.,Ott L.A., Dale M.E.,... Dubey M.K., Love S.P.

Proceedings of SPIE the International Society for Optical Engineering , 11832, 118320F

2021

1

Hide abstract Find It @ UCL View at Publisher Related documents

Hyperspectral imaging with sufficient resolution and sensitivity for scientifically useful space-based mapping of trace gases has long required large and expensive satellite instruments. Miniaturizing this capability to a CubeSat configuration is a major challenge, but opens up more agile and far less expensive observing strategies. A major step in this direction is our development of NACHOS, an ultra-compact (1.5U instrument, 3U complete CubeSat) hyperspectral imager covering the 300-500nm spectral range in 400 channels. Here we describe laboratory and field performance characterization of this new instrument. Laboratory tests demonstrate spatial and spectral resolutions of <0.8 mrad and 1.3 nm, respectively, with good resolution of the spectral lines of our SO2 and NO2 target gases. *Outdoor* field tests under realistic illumination conditions provide real-world signal-to-*noise* benchmarks, and yield hyperspectral images displaying high quality solar and atmospheric spectra. To estimate on-orbit gas retrieval sensitivities, we computationally implanted plumes of varying concentrations into acquired hyperspectral datacubes. Applying our adaptive matched filter gas-retrieval algorithms to the generated scene, we predict NACHOS will be able to distinguish 35 and 7 ppm.m plumes of SO2 and NO2 (respectively) with high sensitivity; a capability well-suited to address scientific goals related to monitoring both passive SO2 degassing from volcanoes and NO2 emissions from anthropogenic sources. Lastly, we will show findings from *thermal* and vibrational environmental tests, performed in preparation for a scheduled early-2022 launch, demonstrating the extremely robust spectrometer design is well-suited for satellite-based deployment.

Article • *Open access*

350 **Effects of living and working in a hot environment on cognitive function in a quiet and temperature-controlled room: An oil and gas industry study**

Girard O.,Gaoua N., Grantham J.,... Walsh A., Racinais S.

Temperature, 8(4)

2021

11

Hide abstract Find It @ UCL View at Publisher Related documents

We investigate the effects of seasonal heat stress on cognitive function in *outdoor* workers. Thirty-nine workers from an oil and gas industry in the Middle-East volunteered for cognitive testing before (5.30 to 7.00 am) and after (3.30 to 5.00 pm) their daily work-shift in hot (August–average daily temperature: ~41°C) and temperate (January–average daily temperature: ~22°C) seasons. While physical activity was reduced in hot compared to temperate season (average normalized acceleration: 96 ± 33 vs. 112 ± 31 × 10^{-3} g; –12.5 ± 4.7%; P = 0.010), the average core temperature during the work-shift was higher in the hot season (37.4 ± 0.2 vs. 37.2 ± 0.2°C; P = 0.002). Peak core temperature was 38.0 ± 0.1°C and 37.8 ± 0.1°C in hot and temperate seasons, respectively. Cognitive performance did not differ between seasons for tests of recognition memory (P = 0.169), working memory (P = 0.797) and executive function (P = 0.145), independent of testing time. Whereas there was no significant main effect of testing time for tests of recognition memory (P = 0.503) and working memory (P = 0.849), the number of problems solved on the first choice for the executive function test was lower in the afternoon than the morning (–9.2 ± 5.3%; P = 0.039). There was no season × testing time interaction for any cognitive tests (P ≥ 0.145). In the absence of hyperthermia, living and working in a hot environment does not alter cognitive function in oil and gas industry workers tested in a quiet and temperature-controlled room, with reduced clothing encumbrance (relative to work). Conclusions should not be extrapolated to more stressful situations (i.e., *thermal* stressor present, pronounced dehydration, *noise*).

Book Chapter

351 **Thermal Insulation of Building Envelope for Ecological Conservation**

Ali M.F.,Natarajan M.M.V.

Ecological and Health Effects of

2021

4

Document title	Authors	Source	Year	Citations
		Building Materials		

Hide abstract Find It @ UCL View at Publisher Related documents

Sustainability has been a rising concern for facility projects since the early 1990s. It is conscious that in terms of carbon emissions, energy usage, water use, utilisation of raw materials, waste production and many other factors, construction and operation of buildings have enormous direct and indirect environmental consequences. Increased economy and population will create more and more challenges for the design, construction and operational community to fulfil the newrequirements for infrastructure that are affordable, safe, resilient and sustainable while reducing their environmental impact. The building envelope's key feature is to offer shelter, safety, solar and *thermal* control, control of moisture, control of indoor air quality, access to sunshine, and *outdoor* views, fire resistance, *acoustics*, cost-effectiveness, and aesthetics. This study seeks to promote awareness on the energy friendly design strategies which economically minimize operational expenditure, while enhancing comfort of the occupants in the building. Thisworkwould allowarchitects, engineers and builders to choose the most efficient or climate-reactive envelope alternative in order to improve *thermal* environment efficiency of the construction design being proposed.

Article

352 **Potential use of coconut fibre modified mortars to enhance thermal comfort in low-income housing** — Quinones-Bolanos E., Gomez-Oviedo M., Mouthon-Bello J., ... Berardi U., Bustillo-Lecompte C. — Journal of Environmental Management, 277, 111503 — 2021 — 60

Hide abstract Find It @ UCL View at Publisher Related documents

Enhancing the *thermal* comfort of low-income housing in developing countries located in tropical areas is one of the main challenges for engineers and architects worldwide. The coconut mesocarp fibre (coir) has shown enormous potential for improving some properties of modified concretes or mortars, such as low-weight and high-*acoustic* isolation. In this study, the potential use of modified mortars by coconut fibres as a facade coating layer to enhance *thermal* comfort in low-income housing structures was evaluated for the city of Cartagena de Indias, Colombia. An actual typical low-income house of 42 m^2 was monitored. Temperature and humidity variations were monitored for 39 days, *thermal* characteristics of coir-modified mortars were also investigated using differential scanning calorimetry (DSC) and an adaptation of the standard test method of the guarded-hot-cartridge apparatus. The EnergyPlus™ software was used to simulate indoor temperature variations in the studied house. Results show that during the period of 4 h of maximum sunlight radiation with *outdoor* temperatures in the range of 29–34 °C, coating the cement-sand hollow block structure with a layer of coir-modified mortar could reduce indoor room temperatures by 0.5–1.5 °C, approximately. Thus, there is a potential to enhance the *thermal* comfort in low-income housing structures with coconut fibre modified mortars while reducing annual energy costs of cooling by 16%, making it affordable for low-income families in the Caribbean region of Colombia.

Article • *Open access*

353 **Building performance evaluation of a new hospital building in the uk: Balancing indoor environmental quality and energy performance** — Jain N., Burman E., Stamp S., ... Mumovic D., Davies M. — Atmosphere, 12(1), 115 — 2021 — 40

Hide abstract Find It @ UCL View at Publisher Related documents

Hospitals are controlled yet complex ecosystems which provide a therapeutic environment that promotes healing, wellbeing and work efficiency for patients and staff. As these buildings accommodate the sick and vulnerable, occupant wellbeing and good indoor environmental quality (IEQ) that deals with indoor air quality (IAQ), *thermal* comfort, lighting and *acoustics* are important objectives. As the specialist nature of hospital function demands highly controlled indoor environments, this makes them energy intensive buildings due to the complex and varying specifications for their functions and operations. This paper reports on a holistic building performance evaluation covering aspects of indoor air quality, *thermal* comfort, lighting, *acoustics*, and energy use. It assesses the performance issues and inter-relationships between IEQ and energy in a new building on a hospital campus in the city of Bristol, United Kingdom. The empirical evidence collated from this case study and the feedback received from the hospital staff help identify the endemic issues and constraints related to hospital buildings, such as the need for robust ventilation strategies in hospitals in urban areas that mitigate the effect of indoor and *outdoor* air pollution

	Document title	Authors	Source	Year	Citations

and ensuring the use of planned new low-carbon technologies. Whilst the existing guidelines for building design provide useful instructions for the protection of hospital buildings against ingress of particulate matter from outdoors, more advanced filtration strategies may be required to enact chemical reactions required to control the concentration levels of pollutants such as nitrogen dioxide and benzene. Further lessons for improved performance in operation and maintenance of hospitals are highlighted. These include ensuring that the increasingly available metering and monitoring data in new buildings, through building management systems, is used for efficient and optimal building operations for better IEQ and energy management. Overall, the study highlights the need for an integrated and holistic approach to building performance to ensure that healthy environments are provided while energy efficiency targets are met.

Article • *Open access*

354 **Health risk for older adults in Madrid, by outdoor thermal and acoustic comfort** — Baquero Larriva M.T., Higueras E. — Urban Climate, 34, 100724 — 2020 — 82

Hide abstract Find It @ UCL View at Publisher Related documents

Cities must adapt to aging populations and mitigate the effects of climate change and urbanization on health. This study analyses the *outdoor thermal* and *acoustic* comfort of older adults in public spaces in Madrid. We compared the subjective perception with real environmental conditions measured in-situ and two *thermal* comfort indices Physiological Equivalent Temperature (PET) and Universal *Thermal* Index (UTCI). Additionally, use and stay of older people in those public spaces was assessed. Results showed that older adults represent 26.35% of the users, environmental variables such as mean radiant temperature, air temperature, and *noise* levels are the most relevant variables for them to decide to stay in these places. Although most of the 413 interviewees perceived the environment as comfortable, this research shows that in dense urban areas there is a significant health risk due to *noise* pollution and extreme temperatures. Average *noise* levels measured exceed the maximum threshold recommended by the World Health Organization and according to PET and UTCI indices around 73% of the interviewees would be in risk of *thermal* stress in winter and 98.2% in summer. The need for further research to find strategies to mitigate the environmental risks of older people in public spaces is evident.

Article

355 **Implications of indoor air temperature variation on the health and performance of Brazilian students** — Silva L.B.D.,de Souza E.L., de Oliveira P.A.A., Andrade B.J.M. — Indoor and Built Environment , 29(10) — 2020 — 13

Hide abstract Find It @ UCL View at Publisher Related documents

The aim of the present study was to evaluate the relationship between cognitive performance, health and environmental comfort as a function of indoor air temperature (T_a) variation. A total of 360 undergraduate students were subjected to the variation of the T_a at 20, 24 and 30 °C; their *thermal* responses were evaluated over three consecutive days. Performance variables measured in the study were cognitive performance, blood pressure, heart rate (HR) and comfort. The environmental variables measured were T_a, globe temperature (T_g), illumination, *noise*, airflow velocity and air quality. The variation in HR was influenced by the variables, relative air humidity and mean radiant temperature (T_{rm}) during the three days of observation, where HR was higher than 100 bpm when T_g was greater than T_a. T_{rm} increased proportionally to the increase in T_g, thus characterising heat exchange by radiation. The number of correct answers and test response time were also positively influenced by T_{rm} when T_a was 20 °C. Teaching environments (TEs) with increased heat load due to the individual body heat of students, increased *outdoor* T_a and urban morphology associated with the building of the TEs result in increasing in T_{rm} due to the T_g being higher than the air temperature, with possible impacts on health and performance variables.

Article • *Open access*

356 **Outdoor light at night, air pollution and depressive symptoms: A cross-sectional study in the Netherlands** — Helbich M., Browning M.H.E.M., Huss A. — Science of the Total Environment , 744, 140914 — 2020 — 98

Hide abstract Find It @ UCL View at Publisher Related documents

Background: Artificial light at night (ALAN) may be an anthropogenic stressor for mental health disturbing humans' natural day–night cycle. However, the few existing studies used satellite-based measures of radiances for *outdoor*

	Document title	Authors	Source	Year	Citations

ALAN exposure assessments, which were possibly confounded by traffic-related air pollutants. Objectives: To assess 1) whether living in areas with increased exposure to *outdoor* ALAN is associated with depressive symptoms; and 2) to assess the potential confounding effects of air pollution. Methods: We used cross-sectional data from people (N = 10,482) aged 18–65 years in the Netherlands. Depressive symptoms were assessed with the Patient Health Questionnaire (PHQ–9). Satellite-measured annual ALAN were taken from the Visible Infrared Imaging Radiometer Suite. ALAN exposures were assessed at people's home address within 100 and 600 m buffers. We used generalized (geo)additive models to quantify associations between PHQ–9 scores and quintiles of ALAN adjusting for several potential confounders including $PM_{2.5}$ and NO_2. Results: Unadjusted estimates for the 100 m buffers showed that people in the 2nd to 5th ALAN quintile showed significantly higher PHQ–9 scores than those in the lowest ALAN quintile (β_{Q2} = 0.503 [95% confidence intervals (CI): 0.207–0.798], β_{Q3} = 0.587 [95% CI: 0.291–0.884], β_{Q4} = 0.921 [95% CI: 0.623–1.218], β_{Q5} = 1.322 [95% CI: 1.023–1.620]). ALAN risk estimates adjusted for individual and area-level confounders (i.e., $PM_{2.5}$, urbanicity, *noise*, land-use diversity, greenness, deprivation, and social fragmentation) were attenuated but remained significant for the 100 m buffer (β_{Q2} = 0.420 [95% CI: 0.125–0.715], β_{Q3} = 0.383 [95% CI: 0.071–0.696], β_{Q4} = 0.513 [95% CI: 0.177–0.850], β_{Q5} = 0.541 [95% CI: 0.141–0.941]). When adjusting for NO_2 per 100 m buffers, the air pollutant was associated with PHQ–9 scores, but ALAN did not display an exposure-response relationship. ALAN associations were insignificant for 600 m buffers. Conclusion: Accounting for NO_2 exposure suggested that air pollution rather than *outdoor* ALAN correlated with depressive symptoms. Future evaluations of health effects from ALAN should consider potential confounding by traffic-related exposures (i.e., NO_2).

Conference Paper

357 **AcuTe: Acoustic thermometer empowered by a single smartphone** — Cai C.,Chen Z.,Pu H., ... Hu M.,Luo J. — Sensys 2020 Proceedings of the 2020 18th ACM Conference on Embedded Networked Sensor Systems — 2020 — 42

Hide abstract Find It @ UCL View at Publisher Related documents

Though measuring ambient temperature is often deemed as an easy job, collecting large-scale temperature readings in real-time is still a formidable task. The recent boom of network-ready (mobile) devices and the subsequent mobile crowdsourcing applications do offer an opportunity to accomplish this task, yet equipping commodity devices with ambient temperature sensing capability is highly non-trivial and hence has never been achieved. In this paper, we propose *Acoustic Thermometer* (AcuTe) as the first ambient temperature sensor empowered by a single commodity smartphone. AcuTe utilizes on-board dual microphones to estimate air-borne *sound* propagation speed, thereby deriving ambient temperature. To accurately estimate *sound* propagation speed, we leverage the phase of chirp signals to circumvent the low sample rate on commodity hardware. In addition, we propose to use both structure-borne and air-borne propagations to address the multipath problem. Furthermore, to prevent disruptive audible transmissions, we convert chirp signals into white *noises* and propose a pipeline of signal processing algorithms to denoise received samples. As a mobile, economical, highly accurate sensor, AcuTe may potentially enable many relevant applications, in particular large-scale indoor/*outdoor* temperature monitoring in real-time. We conduct extensive experiments on AcuTe; the results demonstrate a robust performance, a median accuracy of 0.3° C even at a varying humidity level, and the ability to conduct distributed temperature sensing in real-time.

Article

358 **Indoor environmental quality in existing public buildings in China: Measurement results and retrofitting priorities** — Miao Y.,Ding Y. — Building and Environment , 185, 107216 — 2020 — 32

Hide abstract Find It @ UCL View at Publisher Related documents

Through a large-scale questionnaire survey across China and indoor physical measurements of typical cities in the four climatic zones, this study investigates the current status and retrofitting priorities of the IEQ of existing public buildings in China. The public buildings in China mainly present uneven indoor hot and cold distributions and draft sensations. The utilization ratio of natural light is low, and glare often exists. HCHO and TVOC are not major concerns

	Document title	Authors	Source	Year	Citations

of indoor air quality, but affected by the region and season, the $PM_{2.5}$ and PM_{10} concentrations exceed the limits of the Chinese standard. All the buildings present excessive *noise*. Respondents generally demand improved indoor air quality and *thermal* environment, and different control strategies should be adopted across regions with different local climatic conditions and behavioural habits. For the *thermal* environment, the indoor air distribution should be optimized and the indoor humidity ratio increased in winter in northern regions of China. For the luminous environment, the indoor illuminance value should be decreased and the lighting quality improved. For the *acoustic* environment, the *sound* insulation and *sound* absorption performance of the building envelope should be improved. For indoor air quality, the purification capacity of the air conditioning system and the airtightness of the building envelope should be strengthened. According to the *outdoor* air quality, different channels can be opened to introduce fresh air. For example, when the *outdoor* air quality is good, fresh air can be directly introduced to improve the service life of the filtering system and reduce energy consumption.

Article • *Open access*

☐ 359 **Post-occupancy evaluation of indoor environmental quality in ten nonresidential buildings in Chongqing, China**

Tang H.,Ding Y.,Singer B.C. | Journal of Building Engineering , 32, 101649 | 2020 | 50

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

Indoor environmental quality (IEQ) is an essential element of sustainable building performance. To advance knowledge of indoor environmental quality in Chinese nonresidential buildings, this study conducted post-occupancy evaluation (POE) in varied use areas within schools, hospitals, offices, hotels and shopping malls through environmental measurements and occupant surveys in 10 buildings over four seasons in Chongqing, China. Measured *thermal* conditions were more frequently within comfort standards in winter than in summer according to the predicted mean vote – predicted percentage dissatisfied model; but the model did not reliably predict *thermal* satisfaction. There was insufficient illuminance in 28% of studied areas and high illuminance in shopping malls led to complaints about glare. *Sound* pressure exceeded the Chinese guideline in all buildings with shopping malls having significantly higher levels than other buildings. CO_2 was generally below the Chinese national guideline of 1000 ppm; the two percent of measurements that exceeded the guideline occurred in classrooms, hospital wards and underground supermarkets. Formaldehyde did not exceed the limit level of 0.1 mg/m^3 in any space. Most of the buildings did not provide effective protection from *outdoor* fine particulate matter ($PM_{2.5}$), and infiltration from outdoors was the main factor for exceeding health-based guidelines indoors. Shopping malls had lower IEQ satisfaction than offices, schools and hospitals. *Sound* was the factor with the lowest satisfaction level among the four principal environmental factors. Overall, this study demonstrates value of POE on identifying deficiencies in indoor environment and provides a reference for assessment of IEQ in nonresidential buildings.

Article

☐ 360 **Experimental method to monitor temperature stability of SiPMs operating in conditions of extremely high dark count rate**

Lucchini M.,Musienko Y.,Heering A. | Nuclear Instruments and Methods in Physics Research Section A Accelerators Spectrometers Detectors and Associated Equipment , 977, 164300 | 2020 | 5

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

The use of Silicon Photomultipliers (SiPMs) in high radiation environments (e.g. detectors at collider experiments) or in *outdoor* conditions (e.g. LIDAR applications), requires to deal with a high level of dark count *noise*. An undesired effect emerging under these conditions is the self-heating of the SiPM due to a large power dissipation that causes variations of the breakdown voltage and thus affects operating parameters such as gain and PDE. A method to evaluate the heat dissipation properties of different SiPM packages and the temperature stability of SiPMs during operation under extremely high dark count rates (larger than 30 GHz) is presented in this paper. Starting from a

	Document title	Authors	Source	Year	Citations

condition in which the SiPM is at *thermal* equilibrium with the environment, a change in the current measured under intense illumination with a LED can be attributed to a temperature variation of the SiPM due to self-heating. Under certain assumptions, a quantitative estimate of the local change of temperature can be obtained if certain parameters of the SiPM are known: gain, photon detection efficiency, equivalent charge factor and drift of the SiPM breakdown voltage with temperature. The method is applied to evaluate the *thermal* conductivity of different SiPM package configurations, which is a key requirement for reliable operation in conditions of extremely high dark count rate. Among the SiPM package configurations tested, the one consisting of a ceramic substrate coupled to a copper heat sink provided the best performance in terms of good heat dissipation with less than 0.5 °C temperature variation for 100 mW dissipated power.

Article • *Open access*

☐ 361 **Effects of sound types and sound levels on subjective environmental evaluations in different seasons** — Jin Y.,Jin H.,Kang J. — Building and Environment , 183, 107215 — 2020 — 48

Hide abstract Find It @ UCL View at Publisher Related documents

Human beings live in an environment where various factors act together, therefore, it is essential to study the comprehensive effects of multiple factors, particularly the effects of their interaction on environmental evaluations. In this study, the effects of the interaction between *sound* types and *sound* levels on *acoustic*, *thermal*, and overall evaluations were explored by simulating typical *outdoor* temperatures in different seasons in a controllable environmental chamber. The results indicated that the *acoustic* evaluations were significantly higher for birdsong and slow-dance music than for dog barking, conversation, and traffic *sound*; additionally, the *acoustic* evaluations at the low *sound* level were always higher than those at the high *sound* level. In terms of the *thermal* evaluations, there was a significant variation in different seasons. In summer, birdsong and slow-dance music effectively improved subjects' *thermal* evaluations, while a high *sound* level of dog barking, conversation, and traffic *sound* resulted in a decrease; in the transition season, all types of *sounds* resulted in a decline in the *thermal* evaluations; meanwhile, in winter and summer, dog barking, conversation, traffic *sound* and slow-dance music at the low *sound* level produced higher *thermal* comfort and *thermal* acceptability. In terms of the overall evaluations, birdsong and slow-dance music at the low *sound* level improved overall comfort, while dog barking, conversation, and traffic *sound* resulted in a significant decrease. For dog barking, conversation, traffic *sound* and fast-dance music, the overall evaluations at the low *sound* level were higher than those at the high *sound* level.

Article

☐ 362 **Temperature effects on the mechanical properties of hybrid composites reinforced with vegetable and glass fibers** — Cerbu C.,Wang H., Botis M.F.,Huang Z., Plescan C. — Mechanics of Materials , 149, 103538 — 2020 — 21

Hide abstract Find It @ UCL View at Publisher Related documents

Although there are numerous applications for composite materials reinforced with vegetable textile fibers in *outdoor* parts and components (e.g., *acoustic* or *thermal* insulation panels in construction, furniture for gardens), only a few studies have reported the effects of temperature on hybrid composites reinforced with both glass and vegetable fibers. The main goal of this study was to analyze the effects of temperature on the tensile properties of five types of polymeric composites with different reinforcements: jute fabric, glass and jute fabric, flax fabric, glass and flax fabric, and glass fabric. Our method consisted of tensile testing all five types of composites at room temperature (20 °C) and other different temperatures (50 °C and 70 °C). The greatest values of reduction in tensile strength were recorded for hybrid composites. Reductions of 42.3% and 49.6% at 50 °C and 70 °C, respectively, were observed for the composite reinforced with flax and glass fibers. Reductions of 34.6% and 60% at 50 °C and 70 °C, respectively, were observed for the composite reinforced with jute and glass fibers. Finite element analysis of representative volume elements was conducted to compute the residual *thermal* stresses developed in the fibers and in the matrices at interfaces. It was determined that the *thermal* stresses cause micro-cracks at the interfaces. *Thermogravimetric* analysis and microscopic analysis were also conducted to identify the causes of the significant decreases in tensile properties. The effects of the anisotropy of flax and jute fibers on the *thermal* stresses developed at fiber-matrix interfaces were analyzed as well.

	Document title	Authors	Source	Year	Citations
	Conference Paper				
363	**Performance Analysis of DF Based Cascaded VLC-FSO-VLC System**	Deka R.,Anees S.	3rd International Conference on Advanced Communication Technologies and Networking Commnet 2020, 9199623	2020	6

Hide abstract Find It @ UCL View at Publisher Related documents

This paper investigates the performance of decode-and-forward (DF) based cascaded visible light communication-free space optical communication-visible light communication (VLC-FSO-VLC) co-operative system. In the proposed system, the *outdoor* connectivity is provided by an FSO link and indoor connectivity is provided through VLC environment. The FSO link undergoes the impact of Double Generalized Gamma (DGG) distributed atmospheric turbulence and Rayleigh distributed pointing errors and VLC environment follows the Lambertian radiation pattern of the light-emitting-diode (LED). For this system, using the derived expressions for probability density function and cumulative distribution function of the system's signal-to-*noise*-ratio (SNR), closed-form expressions of outage probability and average bit-error rate (BER) are obtained. The numerical results show that the performance metrics of the system directly depends on LED-PD (photo-detector) distance, field-of-view (FOV), different atmospheric turbulence conditions, and type of detection technique.

	Article				
364	**Accelerated aging laboratory tests for the evaluation of the durability of hydrophobic treated and untreated andesite with respect to salt crystallization, freezing–thawing, and thermal shock**	Celik M.Y.,Sert M.	Bulletin of Engineering Geology and the Environment , 79(7)	2020	37

Hide abstract Find It @ UCL View at Publisher Related documents

The processes of deterioration related to the interaction of the petrographic, physical, and mechanical characteristics of the rocks with physical and chemical actions of water are important. The aim of the article was to study the deterioration of andesite by accelerated weathering tests. For this purpose, laboratory studies were carried out in two stages. First, the main characterization analysis of the andesite was performed. Main technical values and parameters such as mineralogical–petrographic (polarize microscope, XRD, and SEM), chemical (XRF), and mechanical–physical properties were studied. Moreover, calculations on some physical and mechanical tests as well as the resistance to dry weight change caused by weathering were assessed. The change of rock strength and ultrasonic wave velocity measurements were experimented. These experiments were carried out on untreated and hydrophobic treated andesite before and after accelerated alteration tests such as cyclic freezing–thawing, salt crystallization, and *thermal* shock. The ultrasonic wave velocity and uniaxial compressive strength test results on the samples subjected to salt crystallization and freezing–thawing tests showed decreased values for all the andesite samples. However, the physical integrity of the andesite samples did not deteriorate. Laboratory test results showed that all andesite samples were suitable for use as building stones in a humid and salty environment. It was found that treated samples were less affected by the destructive effect of accelerated weathering tests compared with the untreated samples. Therefore, for the *outdoor* use of these stones, it is recommended that the block surfaces are to be treated with chemicals against weathering.

	Conference Review				
365	**3rd International Conference on Sustainable Engineering Techniques, ICSET 2020**	[No Authors Found]	Iop Conference Series Materials Science and Engineering, 881(1)	2020	0

Hide abstract Find It @ UCL View at Publisher

Document title	Authors	Source	Year	Citations

The proceedings contain 188 papers. The topics discussed include: the role of sustainable development approaches in addressing urban tensions in cities; coexistance of form and structure in contemporary architecture; the effect of cooperative learning strategy on the engagement in architectural education; functional transparency in digital architecture; impact of facade material of mass housing on *outdoor thermal* comfort in hot-arid climate; self-sufficiency and its role in the sustainability of exhibition buildings; the effect of parking lots distribution arrangements on environmental *noise* level for the universities campuses; adaptive reuse as an approach to sustainability; the role of addition in leftover places in the sustainability of historical buildings; economic revitalization of historical paths, analytical study of the tourist attraction property; nature simulation in iterative systems of contemporary structures; the effect of the characteristics formation of urban open space on *thermal* comfort for pedestrian; and the effect of smart low emission glass material on reducing energy consumption for office building in hot arid climate.

Article • *Open access*

366 **The challenge of creating age-friendly indoor environments in a high-density city: Case study of Hong Kong's care and attention homes**

Tao Y.,Gou Z.,Yu Z.,Fu J.,Chen X.

Journal of Building Engineering , 30, 101280

2020

46

Hide abstract Find It @ UCL View at Publisher Related documents

Aiming to find out whether the indoor environment quality (IEQ) of Care and Attention (C&A) homes in Hong Kong could meet the need of older people, this research conducted a field measurement and occupancy survey in nine C&A homes to assess their *thermal*, lighting, and *noise* environments according to both building standards and occupant satisfaction. The findings indicated that an operative indoor temperature of 24–27 °C in summer and above 23 °C in winter could be considered to mark a *thermally* comfortable zone. Noticeably, older people preferred a warmer environment in both summer and winter, while the measurement results showed that the indoor operative temperature was lower, and the humidity was higher than the comfort standard for summer, which was confirmed by the occupant survey. The mean illuminance level could meet the national threshold; interestingly, there was a negative correlation between age groups and their satisfaction of natural lighting in winter. The mean street *noise* level was between 74-75 dB, exceeding the maximum 55 dB for neighborhoods. The street *noise* level was significantly negatively correlated with residents' satisfaction. The study made recommendations for regulating both *outdoor* and indoor environments for improving the elderly living conditions in high-density urban areas with hot humid climates.

Article • *Open access*

367 **Housing workers' evaluations of residential environmental quality in south Korean welfare housing for low-income, single-parent families**

Cho M.

Sustainability Switzerland , 12(14), 5599

2020

2

Hide abstract Find It @ UCL View at Publisher Related documents

This study evaluates the spatial conditions of the residential environment in welfare housing built for low-income, single-parent (LISP) families in South Korea to reveal the significant residential environmental (RE) design criteria concerning housing quality and family well-being. The primary data were analyzed by surveying 30 housing directors from 29 welfare residences, in conjunction with facility visits and interviews. The survey data were supplemented by interviewing 11 stakeholders, including government institute officials, project architects, and former residents who exited the program. A statistical analysis examined the RE quality in relation to building and resident features. Most respondents reported insufficient physical quality, specifically due to inappropriate units in size and number; poor indoor *noise* control; and substandard unit rooms, children's rooms, and *outdoor* spaces. Furthermore, adequate rooms, family privacy, and a pleasant indoor environment (i.e., *noise* barrier, *thermal* comfort) were the most critical spatial design criteria. Particularly, the aforementioned housing environmental attributes were found to be different depending on a building's construction year and type, householder type and age, and children's age. Practical and methodological implications and future research directions are discussed to elevate the housing quality and sustainable well-being in welfare housing.

Conference Paper

	Document title	Authors	Source	Year	Citations
368	**A Novel Miniature OCXO Using Hermetically Sealed Ceramic Package**	Hsieh W.-L.,Hsu E.-S., Yen Y.-S.,... Huang Y.-C., Wang M.-H.	Ifcs Isaf 2020 Joint Conference of the IEEE International Frequency Control Symposium and IEEE International Symposium on Applications of Ferroelectrics Proceedings, 9234843	2020	5

Hide abstract Find It @ UCL View at Publisher Related documents

This work presents an ultra-miniature oven controlled crystal oscillator (OCXO) using a hermetically sealed ceramic package with the size of 7.0\ \text{mm} \times 5.0\ \text{mm}. The oven structure combined within a heater-embedded ceramic package and a low *thermal* conductivity bonding adhesive provides excellent *thermal* performance, achieving temperature stability within ±10 ppb across -40 to 95°C. Hermeticity is accomplished with a seam-welded lid to a kovar ring on the ceramic substrate. In addition, a 2.5\ \text{mm} \times 2.0\ \text{mm} SC-cut crystal is utilized to further improve the temperature stability, aging, gravity sensitivity, and phase *noise* performance that can be compliance with Stratum 3E specifications. The proposed ultra-miniature OCXO provides more robust and smaller package alternative for 5G *outdoor* applications, such as remote radio heads (RRHs), active antenna unit (AAU), and small cells.

Conference Paper • *Open access*

369	**New types of windows' forms as a fundamental to reach multifunctional properties of windows in agricultural facilities**	Stetsky S.,Shchelokova T.	E3s Web of Conferences , 175, 11007	2020	1

Hide abstract Find It @ UCL View at Publisher Related documents

The presented article deals with problems of internal environmental characteristics in premises of agricultural facilities in regions with hot and sunny climate. Environmental conditions are being considered in close linkage with forms, proportions and sizes of windows traditionally used in regions in question and with geometrics of new types of windows. It was stated that windows in the said climate have a great effect on the different microclimatic environmental regimes in the premises in question, i.e. on the internal lighting of interiors, *thermal* gains due to solar radiation, *noise* penetration, etc. Also, the paper considers another variable - the informative quality of a window. This one not only dictates the geometrics of window openings, but also needs a vast experimental research to be conducted on subjective appraisal of quantity and quality of *outdoor* information through considered windows.

Review • *Open access*

370	**On the colloidal and chemical stability of solar nanofluids: From nanoscale interactions to recent advances**	Sharaf O.Z.,Taylor R.A., Abu-Nada E.	Physics Reports, 867	2020	125

Hide abstract Find It @ UCL View at Publisher Related documents

The *outdoor* deployment of solar nanofluids in practical solar photothermal conversion devices for prolonged periods of time has been an ongoing challenge. This is due to their exposure to intense solar radiation, elevated temperatures, *thermal* and solar cycling, and other rough operation conditions that compromise their dispersion and chemical stability. This work is an attempt to address this challenge in light of state-of-the-art advances in understanding, characterizing, and mitigating issues pertaining to the stability of solar nanofluids. We first defined the concept of nanofluids, along with their types and applications, merits and limitations, and preparation techniques. This was followed by presenting the basic physical principles that govern the interparticle interactions, clustering and deposition kinetics, and theories of colloidal stability for nanoparticles in aqueous environments. Building on a *sound* understanding of these necessary fundamentals, the destabilization factors and instability effects on solar nanofluids were systematically identified. A critical evaluation of the most recent stability characterization techniques and stabilization strategies for solar nanofluids was then provided. A coverage of selected state-of-the-art works in the literature that target the enhanced stability of solar nanofluids in different photothermal conversion processes based

	Document title	Authors	Source	Year	Citations

on the direct volumetric absorption of solar radiation was presented and comparatively evaluated. Finally, recommendations were drawn regarding current knowledge gaps and future research directions in order to overcome the stability limitations hindering the deployment of solar nanofluids in practical solar photothermal conversion applications.

Article

371 **Effect of turbulence and noise on ultraviolet and mid-infrared spectrum in optical wireless communications**

Biswal M.R.,Arya S.,Chung Y.H.

Photonic Network Communications, 39(3)

2020

8

Hide abstract Find It @ UCL View at Publisher Related documents

Ultraviolet and mid-infrared (MIR) constitute two extreme ends of the optical wireless communication band. Recently, there has been a rise in increased interest for research in these two spectral bands. For successful deployment of these technologies, the performance of these two spectrum bands needs to be thoroughly analyzed. We consider turbulence and *noise* effects over these spectral bands. Specifically, this paper considers separate and combined effects of atmospheric turbulence and various *noise* sources to model the channel impairment. The results show that the system performances are extremely sensitive to the choice of channel model. It is found that the relatively unused MIR spectrum offers excellent characteristics for *outdoor* optical communications in the presence of atmospheric turbulence.

Article • *Open access*

372 **Green Composite of Instant Coffee and Poly(vinyl alcohol): An Excellent Transparent UV-Shielding Material with Superior Thermal-Oxidative Stability**

Lyu Y.,Gu X.,Mao Y.

Industrial and Engineering Chemistry Research, 59(18)

2020

35

Hide abstract Find It @ UCL View at Publisher Related documents

Producing ultraviolet-light-shielding (UV-shielding) materials via green chemistry has been attracting researchers' attention. Herein, a polymer composite produced by hybridizing instant coffee with poly(vinyl alcohol) (PVA) is reported. It has been shown that adding coffee extract can significantly improve the UV-shielding property of the PVA material in the wavelength range between 200 and 400 nm, and the polymer composite remains largely transparent in the visible region. The UV-shielding performance of the composite can be maintained under the accelerated aging condition with a UV light dose of 40 MJ/m^2 for 68 h (approximately equivalent to a 45-day *outdoor* exposure in Miami, Florida). In addition, enhanced *thermal*-oxidative stability has been observed for the composite, as indicated in the *thermal* gravimetric analysis (TGA). The composite material can be used in the form of polymer films processed by solution casting or be served as a protection layer by dip coating. The mechanisms of UV shielding and *thermal*-oxidative stability are discussed based on effective components such as melanoidin, caffeine, and chlorogenic acid. The work demonstrates yet another route toward fabricating biocompatible and environment-friendly UV-shielding materials with *sound* overall performance.

Review • *Open access*

373 **The effect of school design on users' responses: A systematic review (2008-2017)**

Manca S.,Cerina V.,Tobia V.,Sacchi S.,Fornara F.

Sustainability Switzerland, 12(8), 3453

2020

47

Document title	Authors	Source	Year	Citations

Hide abstract Find It @ UCL View at Publisher Related documents

This systematic review focused on the effect of the educational environment design on students and teachers performance, satisfaction, and wellbeing. Starting from a bulk of 1307 articles, a set of N = 68 empirical papers was selected and organized on the basis of four different content clusters, i.e., architectural building design and aesthetic features, indoor environmental features, classroom design, and school green spaces/*outdoor* spaces. From the analysis of research findings, the key role of pleasant, warm, and flexible learning environments emerged, for promoting both wellbeing and performance of users. More specifically, the presence of charming colors and pictures, ergonomic furniture, and adequate *acoustic*, *thermal* comfort, ventilation, and natural lighting have emerged as important features that school designers should care for. Furthermore, an integration of both indoor and *outdoor* learning situations showed to be effective for improving students learning and wellbeing.

Article

374 **Survey study on Chinese commercial kitchen exhaust fume systems in Tianjin** — Zhang A.,Deng N.,Shen C., ... Wang Y.,Long Z. — Building and Environment, 171, 106629 — 2020 — 19

Hide abstract Find It @ UCL View at Publisher Related documents

Exhaust fume system plays an important role in ensuring indoor and *outdoor* environment in the Chinese commercial kitchens (CCKs). To investigate the current status of exhaust fume systems in CCKs (CCKEF), on-site survey of cooking appliances, components of CCKEF including exhaust hoods, purification equipment, exhaust fans, discharge form and replacement air, *outdoor* grease accumulation, and indoor environment, was conducted in 41 CCKs in Tianjin. Results showed that: (1) the diversity and number of cooking appliances was related to kitchen area. 76.19% of stir-frying stoves were located at the end of the appliance lines. (2) Most of the CCKEFs were composed of wall-mounted canopy hood (94.12%), terminal static purifier (98.04%), centrifugal fan (100%), roof-located discharge (66.67%) and transfer air (100%). 47.06% of canopy hoods were equipped with baffle filters, including 29.41% tube-type, 9.80% plate-type and 7.85% strainer-type. The nameplate purification efficiency of purifiers met the fume emission standard. 81.08% of purifier manufacturers were from Beijing. (3) *Outdoor* grease accumulation was serious. 72.73% had much or a lot of grease accumulation. Cleaning period and the number of burners were the major influence factors. For systems with over 4 burners, the cleaning period was better within 2 months. (4) 70.8% of chefs were dissatisfied with the *thermal* comfort in CCKs. *Thermal* and wet sensation varied with seasons and summer was the most unsatisfactory season. Over 80.0% of chefs felt fumes overflowing and interfering *noise*. The indoor environment in CCKs needed improvement. Survey results can provide reference for the designing of CCKEF in China.

Article

375 **Numerical investigation on the stack effect of a general high-rise building model | 一种通用高层建筑模型烟囱效应的数值模拟分析** — Xie X.-F.,Yang Y. — Gongcheng Lixue Engineering Mechanics, 37(3) — 2020 — 6

Hide abstract Find It @ UCL View at Publisher Related documents

The stack effect of super high-rise buildings is a phenomenon of uncontrolled indoor and *outdoor* air infiltrations caused by certain factors such as the indoor and *outdoor* environmental conditions and the architectural features as well. Strong stack effect may lead to the operation failure of the elevator, the aerodynamic *noise* in elevator shaft and the energy waste of air conditioning etc. Currently, the research on this issue is far behind the construction of high-rise buildings. Based on the field investigations on the stack effects of more than 10 super high-rise buildings, their common architectural features were summarized and a general high-rise building model for the analysis of stack effect was designed, which was based on the standard high-rise building model usually used in structural wind engineering field. A multi-zone network model method was employed to analyze the influences of several important factors on the pressure distributions of the elevators, i.e., the airtightness level of building envelope, the design of the entrance door, the indoor space division and the *outdoor* meteorological condition etc. The following conclusions were finally obtained: comparatively, doors of the shuttle elevators were most likely affected by strong stack effect; in addition to the *outdoor* meteorological conditions and the height of the elevator shaft, other factors such as the airtightness of the curtain wall and the indoor space divisions would affect the pressure distributions of the elevator doors as well, improving the airtightness of the curtain wall could effectively reduce the strength of the stack effect

	Document title	Authors	Source	Year	Citations

and the pressure difference on the elevator doors; the design and the status of the entrance door on the first floor would play an important role for the whole stack effect performance, and the opening and closing process of the sliding door would have a great influence on the pressure differences acting on the elevator doors on the first floor; when the elevator lobby door was either installed in the first or the top floor, the pressure difference acting on the elevator doors would significantly decrease; the stack effect would be affected by the *outdoor* wind conditions as well, and the influences of the wind speed and the wind direction on the stack effect were relatively more complex than those of the *thermal* pressure. Different wind conditions would bring different results, thereby, it was necessary to perform individual investigation combined with the local meteorological conditions. Results obtained from the detail parameter analyses based on the general high-rise building model in this paper could provide a useful reference for investigating the stack effect phenomena of real super high-rise buildings, as well as the mitigation measures for strong stack effects.

Review

☐ 376 **Review of parameters used to assess the quality of the indoor environment in Green Building certification schemes for offices and hotels**

Wei W.,Wargocki P., Zirngibl J.,Bendzalova J., Mandin C.

Energy and Buildings , 209, 109683

2020

119

Hide abstract Find It @ UCL View at Publisher Related documents

Fourteen Green Building (GB) certification schemes were reviewed to examine the parameters they used to assess indoor environmental quality (IEQ). Ninety different parameters were identified. They were classified into four major IEQ components defining the *thermal*, *acoustic* and visual environments, and indoor air quality (IAQ). For the *thermal* environment, the most commonly used parameters were PMV, PPD, room operative temperature, room air temperature, room air relative humidity, and air speed. For the *acoustic* environment, the most commonly used parameters were ambient *noise* and reverberation time. For the visual environment, the most commonly used parameters were illuminance level, daylight factor, and spatial daylight autonomy. For IAQ, the most commonly used parameters were ventilation rate (*outdoor* air supply rate), TVOC, formaldehyde, CO_2, CO, PM_{10}, $PM_{2.5}$, ozone, benzene, and radon. Credits are used to rate the importance of different parameters for the overall level of IEQ in the reviewed schemes. Using these credits and the figures published in peer-reviewed papers, it was found out that the average contribution of the *thermal*, *acoustic*, luminous environment and air quality parameters to the overall IEQ rating of a building was respectively 27%, 17%, 22%, and 34%. The present work can be regarded as a reference for selecting parameters that are commonly used to characterize IEQ.

Article

☐ 377 **Studies on the optimum double-skin curtain wall design for high-rise buildings in the Mediterranean climate**

Saroglou T.,Theodosiou T., Givoni B.,Meir I.A.

Energy and Buildings , 208, 109641

2020

51

Hide abstract Find It @ UCL View at Publisher Related documents

The fast growing numbers of high-rise buildings around the world, a product of expanding urbanization and population growth, dictate the implementation of design strategies towards the reduction of their high energy loads. This study focuses on an office high-rise building in the Mediterranean climate of Tel Aviv and the reduction of high cooling loads relevant to this climate. In this process, the design of the building envelope becomes the most important constituent between indoor and *outdoor* environment, by dictating the required use of energy for achieving *thermal* comfort. A double skin façade (DSF) layer is proposed as a more advanced envelope design compared to the single skin curtain walls (controlled ventilation, *acoustic* insulation etc.). A previous study based on *thermal* simulations in Tel Aviv, revealed that a ventilated DSF design with the LowE glazing as the exterior layer of the DSF reduced cooling loads by 15% on average, in comparison with a typical DSF design in temperate climates where the LowE glazing is on the interior layer of the DSF, and by an average of 50% from the option of single skin envelope with LowE glazing. This study draws from the conclusions of previous data, and tests the energy efficiency of different DSF cavities in the Mediterranean climate. Results revealed that by increasing the cavity width from 0.2 m to 0.5 m, cooling loads decrease significantly, while there are also substantial reductions, from 0.5 m to 1.0 m, and from 1.0 m to 2.0 m DSF cavity width.

	Document title	Authors	Source	Year	Citations
378	Article • *Open access* **Comparison of microclimate measurements and perceptions as part of a global evaluation of environmental quality at neighbourhood scale**	Lemonsu A.,Amosse A., Chouillou D., ... Marchandise S., Tudoux B.	International Journal of Biometeorology , 64(2)	2020	12

Hide abstract Find It @ UCL View at Publisher Related documents

The EUREQUA project raises the issue of the definition and evaluation of the environmental quality of neighbourhoods. The approach consists of integrating and cross-referencing observable data characterising the physical environment and people's perception of their quality of life. The study area is a neighbourhood in Toulouse (France) with high social and typo-morphological diversity, subject to *noise* and air pollution nuisances. Three 3-day field campaigns were organised in January, April, and June 2014. Instrumented and commented walks took place three times per day. For each one, measurements of physical environmental parameters and surveys were performed simultaneously at six locations in the neighbourhood. The study focuses on microclimate and *thermal* comfort issues. It aims to compare in situ meteorological data of air temperature, humidity, wind speed, and mean radiant temperature, with quantitative results rating human perception of heat, humidity, wind, and *thermal* comfort. The variability in perception and measurements is mainly driven by seasonal effects, especially for heat and humidity, and, to a lesser extent, for wind. Wind perception and measurement also vary spatially, thus highlighting site effects. Linear models indicate a positive link between heat perception and mean radiant temperature, as well as between wind perception and mean and standard deviation of wind speed (with a higher sensitivity of people to wind under winter climate conditions). Finally, it is found that perception of *thermal* comfort is only slightly linked to the different microclimate dimensions, and is rather driven by other appreciation factors and emotional criteria related to the general environmental quality of the study area.

	Document title	Authors	Source	Year	Citations
379	Article • *Open access* **Assessment practice and thinking of healthy building in China | 我国健康建筑的评价实践与思考**	Ge Y.,Meng C.,Han M., Liu M.	Kexue Tongbao Chinese Science Bulletin , 65(4)	2020	2

Hide abstract Find It @ UCL View at Publisher Related documents

In January 2017, the Assessment Standard for Healthy Building (T/ASC 02-2016) was published. The issue and application of this standard indicates that healthy buildings have developed from labs to field in China. Based on it, the concept of healthy buildings has been gradually implemented in the construction industry and has achieved preliminary results. This paper conducted statistical analysis on the healthy indexes level from 48 certificated projects and quantitatively demonstrated the practical effect of health building projects in China. At the same time, through the analysis of the healthy measures for different types of projects, this study also made a statistical analysis for the operational difficulty of the provisions of the standard in practice, providing practical reference for the continuous updating and improvement of the health building standard and providing reference for the project construction and the manager's strategy formulation. Specific conclusions are as follows: (1) The health effects of the health building projects are reflected in several aspects: The average annual concentrations of indoor $PM_{2.5}$ and PM10 are 14.8 and 23.9 μg/m^3, respectively; the indoor average annual concentrations of formaldehyde and TVOC are 0.043 and 0.228 mg/m^3, respectively; 68.8% of the total hardness of drinking water in the project is less than 150 mg/L (calculated by $CaCO_3$); 85.4% of the main functional rooms reached the *thermal* comfort level of ClassI; the free fitness area takes 1.9% of the total area. (2) The higher the quality of the project is, the better the balance among the healthy elements is. In general, the score of items related to water takes 59% and 55% for comfort, which are significantly lower than that of air (82%), fitness (84%) and humanity (73%). However, the degree of practice completion is relatively low. It is suggested that the standard compilers should refer to the difficulty feedback of project practice in the process of standard revision. (3) According to the difficulty index value from low to high, the provisions are divided into 1-5 categories. Among them, 9 provisions (6.2.15, 8.2.4, 6.2.4, 8.2.3, 8.2.9, 6.2.6, 7.2.6, 7.2.4, 6.2.14) have significant differences in the difficulty index between public construction and residential construction. (4) The non-participation rates of 6 provisions (5.2.3, 6.2.4, 6.2.15, 6.2.16, 7.2.6, 8.2.8) were 64%, 93%, 64%, 64%, 77% and 84%, respectively. There are three types of reasons: Residential buildings do not participate in the evaluation, public buildings do not participate in the evaluation and projects without designated special space do not participate in the evaluation.

	Document title	Authors	Source	Year	Citations

Therefore, it is suggested that in the process of standard perfection, corresponding measures should be taken to improve the effective use rate of provisions. (5) In this paper, measure provisions in the standard are summarized into 16 technical measures, among which intelligent lighting technology, natural lighting technology and *outdoor acoustic* environment construction technology score rate are significantly lower. There are obvious differences in the selection of public buildings and residential buildings in terms of indoor *acoustic* environment control technology, natural lighting technology, fitness site design and child-friendly design.

Conference Paper

380 **The environmental performance of temporary urban interventions: Technical assessment of regeneration initiatives in the city centre of São Paulo, with focus on thermal and acoustic performance**

Michalski R.L.X.N., Rodrigues L., Goncalves J.C.S., ... Bilesky D.F., Guimaraes M.M.

PLEA 2020 35th PLEA Conference on Passive and Low Energy Architecture Planning Post Carbon Cities Proceedings, 1

2020

0

Hide abstract Find It @ UCL View at Publisher Related documents

Temporary urbanism is an approach to reactivate urban spaces through short-term interventions in a range of urban contexts. In central São Paulo, the Luz and Santa Ifigênia neighbourhoods, characterized by deprivation of their physical environments and social structures, were the focus of this investigation. The Mungunzá Container Theatre and the General Osório Square, located within these neighbourhoods, were selected as case-studies. Whilst the *thermal* performance of the container theatre itself was the main interest, in the case of the Square the fundamental issue was the environmental *noise*. The objective was to identify adequate strategies to improve environmental conditions in these locations in order to enhance positive social impact, and, then, contribute to the regeneration of these neighbourhoods. This research was based on fieldwork and analytical procedures of *thermal* and *acoustic* performances. In the container theatre building, the adoption of external shading and wider openings for ventilation reduced its indoor peak temperatures and delivered *thermal* comfort during the warmest period of the year. In the Square, *sound* absorber road surface material and an *acoustic* shell were proposed to reduce *noise* and promote better *acoustic* quality for *outdoor* performances.

Conference Paper

381 **Occupants' evaluation of multi-aspect indoor-environmental exposure: An experimental case study**

Berger C.,Schuss M., Teufl H.,Mahdavi A.

16th Conference of the International Society of Indoor Air Quality and Climate Creative and Smart Solutions for Better Built Environments Indoor Air 2020

2020

0

Hide abstract Find It @ UCL View at Publisher Related documents

The study presented in this paper intends to contribute to the efforts to transcend single-domain assessment approaches that separately treat occupants' perception and evaluation of the different dimensions (e.g., *thermal*, visual, auditory) of indoor environments. For this purpose, a laboratory setting is adapted, involving two identical small office units. *Thermal* and visual conditions can be separately controlled in both units. Moreover, different *outdoor acoustic* circumstances (e.g., traffic *noise*) can be emulated. In the course of the experiments, participants occupy these offices on a short-term basis and are exposed to different combinations of *thermal*, visual, and *acoustical* factors. Specifically, the experiments explore if identical *thermal* conditions are evaluated differently due to interference caused by other exposure variables such as glare and *noise*. The paper investigates if there are differences between *thermal* comfort evaluations of participants, who experience the same *thermal* conditions but different lighting and/or *acoustical* circumstances.

Conference Paper

	Document title	Authors	Source	Year	Citations
382	**Indoor environmental quality evaluation of an institutional building**	Zhong L.	ASHRAE Transactions , 126	2020	1

Hide abstract Find It @ UCL View at Publisher Related documents

Indoor environment in an institutional building, including ventilation, indoor air quality (IAQ), *acoustic* and *thermal* conditions, as well as lighting directly influence the students' study and health performance. In this paper, all five aspects were studied to evaluate the indoor environmental quality (IRQ) in an institutional building. This study examines IEQ parameters, including pressure, illuminance level, *acoustics*, carbon dioxide (CO_2) levels, temperature, and humidity, with appropriate monitors allocated during a lecture duration (50min or 80 min) in four lecture classrooms (59187 ± 1554 ft^3, 1676 ±44 m^3) repeatedly from October 2018 to March 2019 with the objectives of providing a comprehensive analysis and updating the literature. *Outdoor* temperature was-13°F to 54°F (-25°C to 12°C) during two-season measurements, while the classroom environment was maintained at 73 ± 2°F (23 ± 1°C) and 33% ± 3% RH. Indoor CO_2 had mean concentrations of 550-1055 ppm and a mean *sound* level of 58 ± 3 dBA. Using the one-spne steady-state box model, the air change rates were configured at 1.3-6.5 per hour based on continuous CO_2 measurements and occupant loads in the lectures. Results were compared to ventilation guideline for lecture classrooms by ASHRAE. Buildingfeatures and mechanical operation details were discussed to analyse the relationships between building design/operation and IEQ performance.

Conference Paper

	Document title	Authors	Source	Year	Citations
383	**Performance of QPSK modulation for FSO under different atmospheric turbulence**	Janu D.,Janyani V.	Lecture Notes in Electrical Engineering , 546	2020	2

Hide abstract Find It @ UCL View at Publisher Related documents

Free space optical (FSO) communication link provides high-speed data transmission rate within line of sight range for indoor as well as *outdoor* applications. FSO comes with high sensitivity for variation in weather condition as it reflects in the form of change of dielectric properties of medium. The uniqueness of the presented model in this paper is that it achieves 125 GBPS of data rate with QPSK modulation scheme and that too at 1550 nm of wavelength which is compatible with existing optical backbone network. In this paper, authors analyzed the performance of a FSO link with 0.6 km length and modulation scheme QPSK for different atmospheric conditions. Gamma–Gamma distribution model is employed to model the FSO channel link. Performance comparisons are recorded as bit error rate (BER) and signal to *noise* ratio (SNR) with help of simulation tool Optisystem13.

Conference Paper

	Document title	Authors	Source	Year	Citations
384	**Analysing Energy Savings and Overheating Risks of Retrofitting Chinese Suburban Dwellings to the Passivhaus EnerPHit Standard**	Liu C.,Sharples S., Mohammadpourkarbasi H.	PLEA 2020 35th PLEA Conference on Passive and Low Energy Architecture Planning Post Carbon Cities Proceedings, 3	2020	0

Hide abstract Find It @ UCL View at Publisher Related documents

Chinese suburban housing largely consists of uninsulated reinforced-concrete apartment blocks with poor insulation performance. Much energy is required to reach indoor *thermal* comfort in Chinese regions which have hot summer/cold winter climates. Retrofitting existing dwellings to a better energy performance standard might be preferable to new build as existing buildings are mostly structurally *sound* and local demand for new homes is low. This paper examines the viability and benefits of applying the German Passivhaus EnerPHit standard to a suburban dwelling in Hunan, which is in the hot summer/cold winter climate zone of China. The indoor and *outdoor* hygrothermal data of the dwelling was monitored for 12 months, and the dwelling was then modelled and validated against the measured data. Incremental EnerPHit retrofitting measures were applied to the modelled building, and the *thermal* performance of the retrofitted building was evaluated using multiple approaches, including a PMV model, the Passivhaus standard and the Chinese passive house standard. The simulation results suggest that the EnerPHit standard is achievable, producing a high energy saving efficiency. Indoor *thermal* comfort was hugely

	Document title	Authors	Source	Year	Citations

improved after retrofitting, but two of the multiple comfort evaluation approaches highlighted that overcooling was a bigger problem than overheating for the retrofitted dwelling.

Conference Paper

385 **Phase change materials ranking by using the analytic hierarchy process**

Nadeem A.,Rakhman K., Hossain A.

Proceedings of International Structural Engineering and Construction , 7(1), CPM-09

2020

5

Hide abstract Find It @ UCL View at Publisher Related documents

Many phase change materials (PCMs) are available with building envelope use potential. PCM in a building envelope stores heat under warm climate as it changes from solid to liquid phase at the melting temperature. Mainly, the latent heat of fusion is stored, which reduces the daily heat flux from the outdoors into the building. Thus, the indoor cooling requirement is reduced. When the *outdoor* temperature falls, PCM releases the stored heat into the atmosphere over the phase change from liquid to solid that may offset the internal heating needs depending on the design of the PCM application. The PCMs exhibit a great variety in their properties. While many researchers are optimizing the use of PCM for a building application under specific climates, most of them focus on the energy-saving potential of PCMs. Other criteria, such as economic feasibility, environmental hazards, stability, fire resistance, and nonthermal physical properties, should also be considered. Therefore, with many criteria to consider, a *sound* selection method is necessary for choosing a suitable alternative PCM in an application. Some phase change materials were examined and ranked by taking into consideration their *thermal* and non-*thermal* properties. These properties were assessed using a multi-criteria decision-making tool called the Analytic Hierarchy Process (AHP). The results of this paper illustrate a pathway for selecting suitable PCMs and provide support to PCM application in building envelopes.

Conference Paper

386 **Phase change materials ranking by using the analytic hierarchy process**

Nadeem A.,Rakhman K., Hossain M.A.

3rd European and Mediterranean Structural Engineering and Construction Conference 2020 Euro Med Sec 2020, CPM-09

2020

3

Hide abstract Find It @ UCL View at Publisher Related documents

Many phase change materials (PCMs) are available with building envelope use potential. PCM in a building envelope stores heat under warm climate as it changes from solid to liquid phase at the melting temperature. Mainly, the latent heat of fusion is stored, which reduces the daily heat flux from the outdoors into the building. Thus, the indoor cooling requirement is reduced. When the *outdoor* temperature falls, PCM releases the stored heat into the atmosphere over the phase change from liquid to solid that may offset the internal heating needs depending on the design of the PCM application. The PCMs exhibit a great variety in their properties. While many researchers are optimizing the use of PCM for a building application under specific climates, most of them focus on the energy-saving potential of PCMs. Other criteria, such as economic feasibility, environmental hazards, stability, fire resistance, and non-*Thermal* physical properties, should also be considered. Therefore, with many criteria to consider, a *sound* selection method is necessary for choosing a suitable alternative PCM in an application. Some phase change materials were examined and ranked by taking into consideration their *thermal* and non-*Thermal* properties. These properties were assessed using a multi-criteria decision-making tool called the Analytic Hierarchy Process (AHP). The results of this paper illustrate a pathway for selecting suitable PCMs and provide support to PCM application in building envelopes.

Conference Paper

	Document title	Authors	Source	Year	Citations
387	**Welcome to the School of Heat: Assessing heat-perception in schools through participative research and emotional design as drivers for sustainable and healthy approaches in non-adult population**	Dominguez S.,Cuerdo T., Fernandez-Aguera J., Campano M.A.	11th Windsor Conference Resilient Comfort Windsor 2020 Proceedings	2020	0

Hide abstract Find It @ UCL View at Publisher Related documents

Students are exposed during long time to habitually poor environments in schools, and given that their vulnerability and social function makes them a priority, many previous research related Indoor Environment Quality (air-contamination and *thermal* environment) with students-performance and cognitive processes. Most of these approaches focus on cold-period, but Climate Change processes are lengthening the warm-season, particularly in southern Europe, where overheating episodes have evolved from incidental towards a usual risk. Users often have restricted resources to apply adaptive-strategies, such as modifying clothing, attitude and position, as well as denning the natural-ventilation -traditional in these buildings- limited by *outdoor noise*, pollution and others. The efforts of the authorities have mainly been directed towards a timely mitigation, usually circumstantial, and not toward resilience of these facilities, often advising resignation as an answer, thus the “school of heat” concept. Within this study, research participatory-action techniques have been developed, involving schoolchild in the definition of their perceived environment as an empowerment tool, enhancing the evolution of the user role from passive receptor to active drivers. The work identifies the greatest strengths proposed by users, using the environmental, psychological and health impact of school users to generate potential tools for the improvement of the spaces.

Conference Paper • *Open access*

	Document title	Authors	Source	Year	Citations
388	**Green faÇades: Benefits, indicators and design parameters**	Moreira-Zambrano V.A., Moreno-Rangel D.	Wit Transactions on the Built Environment , 195	2020	2

Hide abstract Find It @ UCL View at Publisher Related documents

In the next five years, according to the World Meteorological Organization, the global average temperature will rise between 1.2°C and 1.3°C above the pre-industrial period. For the United Nations, if we support the latest analysis, we can reduce carbon emissions and stop the increase in the average annual temperature to below 2°C. The United in Science Report, issued at the Climate Action Summit, showed the importance of developing actions that could stop global warming and the effects of climate change. In the topics exposed, it proposed acting through local and city actions, which allow the mitigation and resilience at the urban and local levels, focused on low-emission buildings and urban infrastructure; where vertical gardens allow carrying out these actions. The green façades not only reduce the effects of global warming, but also help to improve indoor and *outdoor* comfort. Beyond the aesthetic benefits, they allow the increase of local biodiversity, food production, decrease pollution rates, help *acoustic* and *thermal* insulation of internal spaces and benefit energy savings. The objective of this research is to analyze how vertical gardens act in the *outdoor* areas, comparing the indicators and design parameters that establish the environmental certifications, such as LEED, BREEAM and CASBEE. The methodology for the research is developed in three stages: (i) study of the vertical garden as a mitigating element of the heat island effect and external comfort; (ii) analysis and comparison of environmental certifications, to know what are the indicators and design parameters that they consider for the implementation of green façades; and (iii) quantitative and qualitative benefits of the implementation of green façades in urban environments from the results obtained through a simulation in the ENVI_MET software, applied to a specific case study. According to the analysis based on the simulation performed, the use of these vegetable skins in small percentages (10–20%) minimally improves exterior comfort.

Book Chapter

	Document title	Authors	Source	Year	Citations
389	**Indoor air quality**	Paleologos K.E., Selim M.Y.E., Mohamed A.-M.O.	Pollution Assessment for Sustainable Practices in Applied Sciences and Engineering	2020	36

Hide abstract Find It @ UCL View at Publisher Related documents

Document title	Authors	Source	Year	Citations

This chapter discusses the air pollutants that are found indoors, from what materials and sources they originate, their sampling and measurement, and how they affect human health by presenting case studies with an emphasis on residential buildings and schools. It also presents various indoor air quality (IAQ) guidelines, management, and technologies, as well as the impact of indoor environmental quality (IEQ) on health and comfort. Furthermore, the concept of green buildings is highlighted with its features related to building design and management, IAQ guides, organizations, and practices. The need for optimized buildings ventilation, air conditioning systems design, and indoor air purification to ensure IAQ is also presented. Active versus passive control methods of IAQ are reviewed and examples of residential and schools buildings are discussed in relation to sampling guidelines and IAQ practices. In addition, the IEQ, which in addition to IAQ extends to *thermal* conditions, *noise*, and light of an interior space, is expounded in view of the design of buildings, which need to function effectively in more aspects than those emphasized in current sustainability guidelines. Although significant progress has been made in the construction industry by following sustainability regulations, the emphasis has been on energy, water, and material savings. The current chapter makes the case that health, psychological, and productivity factors are equally important when considering the design, construction, and operation of buildings and, hence, need to be incorporated in current sustainability regulations.

Article • *Open access*

☐ 390 **Climate adaptive design improvement strategies of traditional dwellings in southern Zhejiang for the plum rain season considering comfort conditions**

Zheng R.,Zheng Y.,Cong L., Choi J.-H.,Jung H.

Energies, 13(6), 1428

2020

15

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

This study investigated the adaptations of traditional dwellings to the complex regional microclimate in southern Zhejiang, China. Typical traditional dwellings in a village in the foothills and a village on the mid-slopes of Zhejiang's alpine region were selected to study traditional construction strategies for climate responsiveness and the comfort level of indoor environments during the very humid plum rain season in early summer. Fundamental analysis of the climate and architecture, a response analysis of the dwelling form, an occupants' comfort satisfaction survey, and field measurements of indoor and *outdoor thermal* environmental parameters were performed. The traditional dwellings and their design strategies for various regional environmental factors were explored from the perspective of (1) regional climate-adaptive strategies, (2) the *thermal*, airflow, lighting, and *acoustic* qualities of the indoor environment, and (3) the occupants' indoor environment satisfaction. The results indicated that traditional dwellings in southern Zhejiang incorporate strategies of various effectiveness in ensuring indoor comfort.

Book Chapter

☐ 391 **Pattern Classification for Interpreting Sensor Data from aWalking-Speed Robot**

Dieckman E.A., Hinders M.K.

Intelligent Feature Selection for Machine Learning Using the Dynamic Wavelet Fingerprint

2020

0

Hide abstract ^ Find It @ UCL ↗ View at Publisher ↗ Related documents

In order to perform useful tasks for us, robots must have the ability to notice, recognize, and respond to objects and events in their environment. This requires the acquisition and synthesis of information from a variety of sensors. Here we investigate the performance of a number of sensor modalities in an unstructured *outdoor* environment including the Microsoft Kinect, *thermal* infrared camera, and coffee can radar. Special attention is given to *acoustic* echolocation measurements of approaching vehicles, where an *acoustic* parametric array propagates an audible signal to the oncoming target and the Kinect microphone array records the reflected backscattered signal. Although useful information about the target is hidden inside the noisy time-domain measurements, the dynamic wavelet fingerprint (DWFP) is used to create a time-frequency representation of the data. A small-dimensional feature vector is created for each measurement using an intelligent feature selection process for use in statistical pattern classification routines. Using experimentally measured data from real vehicles at 50 m, this process is able to correctly classify vehicles into one of five classes with 94% accuracy.

	Document title	Authors	Source	Year	Citations
392	Article • *Open access* **Indoor environmental quality evaluation of lecture classrooms in an institutional building in a cold climate**	Zhong L.,Yuan J.,Fleck B.	Sustainability Switzerland , 11(23), 6591	2019	43

Hide abstract Find It @ UCL View at Publisher Related documents

In this paper, ventilation, indoor air quality (IAQ), *thermal* and *acoustic* conditions, and lighting were studied to evaluate the indoor environmental quality (IEQ) in an institutional building at the University of Alberta in Edmonton, Canada. This study examined IEQ parameters, including pressure, illuminance, *acoustics*, carbon dioxide (CO_2) concentration, temperature, and humidity, with appropriate monitors allocated during a lecture (duration 50 min or 80 min) in four lecture classrooms repeatedly (N = 99) from October 2018 to March 2019 with the objectives of providing a comprehensive analysis of interactions between IEQ parameters. The classroom environments were maintained at 23 ± 1 °C and 33% ± 3% RH during two-season measurements. Indoor mean CO_2 concentrations were 550-1055 ppm, and a mean *sound* level of 58 ± 3 dBA was observed. The air change rates were configured at 1.3-6.5 per hour based on continuous CO_2 measurements and occupant loads in the lectures. A variance analysis indicated that the within-lecture classroom variations in most IEQ parameters exceeded between-lecture classrooms. A multilayer artificial neural network (ANN) model was developed on the basis of feedforward networks with a backpropagation algorithm. ANN results demonstrated the importance of the sequence of covariates on indoor conditions (temperature, RH, and CO_2 level): Air change rate (ACR) > room operations (occupant number and light system) > *outdoor* conditions.

393	Conference Paper • *Open access* **Cooling load and noise characterization modeling for photovoltaic driven building integrated thermoelectric cooling devices**	Dehra H.	E3s Web of Conferences , 128, 01019	2019	4

Hide abstract Find It @ UCL View at Publisher Related documents

Photovoltaic driven *thermoelectric* cooling devices are investigated for installation in a modular *outdoor* test-room. Because of Peltier effect in a *thermoelectric* cooling (TEC), heating and cooling is achieved by applying a voltage difference across the *thermoelectric* module. Theoretical design modeling of cooling load and *noise* characterization of building integrated *Thermoelectric* (TEC) Devices is analyzed. System design of photovoltaic driven TEC devices is investigated with varying fresh *outdoor* ventilation rates. Building integrated design of TEC devices inside ceiling suspended duct along with TEC devices mounted on wall driven by rooftop and active façade photovoltaic devices is considered in the analysis. In this way, two-stage dehumidification is achieved by two different sets of TEC devices. The investigation is conducted for effect of voltage, air flow rate and height of fin heat transfer surface. Expressions along with results for *noise* characterization in photovoltaic driven building integrated TEC devices are also provided.

394	Article **A robust air balancing method for dedicated outdoor air system**	Cheng F.,Cui C.,Zhang X., ... Su Y.,Mao T.	Energy and Buildings , 202, 109380	2019	25

Hide abstract Find It @ UCL View at Publisher Related documents

Air balancing is an important means to reduce energy consumption and improve *thermal* comfort and indoor air quality for dedicated *outdoor* air system (DOAS). It effectively distributes the required amount of air to each conditioned zone through dynamic regulation of terminal/branch dampers and fan speed. Up to now, most of the existing air balancing methods are modeled by quadratic models that greatly rely on the accuracy of the collected data and are therefore difficult to obtain satisfactory forecasting results. *Noise*, outlier and model approximation error are likely to cause large prediction errors in air balancing, leading to increased demand for new methods to overcome these problems. In this paper, a robust air balancing method based on data-driven model is proposed to produce a more accurate prediction of pressure difference. With precise prediction, the proposed method obtains a more accurate balancing result and is more robust against *noise* and outlier. The proposed method includes three parts: building mathematical model for duct system based on steady-state pressure balance; training model parameters by robust air balancing learning algorithm; estimating damper positions by the damper characteristic curve. The performance of the proposed method is validated on a laboratory duct test-bed with five terminals. The

	Document title	Authors	Source	Year	Citations

relative error can be controlled with 5.5% in the test cases. Compared with the existing air balancing methods, the proposed method is more accurate in prediction and more robust in the balancing results. The proposed method offers the possibility to obtain a superior balancing performance when sensor accuracy cannot be guaranteed in practical applications.

Article

395 **Performance analysis of liquid-cooled panel system for air conditioner using peltier effect**

Harun M.H., Halim M.F.M.A., Azahar A.H., ... Abidin A.F.Z., Basar M.F.M.

Journal of Advanced Research in Fluid Mechanics and Thermal Sciences, 62(1)

2019

1

Hide abstract Find It @ UCL View at Publisher Related documents

This study reviews the recent advances of *thermoelectric* materials, modelling approaches, and applications. *Thermoelectric* cooling systems have advantages over conventional cooling devices, including compact in size, light in weight, high reliability, no mechanical moving parts, no working fluid, being powered by direct current, and easily switching between cooling and heating modes. In this study, it focused on the replacement of the *outdoor* unit (compressor) for conventional air-conditioner. This system using water as cooling agent in replacing refrigerant system. It involves using Peltier Effect which converts current to rise the temperature between the two junctions of the Peltier. The water will flow through these changes in temperature which results in cooling the air through the process. From this process, it taken 60 minutes in order to achieve 20.2 °C and consumed 1.3 A which is better rather using the conventional method. This can reduce the cost of the project in terms of maintenance, bills, materials and gases. This system can be operated without relying to the compressor which improve the systems in terms of *noise*, low electricity usage, small size and use water as coolant agent.

Conference Paper • *Open access*

396 **Sustainable design of vegetated structures: Building freshness**

Cravero J.,Feraille A., Versini P.-A., ... Tchiguirinskaia I., Baverel O.

Iop Conference Series Earth and Environmental Science , 323(1), 012021

2019

0

Hide abstract Find It @ UCL View at Publisher Related documents

City revegetation strategies seem appealing to mitigate urban heat island effects through shading and transpirational cooling. Moreover, other potential benefits that may derive, e.g. biodiversity enhancement, the reduction in buildings energy consumption, stormwater management, *acoustic* insulation or air purification, earned them the designation 'no-regrets approaches' for adapting to climate change. However, the lack of understanding and quantification of green infrastructures' environmental impacts prevents urban planning policies to be consistent and to turn attractive initiatives to effective implementations. The monitoring of existing green infrastructures is required to evaluate their cooling effect. For this purpose, an elastic gridshell in composite materials has been designed as a support for climbing plants at Ecole des Ponts ParisTech (Champs-sur-Marne, France). The life cycle assessment of the vegetated structure is performed in order to develop sustainable design strategies. Based on an energy balance approach, the collected *thermo*-hydric data can be used to determine which mechanisms are the most suitable for urban vegetation to enhance *outdoor thermal* comfort.

Conference Paper • *Open access*

397 **Classification of Indoor Environment 2018 and Updated Criteria of RTS Environmental Classification**

Sariola L.,Sateri J., Rintala T.

Iop Conference Series Earth and Environmental Science , 297(1), 012045

2019

2

Hide abstract Find It @ UCL View at Publisher Related documents

The Finnish Society of Indoor Air Quality and Climate (FiSIAQ) introduced over twenty years ago in 1995 a Classification of Indoor Climate, Construction Cleanliness, and Finishing Materials. After two revised versions, the

Document title	Authors	Source	Year	Citations

fourth edition Classification of Indoor Environment was published on May 2018. Based on the criteria set in the classifications, The Building Information Foundation RTS sr started M1-labelling of building products in 1996, named Emission Classification of Building Materials and Cleanliness Classification of Air-handling Components. Indoor air quality and instruments in improving all the sectors to concern are in great importance in Finland. First part of The Classification of Indoor Environment 2018 (CIE 2018) includes new Requirements for Technical Target Values. The other part, Guidance for design and Construction, includes surface structural design, construction site design, Classification of Construction and Construction Work Cleanliness (P1), Classification of Building Materials (M1), Cleanliness Class of Air-handling components (M1) and instructions for designing mechanical systems for buildings. The main changes in the 2018 edition are revised target values for indoor operative temperature, inclusion of a PM2.5 criteria, revision of some *acoustical* target values and revision of recommended *outdoor* air flows. Regarding *thermal* comfort, the target values of the individual indoor climate, S1, will be slightly more demanding than the targets of the S2, good indoor climate, class. The criteria for particulate matter, PM2.5, has been included due to the significant health impact of fine particles. The targets for *acoustic* environment in offices have been revised to reflect the importance of speech transmission. The recommended *outdoor* air flows follow the principles of EN15251-2007, considering the common usage of very low-emitting M1-class materials in Finland. The *outdoor* air flow rates have also been aligned with the recently changed building code values. Companies have developed low-emitting building materials, fixtures, furniture and HVAC-devices during the years to meet the requirements. There are almost 5200 M1-classified low emitting building materials, fixture, furniture and HVAC-devices at the moment (May 2019). The Classification of Indoor Environment 2018 (classes S1, S2) and sub-classifications (P1) and (M1) are integrated part of the new Finnish environmental classification system for building processes called RTS Environmental Classification launched in the spring 2017 and updated in September 2018. The Classification of Indoor Environment has been used in multiple office and residential construction projects during the years. By integrating the CIE 2018 requirements to the RTS 2018, third-party audit and certification of the whole construction project is possible. Therefore, the continuous improvement will be one considerable option.

Article • *Open access*

☐ 398 **Differences in occupants' satisfaction and perceived productivity in high- and low-performance offices**

Gocer O.,Candido C., Thomas L.,Gocer K.

Buildings, 9(9), 199

2019

56

Hide abstract Find It @ UCL View at Publisher Related documents

This paper reports the results from a dataset comprising 9794 post-occupancy evaluation (POE) surveys from 77 Australian open-plan offices. This paper specifically focuses on a sub-set of 20 offices (n = 2133), identified from ranking 10 offices each, with the least (n = 1063) and highest (n = 1070) satisfaction scores, respectively. The satisfaction scores were evaluated on the basis of seven factors (i.e., building/office aesthetics and quality, *thermal* comfort and indoor air quality, *noise* distraction and privacy, personal control, connection to *outdoor* environment, maintenance and visual comfort, and individual space). Using the POE survey data from 20 offices, regression analyses and two-way ANOVA tests were carried out to understand the differences in occupants' satisfaction and perceived productivity arising from open-plan offices. According to the statistically significant regression analyses results, it was identified that building/office aesthetics and quality ($\beta = 0.55$, $p < 0.001$) and *noise* distraction and privacy ($\beta = 0.33$, $p < 0.001$) were the two strongest predictors contributing perceived productivity in low-performance offices. Two-way ANOVA test results for the 10 high-performance offices indicate that the perceived productivity was strongly associated with the office's physical configuration, the employees' working experience, and the working hours at that office.

Conference Paper • *Open access*

☐ 399 **Indoor climate and energy performance in nearly zero energy day care centers and school buildings**

Ahmed K.,Kuusk K., Heininen H.,... Lolli N., Kurnitski J.

E3s Web of Conferences , 111, 02003

2019

4

Hide abstract Find It @ UCL View at Publisher Related documents

This study presents the assessment of actual indoor climate condition and energy performance in eight NZEB school and daycare centers of NERO H2020 project. Physical parameters such as indoor temperature, relative humidity, CO2, airflow rate and temperature were measured during heating and cooling seasons, in parallel with an occupants'

Document title	Authors	Source	Year	Citations

questionnaires survey. Besides, calculated and measured energy data was collected from energy performance certificates and energy bills. *Thermal* comfort and IAQ were assessed based on categories in EN15251 standard with color footprints. Results showed that all the buildings had good or excellent indoor climate during the heating season. However, a large percentage of occupied hours were categorized as category IV during the cooling season, which mainly occurred due to too low indoor temperature caused by the low *outdoor* temperature during the measurement period. Also, all buildings achieved low CO2 levels. Moreover, the conducted questionaries' survey showed good correlation with measured results for all buildings except in one building, which had odor and *noise* problems. In contrast, the measured energy use in 5 buildings out of 7 was increased by factor 2.1-3.0 compared to calculated annual energy use due to a full-time operation of the ventilation system and presence of hot kitchens.

Article • *Open access*

400 **Real time monitoring of indoor environment quality and energy consumption in a residential building** — Toderasc M.,Iordache V., Petcu C.,Petran H. — Environmental Engineering and Management Journal, 18(7) — 2019 — 6

Hide abstract Find It @ UCL View at Publisher Related documents

Several theoretical studies were oriented towards the relationship between the main two building design estimators (building energy consumption and indoor environmental quality) but very little experimental research has been carried out in order to underline the correlation between this two parameters. This paper is focused towards the experimental study of the correlation between these two design estimators. Several indoor comfort parameters (air temperature, radiant mean temperature, *sound* pressure level, *outdoor* ventilation rate and lighting level) as well as energy consumption were monitored in different rooms inside a laboratory experimental house at real scale located in Bucharest, Romania. Time variations of these parameters and space distribution maps inside the house were analysed in order to understand the indoor comfort variations. Four comfort indexes (*thermal* comfort index, *acoustic* comfort index, indoor air quality index and visual comfort index) were determined and their time variations were used to emphasize the correlation between the indoor environment quality and the energy consumption. The overall indoor environment quality was found to be inversely correlated with the energy consumption and that this general trend can be easily influenced by the weight of the four comfort types.

Previous 1 2 3 4 Next

Display 200 results

Back to top

About Scopus

What is Scopus

Content coverage

Scopus blog

Scopus API

Privacy matters

Language

日本語版を表示する

查看简体中文版本

查看繁體中文版本

Просмотр версии на русском языке

Customer Service

Help

Tutorials

Contact us

ELSEVIER

Terms and conditions ↗ Privacy policy ↗ Cookies settings

All content on this site: Copyright © 2026 Elsevier B.V. ↗, its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.

RELX™